Since we worked for the last two semesters, I refrain from giving general advice. Shall remind you the following. Please start writing solutions of exercises, read yourself later and see if you understand. if you do not understand, that means you have to practice writing. This will help you not only to understand better but also to express better.

where are we:

So far we learnt some basic properties of real numbers,
convergence of sequences helped to 'approximate' unknown numbers with known fractions; discovered new numbers like e and Euler's constant, and various other things. convergence of series helped us to add infinitely many numbers which in turn helped us to make sense of power series — natural extension of the high school concept of polynomial — gives us a way to discover functions.
We learnt continuity and differentiation of functions from R to R. These are essential because most of the physical processes are described by functions satisfying differential equations. Integration of functions helped in calculating areas; allowed us to define new functions; For example

$$f(x) = \int_0^x \frac{1}{\sqrt{2\pi}}\, e^{-t^2/2}\, dt; \qquad x \in R$$

is a function which has no explicit formula using the functions you know. Integration will soon help us solve some differential equations
We then learnt functions of several variables. This topic will be further continued in your current Calculus course to develop various useful stories.

In our course we shall discuss Analysis. Basically we discuss
(1) some set theory;
(2) We construct real number system, that is, we construct a set and define operations $+$ and $\times$ and define an order $\leq$ and show all axioms we assumed in our first course hold;
(3) We discus some 'topological aspects'. interesting and useful theorems of Cantor, Baire and Banach,
(4) we discuss about sets of continuous functions and an important theorem of Arzela and Ascoli,
(5) Finally we discuss some Fourier series.
Actually, questions concerning Fourier series lead to the discovery and devel-

opment of set theory by Cantor (in fact Fourier series is responsible for the rigorous development of most of analysis).

Sets:
Many of these things below start from things you already know and then develop to new stories. A set is any collection of objects. If S is a set and x is an object in this set then we write $x \in S$. If x is not an object in this set we write $x \notin S$. This definition is good enough for us. The reason for making this statement is that if you literally take the above definition you will end up in problems.

You must have heard about the barber paradox. Consider only males. A barber in a town declared that he shaved those and only those who have not shaved themselves. Decide whether the barber shaved himself or not. Thus if you consider the set S of all persons in the town whom the barber shaved, you are unable to decide whether the barber himself is a member of this set or not. Thus even though S is defined as a collection of objects, we are unable to decide if a specific object (namely, the barber) is in this set or not.

There are other such paradoxes too. Let k be the least natural number that can not be described in less than hundred letters. Does this make sense? After all there are at most 26 natural numbers that can be described using one letter; 26×26 natural numbers at most using two letters and so on. You see that only finitely many natural numbers can be described using at most hundred letters. Thus there must be numbers that can not be so described. Let S be the set of those natural numbers. Since it is not empty, it must have a first element in it. Is this in S or not? Or, Consider the collection S of all sets. This is clearly a collection of objects. Is this a set? If so it should belong to itself. Do you believe it?

such sentences are called self referential, the sentence refers to the object being defined already. Such statements lead to problems (you will see positive use of this kind of reasoning soon when we come to Cantor's diagonal argument). Sometimes you see a definition like: a set is a well defined collection of objects. But then, what is meant by well defined? You have to start from somewhere. Thus for us a set is a collection of objects. We leave like this, but be assured that set theory is a well defined rigorous theory, but our course is not set theory!

Set operations:

If S and T are two sets, then they are same and we write $S = T$; if they have the same objects. In other words $x \in S$ implies $x \in T$ and $x \in T$ implies $x \in S$. We say S is a subset of T and write $S \subset T$; if $x \in S$ implies $x \in T$. Thus every object which is in S is also in T. We also say that T is a superset of S and write $T \supset S$.
If S and T are two sets then their union $S \cup T$ is the collection of those objects which are either in S or in T, they could be in both the sets. Similarly their intersection $S \cap T$ consists of all objects which are in both S and T.

$$S \cup T = \{x : x \in S \text{ or } x \in T\}; \quad S \cap T = \{x : x \in S \text{ and } x \in T\}.$$

We define $S \setminus T = \{x : x \in S; \quad x \notin T\}$, that is, it consists of all objects which belong to S but which do not belong to T.
In case we have a grand set Ω under consideration and if we are only considering objects that belong to Ω, then for $\Omega \setminus S$ we write S^c and is called complement of S. Since objects which are in Ω are under discourse, it is called universe. Sometimes it is also called universal set, but there is nothing universal about it; the word only means that it is the universe now and all objects we are talking about are in this set (whether we say so or not).

Thus, unlike union and intersection which have no reference set, complement must be with reference to something. It may be specified or understood from the context.

Theorem (DeMorgan's laws): $(A \cup B)^c = A^c \cap B^c$; $\quad (A \cap B)^c = A^c \cup B^c$.
Proof:
$x \in (A \cup B)^c$
$\Rightarrow x \notin (A \cup B)$ (def. complement)
$\Rightarrow x \notin A$; and $x \notin B$ (def union)
$\Rightarrow x \in A^c$ and $x \in B^c$ (def complement)
$\Rightarrow x \in A^c \cap B^c$ (def intersection).

These arrows can be reversed and equality is obtained. The second equality can be proved similarly.

Many times we use the word 'similar'. You should not blindly copy it like a faithful elementary school student. This phrase is an abbreviation for the following. 'I have done the proof once, it is similar to the earlier part and so I omit writing it (probably to spare the reader from getting bored).' As a result you should verify and earn the eligibility to use this phrase.
A set which has no objects is called empty set (that such a thing exists is an axiom). It is denoted by $\emptyset$.
Let A and B be two sets. Their Cartesian product $A \times B$ is defined as the set of all ordered pairs, with first element of the pair being an object from A and

the second element being an object from B. $A \times B = \{(x, y) : x \in A; y \in B\}$. Of course I have used the word ordered pair (x, y) and you might wonder what the hell it is. If you are wondering, here it is $(x, y) = \{x, \{x, y\}\}$. It is a set consisting of two objects. First object is x. Second object is the set consisting of x and y. Of course you can keep your mental picture in tact and ignore this definition. This definition of ordered pair is only to reinforce the feeling that ordered pair can be defined precisely. (everything is a set!)

functions and cardinality:

Let X and Y be sets. A function on X to Y is a rule that associates with every element of X one element of Y.

You can keep this in mind and there is nothing wrong with this definition. However you may suddenly get a doubt: what is this 'rule' and 'association' business? What is rule and what is not rule? What is association and what is not association? If you get such a doubt, there is no need to worry. Here is something that helps. A function is a subset $R \subset X \times Y$ such that

$$\forall (x \in X)\ \exists!\ (y \in Y)\ \ (x, y) \in R.$$

Here $\exists!$ is an abbreviation for 'there is a unique'. Thus for each $x \in X$ there is exactly one y such that $(x, y) \in R$. If you want to denote your function by f then for a given $x \in X$, this unique point y is denoted by $f(x)$, value of the function f at the point x. Thus R associates with every x one y. If you go back in time, you realize that a function f is a rule as said earlier and 'graph' of the function f is the set of all points (x, y) such that $y = f(x)$. Now if you want to be rigorous and not use unnecessary words like rule/association etc, then you turn things around. By definition the graph is regarded as function. Think about it.

But let me assure you that you should keep in mind the same good old idea of function. What all I said is an explanation, in case you feel you are cheated.

I should say two things. Firstly, as already said, the symbol $\exists! y$ is an abbreviation for there exists unique y. Thus $\forall x \exists! y\ \ (x, y) \in R$ means

$$\forall x[\{\exists y\ \ (x, y) \in R\}\ \ \wedge\ \ \{(x, y_1) \in R, (x, y_2) \in R \Rightarrow y_1 = y_2\}].$$

Second point is the following. Sometimes you see a different (not equivalent to ours) definition of function. It is a subset $R \subset X \times Y$ such that

$$(x, y_1) \in R, (x, y_2) \in R \Rightarrow y_1 = y_2.$$

Thus for any point x there can not be two y. However it is quite possible there is no y at all. Thus for each x there is at most one y such that

$(x, y) \in R$. The set of x for which there is such y is called the 'domain' of the function and for x in the domain the unique y is thought of as $f(x)$. In a sense these are 'partial' functions. What we defined are 'total' functions. We stick to our definition.
f is one-to-one, or 1-1 or injective if two different points have different images. That is $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$.
f is onto or surjective if image is all of Y. That is $\forall y \exists x\ (x, y) \in R$.
f is bijective if it is both one-one and onto.
Since you have seen several examples of functions, there is no need for me to give examples here in a text book style.

Cardinality:
The number of elements or cardinality of empty set is 0; $|\emptyset| = 0$. If X is a non-empty set and there is a bijection from X to $\{1, 2, \cdots, n\}$ then we say that X has n elements; $|X| = n$
For this definition to make sense you should first convince yourself that the following can not happen: X has bijection with $\{1, 2, \cdots, 25\}$ and also X has a bijection with $\{1, 2, \cdots, 33\}$. Because if this happens then we would be saying $|X| = 25$ and also $|X| = 33$. Yes, such a thing as above can not happen. (Why?)
A set which is either empty set or is in bijection with $\{1, 2, \cdots, n\}$ for some $n \in N$ is said to be a finite set. A set which is not finite is said to be infinite. If X is in bijection with N, we say that X is countably infinite. If X is either finite or countably infinite, we say that X is countable. If X is not countable then we say that X is uncountable.
Several properties of countable sets were discussed last year. We showed that the set of real numbers R is uncountable. You should recall those.
We have only associated 'numbers' with finite sets. There are numbers associated with infinite sets too. If X is countably infinite, we say $|X| = \aleph_0$ (aleph zero). We say $|R| = c$ (c stands for 'continuum').
How do you compare sets? The good old primitive way. Associate with each student the chair he/she is sitting. If some 'unassociated chairs' are leftover after associating chair to each student, then we say there are more chairs than students. If 'unassociated students' are leftover after associating a student to each chair, then we say there are more students than chairs.
Of course, you might think we have made so much progress, we can as well 'count' the number of students and number of chairs and then compare these numbers. But the point is you are using too many unnecessary words in this counting. Moreover this (so called) progress stops with finite sets and is useless while comparing infinite sets.

Let X and Y be any non-empty sets, finite or countable or uncountable; anything. We say Y has more elements than X (or X has less elements than Y); write $|X| \leq |Y|$; if there is an injection (one-one function) on X into Y. It is a peculiar situation. We have not defined 'number of elements' in X or Y. But we defined what it means to say X has less elements than Y.
Common sense suggests

$$|X| \leq |X|$$
$$|X| \leq |Y|; \quad |Y| \leq |Z| \Rightarrow |X| \leq |Z|$$
$$|X| \leq |Y| \quad |Y| \leq |X| \Rightarrow |X| = |Y|.$$

The first statement is seen to be true using the identity map, second is seen true using composition of functions. The last statement is true but not trivial. It is known as Cantor-Schroder-Bernstein Theorem.

Cantor-Schroder-Bernstein Theorem:
Let X and Y be sets. Let $f : X \to Y$ be an injection and $g : Y \to X$ be an injection. Then there is a bijection $\varphi : X \to Y$. In other words, if $|X| \leq |Y|$ and $|Y| \leq |X|$ then $|X| = |Y|$.
The original proof of Cantor used well-ordering principle. After several refinements, here is a proof.
Let us think of $f(x)$ as child of x. Thus if $f(x) = y$, then y is child of x or equivalently x is immediate ancestor of y. Similarly, $g(y) = x$ would mean child of y is x or equivalently y is immediate ancestor of x. Since f and g are injective an immediate ancestor, if exists, is unique.
Suppose that $x \in X$ has an immediate ancestor y_1 and y_1 has an immediate ancestor x_2 and x_2 has an immediate ancestor y_3 and so on. In other words, we start with $x \in X$;

$$g(y_1) = x; \quad f(x_2) = y_1; \quad g(y_3) = x_2; \cdots .$$

This is the ancestral chain for x. It is quite possible that the ancestral chain of x continues for ever or the ancestral chain of x stood at a finite stage. Again there are two possibilities in the later case, x may have an even number $(0, 2, 4, \cdots)$ of ancestors or an odd number $(1, 3, 5, \cdots)$ of ancestors. Similarly if you start with $y \in Y$ its ancestral chain may continue for ever or may stop with even number of ancestors or odd number. Accordingly we partition the sets

$$X = X_\infty \cup X_e \cup X_o; \quad Y = Y_\infty \cup Y_o \cup Y_e.$$

We define $\varphi(x) = f(x)$ in case $x \in X_\infty$ or $x \in X_e$. And $\varphi(x) = g^{-1}(x)$ if $x \in X_o$. This last clause makes sense because x has an immediate ancestor.

We show that this does.
We show that φ maps in a bijective way X_∞ to Y_∞; X_e to Y_o; X_o to Y_e. This shows that φ is globally bijective too — since we have partitions of the sets X and Y. To realise there is something to be justified here, remember the x^2 map is injective on $(-\infty, 0]$; injective on $[0, \infty)$ but not injective on $(-\infty, \infty)$.
Note that if the ancestral chain of x is infinite then so is that of $f(x)$. Indeed the ancestral chain of $f(x)$ has x and all its ancestors. Hence $f(x) \in Y_\infty$. Conversely if $y \in Y_\infty$, it has an immediate ancestor x. Clearly $x \in X_\infty$ and $f(x) = y$ showing that φ — which is same as f — is a bijection between X_∞ and Y_∞.
If $x \in X_e$ then x has an even number of ancestors. Since ancestors of $f(x)$ consists of x and ancestors of x we see $f(x) \in Y_o$. conversely if $y \in Y_o$, then y has an immediate ancestor, say x. The ancestors of x consists of ancestors of y except x itself. Thus $x \in X_e$ and of course $f(x) = y$. Thus φ, which is same as f is again a bijection between X_e and Y_o.
Similar argument as above shows that $x \in X_o$ implies that $g^{-1}(x) \in Y_e$. Conversely if $y \in Y_e$ then $g(y) \in X_o$ and $\varphi(g(y)) = y$; showing that g^{-1} is a bijection between X_o and Y_e.
This completes the proof. ∎

The proof is tricky, proof using well-ordering principle is straightforward. However to get a feel for the proof, it is better to look at instructive examples. try $X = Y = \{0, 1, 2, 3, \cdots\}$; $\quad f(x) = 2x$; $\quad g(y) = 2y$. and see.

But are there larger and larger sets? Yes, given any set X we can cook up a set Y such that $|X| < |Y|$. This means $|X| \leq |Y|$ and $|X| \neq |Y|$. Here is how. Let X be any set. Then $P(X)$ denotes the collection of all subsets of X. There is an injection on X into $P(X)$, namely, associate with $x \in X$ the singleton set $\{x\}$ in $P(X)$. Thus $|X| \leq |P(X)|$. However there is no bijection. Indeed, if there is a bijection, say $f : X \to P(X)$ then let us define the following subset of X

$$S = \{x \in X : x \notin f(x).\}$$

Since S is clearly a subset of X, there must be a $s \in X$ such that $f(s) = S$.

Assume $s \in S$. The definition of the set S tells you $s \notin f(s)$. But $f(s) = S$. So $s \notin S$. contradiction to the assumption.

Assume $s \notin S$. The definition of S tells you $s \in f(s)$. But $f(s) = S$. Thus $s \in S$. contradiction to the assumption.

But s must be either in S or in S^c. The above argument says none of

these happen. This shows there is no bijection of X with $P(X)$. Thus we proved

Theorem (Cantor) : $|X| < |P(X)|$. ∎

The above argument is known as Cantor's diagonal argument. If you use a symbol $\aleph$ for $|X|$ then it is customary to denote $|P(X)| = 2^{\aleph}$.

Of course, we should make sure that such notations do not contradict existing notations. For example suppose X is finite and $|X| = 5$. Then the above notation tells us $|P(X)| = 2^5$. But we already have a meaning for 2^5 and it is 32. So we must convince ourselves that indeed $|P(X)| = 32$. Yes, you can do it.

(You can ignore this para, if it is confusing) Generally empty set needs special consideration (thought). In the above proof what happens if X is empty set? Same argument works. The only function from X to anything is $R = \emptyset$, simply because whatever be Y, we have $X \times Y = \emptyset$. Recall $P(X) = \{\emptyset\}$. In other words $P(X)$ has one element (namely, empty set). Obviously, if you take this element as y, there is no x such that $(x, y) \in R$. Handling empty set is a little troublesome. You are encouraged to think and free to ask me but not encouraged to worry about empty set.

There is a very very interesting issue now. You know N is a countable set and R is an uncountable set. Question: Is there a subset $S \subset R$ such that $|N| < |S| < |R|$. that is, $\aleph_0 < |S| < c$? that is, S is not in bijection with a subset of N and also not in bijection with R. The following statement is known as Continuum hypothesis:

(CH) if $S \subset R$, then $|S| \leq \aleph_0$ or $|S| = c$. In other words there is no subset $S \subset R$ such that $|N| < |S| < |R|$.

Kurt Godel showed that the above statement cannot be disproved. In other words if you assume the above statement is true you do not end up in any contradiction.

Paul Cohen showed that the above statement cannot be proved. In other words if you assume the above statement is false, you do not end up in any contradiction.

posets and losets:

Let X be a set (not empty). A relation on X is simply a subset of $X \times X$. These are actually called binary relations, but we need not bother. We call them relations.

We say a relation is a partial order if

$$\forall x \quad (x, x) \in R$$

$$\forall x, y, z \quad (x, y) \in R \ \ (y, z) \in R \Rightarrow (x, z) \in R$$

$$\forall x, y, \quad (x, y) \in R \ \ (y, x) \in R \Rightarrow x = y.$$

If the first statement holds, we say that the relation is reflexive. If the second statement holds we say that the relation is transitive. If the third statement holds we say that the relation is anti-symmetric. But you need not remember these technical words at this moment.
If xRy holds, we say x and y are comparable and write $x \leq y$. Thus with this new notation we have (i) $x \leq x$; (ii) $x \leq y$ and $y \leq z$ implies $x \leq z$ and finally (iii) $x \leq y$ and $y \leq x$ implies $x = y$.
In some books, a different definition is used. They, in essence, define $<$ instead of $\leq$. Thus you may see the following definition: a relation R' is a partial order if it satisfies
(i) $\forall\ x, y, z \quad (x, y) \in R', (y, z) \in R' \Rightarrow (x, z) \in R'$
(ii) $\forall\ x \quad (x, x) \notin R'$.

Both these definitions are equivalent. If there is R' as above then first observe that $(x, y) \in R'$ and $(y, x) \in R'$ can not hold simultaneously for any two points x and y. If they both did, then condition (i) implies $(x, x) \in R'$ contradicting (ii). Define $(x, y) \in R \longleftrightarrow x = y \quad \vee \quad (x, y) \in R'$.Then this relation obeys our three rules.

Conversely, suppose we have a binary relation R satisfying our rules. If we define R' by $(x, y) \in R' \longleftrightarrow x \neq y \ \ \wedge \ \ (x, y) \in R$. Then this satisfies the two conditions of the alternative definition. We follow our definition, thus keep in mind, we are talking of $\leq$.
If the following condition also holds then the partial order is called a linear order. $\forall x, y \quad x \leq y \ \ \vee \ \ y \leq x$. Thus any two elements are comparable.

A set with a partial order is called a partially ordered set or simply a poset. A set with a linear order is called a linearly ordered set or simply a loset.

The set of all subsets of the real line with $A \leq B$ if $A \subset B$ is a poset but not a loset.

The following sets are losets. I give only sets, the order is the usual order coming from real numbers.

(i) $X = R$ (ii) $X = [0, 1]$ (iii) $X = [0, 1] \ \cup \ [2, 3]$
(iv) $X = [0, 1) \ \cup (2, 3]$. (v) $X = [0, 1) \ \cup [2, 3]$
(vi) $X = Q$ the set of rationals .

You might wonder why I am giving these examples rather than saying: take any subset of R. Each set above illustrates a property.
In the first example, there is no first element (something smaller than every

other thing is first element) and it has no last element (some thing which is larger than every other thing is called last element). The second set has both of these.
In the first two sets given any two different elements, there is an element strictly in between them. The third set does not have this property. For example between 1 and 2 which are in this set, there is none in between.
In all these three examples, every subset which is bounded above has a supremum. In the fourth example the subset $[0, 1)$ is bounded above but it has no sup. Just to make clear the terms we used, recall from calculus course, $A \subset X$ is bounded if there is an $s \in X$ such that $a \leq s$ holds for all $a \in A$. Such an element s is called an upper bound. An element s is supremum of A if it is an upper bound and it is smallest, that is, if t is an upper bound of A then $s \leq t$.
The fifth example appears deceptively similar to the fourth example. It appears that the subset $[0, 1)$ has no sup, In fact, this subset has sup.
The sixth set is countable and also has interesting properties which we shall discuss next.

Theorem (Cantor's Characterization of Q):
$(Q, \leq)$ has the following three properties: (a) non-empty countable loset (b) has neither first nor last element and (c) between two distinct elements, there is another element.
Conversely, if $(X, \preceq)$ has the above three properties, then it is order isomorphic to Q. that is, there is a bijection $f : X \to Q$ such that $x \preceq y$ iff $f(x) \leq f(y)$. ∎

The proof is a beautiful 'back and forth' argument. We first fix any enumeration.

$$X = \{x_1, x_2, x_3, \cdots\}; \quad Q = \{q_1, q_2, q_3, \cdots\}.$$

We shall re-enumerate the sets

$$X = \{a_1, a_2, a_3, \cdots\}, \quad Q = \{b_1, b_2, b_3, \cdots\},$$

in such a way that the map $f(a_i) = b_i$ is the required isomorphism.

Proof of Cantor's characterization of Q:
So we have X – a (nonempty) countable linearly ordered set, without any first and last points (that is for all x there is y such that $y < x$ and for all x there is z such that $x < z$)and between any two there is one more point (that is given $u < w$ there is a v such that $u < v < w$). Also remember we use $x < y$ to convey $x \leq y$ and $x \neq y$; we use $y \geq x$ to mean $x \leq y$; we use $y > x$ to mean $y \geq x$ and $y \neq x$ which is same as $x < y$. We are denoting the order on X also by $\leq$ and not by $\preceq$. there is no confusion since when you compare you know where the points are.

We start with some observations.
1. X is countably infinite. That it is countable is hypothesis. Since it is assumed to be non-empty, pick $x_0 \in X$, repeatedly use nonexistence of last element to get $x_0 < x_1 < x_2 < \cdots\cdots$. This does. By the way you do it by induction, if you obtained $x_1 < \cdots < x_n$, use the fact that x_n is not last element to get $x_{n+1} > x_n$.
2. Given a finite non-empty set S there is a such that $S < a$, that is $x < a$ for all $x \in S$. This is because S being finite, there is $z \in S$ which is largest, that is, $x \leq z$ for all $x \in S$. Now use z can not be last element. Similar argument shows: Given a finite non-empty set T there is a such that $a < T$, that is $a < x$ for all $x \in T$.
3. Let S and T be two finite non-empty subsets of X with $S < T$. This means $s < t$ for every $s \in S$ and every $t \in T$. Then there is $a \in X$ with $S < a < T$, that is $x < a < y$ for every $x \in S$ and $y \in T$. Since S and T are finite sets, take maximum of S and minimum of T and pick an element between these. Note that finiteness of the sets is important.
First let us fix an enumeration.

$$X = \{x_1, x_2, x_3, \cdots\}; \quad Q = \{q_1, q_2, q_3, \cdots\}.$$

4. We produce, by induction, increasing sets

$$X_n = \{a_1, \ldots, a_n\} \subset X \qquad Q_n = \{b_1, \ldots, b_n\} \subset Q$$

such that (i) $f(a_i) = b_i$ for $1 \leq i \leq n$ is order preserving from X_n onto Q_n and (ii) $\{x_1, \ldots, x_n\} \subset X_{2n}$ and $\{q_1, \ldots, q_n\} \subset Q_{2n}$.
If this is done the proof is complete. Firstly, condition (ii) implies $X_n \uparrow X$ and $Q_n \uparrow Q$; so that $f(a_i) = b_i$ is defined on all of X with range all of Q. And condition (i) implies that $f(a_i) = b_i$ is an order isomorphism between

X and Q.
Here is the back and forth argument. This trick nicely captures every x and every q. We repeatedly use the observations made above.

Step1: Put $a_1 = x_1$ and $b_1 = q_1$. $X_1 = \{a_1\}$ and $Q_1 = \{b_1\}$. (Remember, whatever be your enumeration, we just do this!)
Step 2: Put $b_2 = q_2$. and $Q_2 = \{b_1, b_2\}$. If $b_2 < b_1$ consider the first i such that $x_i < x_1$ and declare this x_i as a_2. If $b_1 < b_2$ consider the first i such that $x_1 < x_i$ and declare this x_i as a_2. $X_2 = \{a_1, a_2\}$. check this does.
Step 3: Put a_3 to be the first unused x_i. That is first i such that $x_i \notin X_2$. (Explanation: I do not know what was my a_2 above. If it was x_2 then my a_3 would be x_3. If that a_2 was not x_2, then my a_3 would be x_2)

$$S = \{b_i : i \leq 2, a_i < a_3\}; \quad T = \{b_i : i \leq 2, a_3 < a_i\}.$$

Choose the first unused q_i such that $S < q_i < T$ and set this q_i as b_3. This is possible by the observation made at the beginning. $X_3 = \{a_1, a_2, a_3\}$, $Q_3 = \{b_1, b_2, b_3\}$ is an order preserving listing.

In general, if we have $X_{2k} = \{a_1, a_2, \cdots, a_{2k}\}$ and $Q_{2k} = \{b_1, b_2, \cdots, b_{2k}\}$ satisfying induction hypotheses, we proceed as follows..
step $(2k+1)$: Put a_{2k+1} to be the first unused x_i. Let

$$S = \{b_i : i \leq 2k, a_i < a_{2k+1}\}; \quad T = \{b_i : i \leq 2k, a_{2k+1} < a_i\}.$$

Note that if $b_i \in S$ and $b_j \in T$, then $a_i < a_{2k+1} < a_j$ so that $b_i < b_j$ — remember the existing listing is order preserving. Choose the first unused q_i such that $S < q_i < T$ and set this q_i as b_{2k+1}. Clearly, $X_{2k+1} = \{a_1, a_2, \cdots, a_{2k}, a_{2k+1}\}$ and $Q_{2k+1} = \{b_1, b_2, \cdots, b_{2k}, b_{2k+1}\}$ will do the job, that is, satisfy (i) and (ii) of induction promise.
step $(2k+2)$: Put b_{2k+2} to be the first unused q_i. Let

$$S = \{a_i : i \leq 2k+1, b_i < b_{2k+2}\}; \quad T = \{a_i : i \leq 2k+1, b_{2k+2} < b_i\}.$$

As earlier if $a_i \in S$ and $a_j \in T$, then $a_i < a_j$. Choose the first unused x_i such that $S < x_i < T$ and set this x_i as a_{2k+2}. Clearly, $X_{2k+2} = \{a_1, a_2, \cdots, a_{2k}, a_{2k+1}, a_{2k+2}\}$ and $Q_{2k+2} = \{b_1, b_2, \cdots, b_{2k}, b_{2k+1}, b_{2k+2}\}$ will do the job; satisfy (i) and (ii) of induction promise.
Pause and think. This completes the proof. ∎
The proof actually shows something more. Let X be a set as in the theorem with a linear order. Given any two elements $x \in X$ and $q \in Q$ we can get an order preserving bijection f so that $f(x) = q$. In particular, such an order

preserving bijection is not unique. You have freedom.

A subset D of the set of real number R is said to be dense if every non-empty interval (a, b) contains a point of D. For example Q the set of rationals, A the set of algebraic numbers and R the set of all real numbers are examples.
Notice that any such set is non-empty, has no first and last element (usual order of R) and between any two different elements of D there is another element.
Theorem 2 (countable dense subsets of R): Let A and B be two countable dense subsets of R. Then we can get a bijection φ of R to itself so that both φ and its inverse are continuous which moreover sends A to B. More precisely, there is $\varphi : R \to R$ such that $\varphi(A) = B$; φ is a bijection, φ is continuous; φ^{-1} is continuous.
A function $\varphi : R \to R$ which is a bijection; continuous, inverse is also continuous is called a homeomorphism of R. In other words, it is an isomorphism for the 'concept of continuity'.

Proof: Since A and B are sets satisfying the hypotheses of Theorem 1, fix an isomorphism $f : A \to B$. Define, for each x and y in R,

$$A_x = \{a \in A : a < x\}; \quad B_y = \{b \in B : b < y\}.$$

(i) For each x, $A_x \neq \emptyset$. This is because A being dense, there are points in a with $a < x$. Also $a \in A_x$ and $a' \in A$ and $a' < a$ imply that $a' < x$ so that $a' \in A_x$. Further, $\sup A_x = x$. This is because if $x' < x$, then by denseness of A, there are points a with $x' < a < x$. Thus nothing smaller than x is an upper bound of A_x. Of course x is an upper bound and hence it is $\sup A_x$. Finally, $x \notin A_x$.

Similar statements holds for B_y.
Define

$$\varphi(x) = \sup f(A_x) = \sup\{f(a) : a \in A_x\} = \sup\{f(a) : a < x\}.$$

This supremum is sensible because the set under consideration is non-empty from (i). Also A being dense, we can get $\alpha \in A$ with $x < \alpha$ and clearly $f(\alpha)$ is an upper bound for the set under consideration. We have used the fact that every non-empty set of reals which is bounded above has a supremum.
(ii) φ is a strictly increasing function.
Indeed if $x < x'$ then by successively using denseness of A, we get points $a_1, a_2 \in A$ such that $x < a_1 < a_2 < x'$. Thus every point of $f(A_x)$ is smaller than $f(a_1)$ so that $\varphi(x) \leq f(a_1) < f(a_2) \leq \varphi(x')$.

(iii) If $x \in A$, then $\varphi(x) = f(x)$.
Indeed, for every $a \in A_x$, we have $a < x$ so that $f(a) < f(x)$. Hence $\varphi(x) \leq f(x)$. If $y < f(x)$, we can choose $b \in B$ with $y < b < f(x)$. Then $a = f^{-1}(b) \in A_x$ and $y < f(a)$. Thus nothing smaller than $f(x)$ is an upper bound of $f(A_x)$, So $\varphi(x) = f(x)$.
(iv) Given any number y, there is an x such that $\varphi(x) = y$.
Indeed put $x = \sup f^{-1}(B_y) = \sup\{f^{-1}(b) : b < y\}$. By (i) B_y and hence $f^{-1}(B_y)$ is non-empty. It is bounded above because, B being dense there are points $b \in B$ with $y < b$ and $f^{-1}(b)$ is an upper bound for $f^{-1}(B_y)$. We show that $\varphi(x) = y$.
Need to show that $\sup f(A_x) = y$. If $a \in A_x$, then $a = f^{-1}(b)$ for some $b < y$ so that $f(a) = b < y$ and hence y is an upper bound for $f(A_x)$. If $y' < y$, then by denseness of B get $b \in B$, with $y' < b < y$. Then, definition of the point x tells that, $a = f^{-1}(b) \in A_x$ and $y' < b = f(a)$ showing that anything smaller than y is not an upper bound for $f(A_x)$.
Thus φ is a strictly increasing map of R in view of (ii). It is onto R in view of (iv). And $f(A) = B$ in view of (iii). This is the required map.
Just note that any increasing bijection φ of R is continuous and its inverse is also continuous. Being increasing bijection it is first of all strictly increasing. For reals $a < b$, we have

$$\varphi^{-1}(a, b) = \big(\varphi^{-1}(a), \varphi^{-1}(b)\big); \quad \varphi(a, b) = \big(\varphi(a), \varphi(b)\big).$$

These equalities are enough to show required continuity. Please think and argue. ∎

For example, there is a homeomorphism of R to itself which transports set of rationals to the set of algebraic numbers. There is a homeo that transports the set of rationals to the set of non-rational algebraic numbers.

Axiom of Choice:
There are several equivalent formulations of this axiom. Without converting this course into a course in Set Theory, here are some.

(1) (Existence of choice set) Given any (nonempty) family $\mathcal{A}$ of disjoint non-empty sets, there is a set S such that $S \cap A$ is singleton for all $A \in \mathcal{A}$. In other words, you can make a set picking exactly one point from each of the given sets.

(2) (Existence of choice function) Given any family of non-empty sets $\mathcal{A}$, there is a function f with domain $\mathcal{A}$ such that $f(A) \in A$ for each $A \in \mathcal{A}$.

That is, given any family of nonempty sets we can associate with each set in that family one point from that set.

Before stating the next one, we need a definition. Let P be a poset. A set $C \subset P$ is called a *chain* if given any two elements $p, q \in C$ either $p < q$ or $p = q$ or $q < p$. In other words $(C, <)$ is a loset. An upper bound for a chain C means an element $s \in P$ such that for each $p \in C$ either $p < s$ or $p = s$. In other words, it is just an upper bound in the sense we have defined for subsets of losets. The only difference is that, now we have a poset and the subset we are talking about is linearly ordered. Note that the upper bound itself need not belong to the chain, it belongs to the poset.
An element m of the poset is said to be a maximal element if $\neg(m < p)$ for all $p \in P$. Observe that we are not saying that $p \leq m$ for each p. We are only saying that there is nothing larger than m. After all, some elements p may not be 'comparable' with m – We only have a poset.

For instance, let P be the collection of subsets of R having at most four elements, with usual inclusion order. The elements $\{1, 2, 3, 4, \}$, $\{8, \sqrt{2}, 3/4, 49\}$ are maximal elements in P. In fact any four element set is a maximal element. Here is a chain: $\{1\}, \{1, 2, 3\}$. Here is another chain:
$\emptyset, \{3\}, \{3, 49\}, \{3, 49, 99\}, \{3, 49, 99, -31\}$.

The collection of all finite subsets of R is another poset. It has no maximal elements. Here is a chain in this poset.
$\emptyset, \{1\}, \{1, 2\}, \{1, 2, 3, \}, \{1, 2, 3, 4, \}, \cdots\cdots\cdots$.
This chain has no upper bound, because then that set must include all these points and hence can not be finite.
You can also consider the collection of countable subsets of R.

Here is an equivalent of the Axiom of choice.
(3) (Existence of maximal elements) Let $(P, <)$ be a (non-empty) poset. Suppose that every chain $C \subset P$ has an upper bound. Then P has a maximal element.
This statement is called Zorn's lemma. It was actually (in an equivalent form) proposed by Hausdorff in 1914, Kuratowski in 1922 but popularized by Zorn in 1935.

Here is another equivalent of the axiom of choice. We say that a linear order is a well order if every non-empty subset has a first element. In other words, let $(X, \leq)$ be a loset. it is called a well-ordered set, simply woset, if $A \subset X$ and $A \neq \emptyset$ implies there is an element $a \in A$ such that $a \leq x$ for

all $x \in A$. Note that $a \in A$ and also note that such an element (if exists) is unique. Here is another form of the axiom of choice.
(4) (Well ordering principle) Every non-empty set can be well-ordered, that is, given a non-empty set X, there is a binary relation $\leq$ on X so that $(X, \leq)$ is a woset.

Here is an application of the Zorn's lemma which you already know. Every non-trivial vector space (over R or C or Q) has a basis, that is, linearly independent set B which is maximal — if you put in another vector in B then the resulting set is no longer independent. equivalently, every non-zero vector v can be uniquely expressed as a finite sum $v = \sum c_i v_i$ such that (i) the vectors v_i are distinct; (ii) $v_i \in B$ for each i and (iii) $c_i \neq 0$ for each i. Here non-trivial vector space means it is not just $\{0\}$.

Consider the collection $\mathcal{P}$ of all independent subsets of the vector space. Order this collection by saying $A \leq B$ if $A \subset B$ for $A, B \in \mathcal{P}$. This collection is non-empty because singleton set consisting of a non-zero vector is an independent set. This is a poset (Any collection of sets with inclusion as order is a poset). If you take a chain $\mathcal{C} \subset \mathcal{P}$, then union S of all sets in this chain is an upper bound. It obviously includes all sets in the chain and hence is an upper bound, if only we verify that this $S \in \mathcal{P}$. If you take finitely many vectors in this set S, then each of these vectors is in one set in the chain, In other words these finitely many vectors belong to one independent set. So the are independent.

You should get feeling for the feature 'independence is a finitary property'. that is, a set of vectors is independent iff every finite subset is independent. There are several applications of AC in diverse contexts. you learnt groups and rings. AC shows Every (non-trivial) unital ring has a maximal ideal. If you travel a little more on group theory road, you will see the Neilsen-Schreier theorem which says that every subgroup of a free-group is itself a free-group. This is proved using AC.

Intuitively we feel that an axiom is unnecessary for choice set — after all pick any element from each of the disjoint sets; why not. Alright, get it and name it C. If I take a set from our collection and a point x from it and ask you 'is this point x in C?' you can not answer because you do not know which point was selected. In other words there was no rule prescribed to make the set C and hence you can not answer if a specific given point is in C.

This is brought out very beautifully by Bertrand Russel: Given lots of

pairs of shoes you can provide an algorithm to select one shoe from each pair; but given lots of pairs of new socks you have no algorithm to select one from each pair. Think.
Further there are unexpected consequences from AC! you can take a (solid) foot ball, cut it into pieces and reassemble to get two (solid) footballs of the same size; looks counter-intuitive. Leaving unnecessary discussion; here is the main theorem regarding the four principles above.

Theorem: (1) existence of choice set (2) existence of choice function (3) Zorn's Lemma and (4) Well-ordering principle; are equivalent statements. ∎
Brief idea why this is true:
(1) implies (2): given any family of sets $(A_\alpha; \alpha \in \Delta)$ put

$$B_\alpha = \{(\alpha, x) : x \in A_\alpha\}; \qquad \alpha \in \Delta.$$

These are disjoint because every point in B_α is a pair and its first coordinate is α. Take a choice set S, thus for each α we have $f(\alpha) \in B_\alpha$. Define $f(\alpha)$ to be the second coordinate of the unique point in $S \cap B_\alpha$.
If you are confused about my using α, denote the given collection (of not necessarily disjoint sets) by $\mathcal{F}$. You can safely assume that the sets are distinct (not saying disjoint, just different). For each set $F \in \mathcal{F}$ denote by F^* the set of all pairs (F, x) with $x \in F$. Thus remember F^* is a set consisting of pairs. In each pair here first coordinate is F and second coordinate is a point of F. Observe these sets F^* as F varies in $\mathcal{F}$ are disjoint — points in F^* always have first coordinate F where as points in G^* always have first coordinate G. Get choice set and get confused and sort it out. If you can not, ask me, but do not accept!
(2) implies (1): Range of the function is the choice set (Remember, now our sets are disjoint).
Thus (1) and (2) are equivalent.
(3) implies (1): Take a collection $\mathcal{F}$ of nonempty disjoint sets. Let G be the union of all the sets in $\mathcal{F}$. Let

$$P = \{S \subset G : |G \cap A| \leq 1 \quad \text{for each} \quad A \in \mathcal{F}\}.$$

This is a poset by defining $S_1 \leq S_2$ if $S_1 \subset S_2$. It is non-empty because if you take one of the sets and a point x from that set then S consisting of this single point is in P. Every chain has upper bound, namely, union of sets in the chain. So let T be a maximal element. If there is an $A \in \mathcal{F}$ such that T has no point of A, pick one point from this A and let T' be the set consisting of points in T along with this extra point. Then $T' \in P$ and contradicts maximality of T. Thus for each α, we have $|S \cap A_\alpha| = 1$ showing that T is

a choice set.
(2) implies (3): Let us say that a chain C is (pre)maximal if there is no element of P larger than every element of C. That is

$$\neg \ \exists p \in P \ \ \forall x \in C \ \ (x < p).$$

The idea is to show that there is a (pre)maximal chain. For this several methods are available. Simplest (not necessarily the best) method is the following. Fix a choice function for subsets of P. That is a function f which associates with each non-empty subset of P a point in that set. Consider

$$S_0 = P; \qquad p_0 = f(S_0);$$

$$S_1 = \{x : p_0 < x\} \qquad p_1 = f(S_1)$$
$$S_2 = \{x : p_1 < x\}; \qquad p_2 = f(S_2)$$

and so on

$$S_\infty = \{x : p_n < x \ \text{ for all } \ n\}; \qquad p_\infty = f(S_\infty)$$
$$S_{\infty+1} = \{x : p_\infty < x\}; \qquad p_{\infty+1} = f(S_{\infty+1})$$

etc. The points p so collected form a chain because at any time we are selecting a point larger than what we already have. if at some stage the set $S_?$ is empty then collect all the points p we got so far. This is a (pre)maximal chain. The fact that we have arrived at empty set signals that there is nothing larger than all the selected p s. And of course, at some stage you do arrive at empty set. The only issue is we have to make precise the meaning of $\infty + 1, \infty + 2, \ldots, \infty + \infty, \infty + \infty + 1, \ldots\ldots$.
Now this (pre)maximal chain has a upper bound, let it be a. This is a maximal element of P. Indeed if there is an element b with $a < b$ we have $x \leq a < b$ for every $x \in C$ contradicting the fact we reached empty set.

[The word premaximal may be confusing to you at first sight. If it is not, ignore this part. If I use the word maximal, then I must say that there is no chain larger than this. But unfortunately this may not be so. For example if you are considering all four element subsets of R, then $C = \{(2), (2, 3, 4, 5)\}$ is a premaximal chain, its upper bound is the set $(2, 3, 4, 5)$ and it is a maximal element. This is not biggest chain because both $C_1 = \{(2), (2, 3), (2, 3, 4), (2, 3, 4, 5)\}$and $C_2 = \{(2), (2, 4), (2, 4, 3), (2, 3, 4, 5)\}$ are larger chains. But no issues, all have the same upper bound, namely, $(2, 3, 4, 5)$]
Thus (1) and (2) and (3) are equivalent.
(4) implies (1)/(2): Given a family of sets consider their union and well order

the union. Now from each set choose its first (in this well order) element.
(2) implies (4): Again several ways of doing it. The best is to establish a bijection of the given set with a well ordered set and bring that well order here. This bijection is to inductively associate with each point a large well ordered set (such sets exist) a point of the given set by using a choice function. We shall leave it at this.
Thus all the four are equivalent. ∎

Here are some consequences of AC:
(1) Countable union of countable sets is countable.
Of course we have already done this rather quietly in an earlier course. No loss to Assume all sets are non-empty. $(A_n : n \geq 1)$ are the sets, then using hypothesis, for each n, pick one say $f_n : N \to A_n$ surjection. AC allows you this. Now define $f : N \times N \to \cup A_n$ by $f(i,j) = f_i(j)$. This is enough. Think. ∎

(2) For any two sets X and Y, either $|X| \leq |Y|$ or $|Y| \leq |X|$.
In other words there is an injection from X to Y or from Y to X. equivalently, any two sets can be compared. To prove this, we assume that both are non-empty. Let P be the set of all pairs (S, f_S) where $S \subset X$ and f_S is an injection on S into Y. This is non-empty because you can take any $x \in X$ and any $y \in Y$; put $S = \{x\}$ and $f_S(x) = y$. Then $(S, f_S) \in P$.
Define $(S, f_S) \leq (T, f_T)$ if $S \subset T$ and f_T extends f_S. The last phrase means that for $x \in S$ we have $f_S(x) = f_T(x)$. This is a partial order and so P is a poset. Every chain has an upper bound. Indeed if $C = \{(S, f_S)\}$ is a chain take T to be the union of all S in the chain and define f_T as: if $x \in T$, say $x \in S$, then $f_T(x) = f_S(x)$. This is well defined, if $x = S$ and also $x \in S_*$, then $f_S(x) = f_{S_*}(x)$ because we have a chain. Clear that (T, f_T) is an upper bound for the chain.
So by Zorn, get a maximal element (T, f_T). In case $T = X$ then you got an injection on X into Y showing $|X| \leq |Y|$. In case range f_T is all of Y, then f_T^{-1} provides an injection on Y into X showing $|Y| \leq |X|$. If neither happens, then you contradict maximality of (T, f_T) by picking a point in $X \setminus T$ and a point from complement of range f_T. This completes proof. ∎

(3) Every infinite set contains a copy of N, that is, contains a countably infinite set.
Fix a choice function f for subsets of X, the given infinite set. Thus for each non-empty subset $S \subset X$ we have $f(S) \in S$. Define $x_1 = f(X)$ and $x_2 = f(X \setminus \{x_1\})$ and $x_3 = f(X \setminus \{x_1, x_2\})$ and in general $x_{k+1} = f(X \setminus \{x_1, \cdots, x_k\})$. The set $(x_i : i \geq 1)$ does the job. Carefully note that

these are distinct elements. ∎

For any non-empty set X let $seq(X)$ denote the set of all finite sequerces of points from X. That is the set of all $(x_1, x_2, \ldots, x_k)$ where $k \geq 1$ and x_i are points from X.
(4) For any infinite set X we have $|X \times X| = |X| = |seq(X \times X)|$ We shall not prove this, not that it is difficult but we need to a long string of simple arguments. ∎

(5) if B basis of R as a vector space over the field Q (rationals), then $|B| = |R|$.
Firstly $|Q| < |B|$ because if B were countable then its Q-span is also countable. Clearly $|B| \leq |R|$. On the other hand

$$|R| \leq |seq(Q \times B)| \leq |seq(B \times B)| = |B|.$$

Here the first inequality is because we have an injective map from R to $seq(Q \times B)$ via the unique linear combination of basis elements (for zero associate any other element); second is because $|Q| \leq |B|$; third is from earlier result. Now Cantor- Schroder-Bernstein completes the proof.
Same argument also shows if B^* basis of $R \times R$ as a vector space over the field Q (rationals), then $|B| = |R \times R| = |R|$. ∎

(6) The groups $(R, +)$ and $(R^2, +)$ are isomorphic. There is a function $f : R \to R$ such that $f(x+y) = f(x)+f(y)$ so that it is a group isomorphism. Setup a bijection between Q-bases. Possible from above. Extend this to all of the spaces using routine argument you know. ∎

(7) There are discontinuous functions $f : R \to R$ such that $f(x + y) = f(x) + f(y)$.
Take a basis B as above and fix an element $v \in B$. Define $f(x)$ to be the coefficient of v when you express x as rational combination of elements of B. This takes only rational values and it is clearly not constant. Hence it can not be continuous.
Remember last year you learnt about such functions: if continuous, should be of the form $f(x) = ax$ for some number a.

Any basis for R as a vector space over Q is called a Hamel basis of R.
Before leaving set theory, you should understand that not always you need AC. For example to show $|N \times N| = |N|$ you can do as follows. $f(n) = (n, 0)$ is injection on N into $N \times N$; and $f(i, j) = 2^i 3^j$ is injection from $N \times N$ to

N. Now CSB does. To show $|(0,1)| = |(0,1) \times (0,1)|$ you do not need AC. $f(x) = (x, 0)$ is injection one way and $g(x, y) = 0.x_1 0 y_1 0 x_2 0 y_2 0 x_3 0 y_3 \cdots\cdots$ is injection the other way. Here $x = 0.x_1 x_2 x_3 \cdots\cdots$ and $y = 0.y_1 y_2 y_3 \cdots\cdots$ are their decimal expansions (you fix once and for all whether you want terminating or nonterminating). From this you can show $|R| = |R \times R|$ by CSB, not using AC.
Just like CH, for AC also we have the following: Kurt Godel showed that if you assume AC, you will not end up in any contradicion. Paul Cohen showed that if you assume AC is not true (there is a family of sets without choice function) then you will not end up in any contradiction.

You must have noticed that I have not defined cardinality, I have not defined $|X|$ for all X. Only when X is finite I defined and it is the number of elements. I also mentioned $|N| = \aleph_0$ and $|R| = c$. That is all. Without defining $|X|$, I have defined when to say $|X| = |Y|$; that is, X and Y have same number of elements or equipotent. There are several other beautiful results in set theory, but we need to return to Analysis.

construction of Real numbers:
Our objective is to construct the set of real numbers —- this means to show a set with operations and relation satisfying whatever we assumed last year. We need to exhibit a set R and define operations $+$, $\cdot$ along with elements 0, $1 \neq 0$ and relation $\leq$ such that those axioms hold. We shall recall those axioms only briefly now:
(I) deals with $+$ and 0; says we have a group;
(II) deals with $\cdot$ and $1 \neq 0$; says nonzero things form a group;
(III) deals with $+$ and $\cdot$; says these two things are friendly;
(IV) deals with $\leq$; says we have a loset;
(V) deals with $\leq$ and $+$ and $\cdot$; says they are friendly;
(VI) deals with subsets: says bounded non-empty subset has sup.

Of course, any construction work needs using cement, bricks, water and so on and more important, dirtying our hands. It is boring too. Try to understand the construction. But at the end return to the same picture we had last year of real numbers. You can ask: if we have the same good old picture, then why do we construct at all? Two reasons: Firstly none of us have seen real numbers and we want to convince ourselves that they exist. Secondly, the techniques are useful in other contexts (not in our course). For example the road from N to Z helps in embedding some semigroups into groups; the road from Z to Q helps us in embedding some rings in fields; the road from Q to R helps us in completing some metric spaces which are not complete,

a desirable property. Of course, this exercise should also inspire you to see what can be done by pure thought process.

Peano Axioms:
There is a set N_0 and a function $s : N_0 \to N_0$ satisfying the following three rules:
(i) $x = y$ if and only if $s(x) = s(y)$.
(ii) there is a unique element θ which is not in the range of s. This θ, being unique, shall be denoted by 0.
(iii) If $A \subset N_0$, and if $0 \in A$, and if $x \in A$ implies $s(x) \in A$; then $A = N_0$.
The function s is called successor function.

If you want you can restate axiom (i) as $x = y \leftrightarrow s(x) = s(y)$. There is no need to put 'iff', we could have said: $s(x) = s(y) \to x = y$.The definition of function already tells you that when $x = y$, then you must have $s(x) = s(y)$.

Second axiom means there is no x such that $s(x) = \theta$. if you want you can restate this as $(\exists!\theta)(\forall x)\ \ \neg(s(x) = \theta)$.

The last axiom is called induction axiom, because it allows to use mathematical induction. This is also called well-ordering axiom, because it makes N_0 as well-ordered set; with a natural order to be defined shortly.

It is also possible not to demand uniqueness of θ in (ii). Take any one such θ and name it zero and proceed to axiom (iii). Then it follows that there can not be another such θ. Suppose that $\theta' \neq \theta$, also satisfies (ii) then the set $N_0 \setminus \{\theta'\}$ satisfies conditions of (iii) and hence must equal N_0.

Let us not bother with minor details of no interesting consequence. The above are our axioms. These axioms are called Peano axioms.
You should note something very very interesting about axiom (iii). Usually axioms about a system tell about rules to be satisfied by elements of the system. For example axioms of groups tell you the rules to be satisfied by the elements of the group. Similarly about rings or fields. None of these tell you 'rules that subsets should obey'.
axiom (iii) tells you a rule to be obeyed by subsets of N_0, not by elements of N_0. This is a fine distinction, difficult to appreciate for you, but you should not ignore. The same remark applies to the 'least upper bound axiom' for R. all other axioms tell you rules to be obeyed by real numbers, or existence of special real numbers like zero and one. But the lub axiom is a rule that imposes a condition to be satisfied by subsets of R.

We denote $s0 = 1$ and $s1 = 2$ and $s2 = 3$ etc. Thus

$$N_0 = \{0, 1, 2, 3, 4, \ldots\ldots\}$$

For those interested, I add that such a system exists. Axioms of Set theory assert that there is a set with no elements and it is denoted by $\emptyset$ called empty set. Now consider

$$\begin{aligned} \emptyset & \qquad 0 \\ \{\emptyset\} & \qquad s0 = 1 \\ \{\emptyset, \{\emptyset\}\} & \qquad s1 = 2 \\ \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}. & \qquad s2 = 3 \end{aligned}$$

and so on. You can define the operation s of 'set consisting of all previous sets' starting from empty set and describe N_0 as the smallest containing empty set and closed under the operation s. We then name inductively the elements as mentioned above. Observe the number on the right is the 'number of elements' in the set on the left.

Peano arithmetic: from (N_0, s) to $(N_0, +, \cdot, \leq)$:
We shall use the successor map to define addition and multiplication on N_0 as follows. These are defined using induction.

$$x + 0 = x; \qquad x + s(y) = s(x + y).$$

In other words, we first define $x + 0 = x$ for every x. Then we define $x + 1 = x + s(0) = s(x + 0) = s(x)$ which we already named $x + 1$. and then define $x + 2$ etc. Did we define $x + y$ of every y? Yes. the set of y for which we defined $x+y$ includes 0 and if it includes y then it includes $s(y)$ and hence this is all of N_0. For example to calculate $x + 5$, you need to calculate $x + 0$, then $x + 1$, then $x + 2$ etc and finally $x + 5$.
One can show again several rules like: (commutativity) $x + y = y + x$ for all x and y; (associativity) $(x + y) + z = x + (y + z)$ for all x, y, z; (cancellation law) $x + z = y + z$ implies that $x = y$
We define multiplication

$$x \cdot 0 = 0; \qquad x \cdot s(y) = x \cdot y + x.$$

Thus to calculate $x \times 5$, you need to calculate $x \times 0$, then $x \times 1$ and finally $x \times 5$. This satisfies analogues of the above three rules: commutativity; associativity and cancellation law.

Now we can define order:

$$x \leq y \leftrightarrow (\exists z)\ \ x + z = y.$$

You might think this is not right because for every x and y there is always such an z, namely $y - x$. But this is a wrong thought because at this moment we do not know this minus sign; for example there is nothing like $3 - 4$. At this moment the existential quantifier refers to existence of z in N_0. Since we have nothing else I did not have to say $\exists(z \in N_0)$. We have only N_0 at this stage before us, no more.
Very interesting, the proof of transitivity uses associativity of addition: suppose $x \leq y$ and $y \leq w$ then we have z and u such that $x + z = y$ and $y + u = w$. Now $x + (z + u) = (x + z) + u = y + u = w$. As I said we are being brief, but you can verify all the rules.

From Peano Arithmetic to integers, N_0 to Z:
As already noticed, given x and y there may not always exist u such that $y + u = x$; in other words I may not be able to talk about $x - y$ for some pairs. Of course, the cancellation law and commutativity tell us that if there is one such u then there is only one u.
Since we are unable to do subtraction, for some pairs we cook up some ideal elements and put them in our set to make this possible. Basic philosophy is this: The very phenomenon that is not possible is put in as an element and this element represents that phenomenon. But then we should be careful, $3 - 4$ and $8 - 9$ should be same.
This is achieved by considering all possible phenomena; identify two phenomena if there are reasons for you to believe that they are same. Then hopefully this set would make all phenomena possible. All this is abstract nonsense. Let us actually do it. Let

$$Z_0 = \{(a, b) : a, b \in N_0\}$$

Thus when I say (a, b) I am thinking of $a - b$, but I can not say so because you will question me: what is this minus sign? Till I introduce this sign I should be careful not to go beyond my vocabulary.
Some people think of $b - a$ when they write (a, b). It makes no difference as long as you think same way throughout. We identify some pairs. Say

$$(a, b) \sim (m, n) \leftrightarrow a + n = b + m.$$

Actually I wanted to identify when $a - b = m - n$, I said the same thing without using minus sign, that is, using only the vocabulary I have now. It

is a routine matter to verify that this is an equivalence relation.
Let the space of equivalence classes be denoted by Z. We have enlarged the set N_0. We can identify N_0 as a subset of Z. Take $n \in N_0$ and let $\varphi(n)$ be the equivalence class containing $(n, 0)$ — remember, I have $a - b$ at the back of my mind when I wrote (a, b).
Actually, I can make a nice selection of 'one thing' from each equivalence class and use them rather than thinking of the huge equivalence class all the time. Each equivalence class contains exactly one pair where one of the coordinates is zero. if $a = b$ then easy to see $(a, b) \sim (0, 0)$. if $a < b$, then there is c such that $a + c = b$, then easy to see $(a, b) \sim (0, c)$. if $b < a$, then there is c such that $b + c = a$, then easy to see $(a, b) \sim (c, 0)$.
These are the only possibilities because we have linear order. Clearly a pair $(a, 0) \sim (0, b)$ implies $a + b = 0 + 0$ so that $a = b = 0$. If $(a, 0) \sim (b, 0)$ then $a = b$. If $(0, a) \sim (0, b)$ then $a = b$.

Thus there is exactly one pair (a, b) in each equivalence class with one coordinate zero. Thus we could think of this element instead of the equivalence class. In other words we could have defined

$$Z = \{(a, b) : a = 0 \quad \text{or} \quad b = 0\}$$

And we could have though of $(a, 0)$ as $a - 0 = a$ and $(0, a)$ as $0 - a = -a$. Of course $(0, 0)$ is identified as 0.
Yes you are right, we did complicate life. We could have just said 'add a symbol minus n" for every n, then these symbols along with N_0 is precisely our Z. Such a prescription will serve the present purpose. However it misses the general philosophy, it looks artificial, serves only immediate purpose. The philosophy 'if you can not achieve certain phenomenon, consider the set of phenomena' gives a unity for mathematical constructions.
But you can keep both pictures in mind. you will see first picture looks complicated but 'smooth' to operate; second picture looks smooth but complicated to operate.
On this set Z (of equivalence clases) we define addition multiplication and order:

$$(a, b) + (c, d) = (a + c, b + d).$$
$$(a, b) \cdot (c, d) = (ac + bd, ad + bc)$$

These prescriptions respect equivalence relation, so that, this definition is indeed a deinition on the space of equivalence classes. Under addition it becomes a group and multiplication obeys all the three rules we had earlier. We define order as follows.

$$(a, b) \leq (c, d) \leftrightarrow ad + bc \leq ac + bd.$$

This definition makes Z a loset and the order is friendly with addition and multiplication.

In the second picture, namely when we selected one specific point from each equivalence class as above, these definitions take the following shape. Of course, you can repeat the same earlier formulae.

$$(n,0)+(m,0)=(n+m,0); \quad (0,n)+(0,m)=(0,n+m)$$

To define $(n,0)+(0,m)$ we need to consider two cases. If $n>m$, say, $n=k+m$ then the sum equals $(k,0)$. But if $m>n$ and say $m=k+n$ then the sum equals $(0,k)$. Think about it. Multiplication is as follows.

$$(m,0)\cdot(n,0)=(mn,0); \quad (0,m)\cdot(0,n)=(mn,0)$$

$$(m,0)\cdot(0,n)=(0,mn).$$

selecting specific points from equivalence class may have made you feel better but defining the operations looks definitely messy. Think about it. However order takes the following simpler shape. Each $(0,n)$ is smaller than each $(m,0)$. Among $(m,0)$ kind it becomes as earlier on N_0, whereas among $(0,m)$ kind it becomes the 'reverse'.
Of course all of it just provides you the picture you already have in mind

$$\cdot\cdot\,, -3,\ -2,\ -1,\ 0,\ 1,\ 2,\ 3,\ \cdots\cdots\,.$$

This is how we think of Z from now on. Once constructed, no matter how, Z has existence independent of how it is constructed and by whom.

From integers to rationals; from Z to Q:
The trouble with Z is that while we can add and subtract, we can multiply but not divide. In other words we have additive inverses but not multiplicative inverses. By the way, using known rules, we understand that anything multiplied by zero has to be zero, so when we say multiplicative inverse, we mean only for non-zero things. We know that we can never have multiplicative inverse for zero.
If I could not subtract earlier $3-4$ then we added $(3,4)$ and when we see this 'symbol' we think of $3-4$. In the same way, if I can not divide $3/4$, I add the symbol $(3,4)$ and when I see this I think of $3/4$.
Of course there is a confusing point. When you see $(3,4)$ do you have at your back of the mind $3-4$ or $3/4$? You can not have both. That is why I said earlier how we think of Z. We do not think of $3-4$ now, it is represented by the symbol -1, that is all. There is no pairs. elements of Z are as described

at the end of the discussion above. First understand this point.

So since division between some pairs of points of Z is not possible, let us repeat the earlier process. Let

$$Q_0 = \{(a, b) : a, b \in Z, b > 0\}.$$

But as you see I can not think of 3/4 and 6/8 as two separate quantities. Define in Q_0 an equivalence relation

$$(a, b) \sim (c, d) \leftrightarrow ad = bc.$$

I wanted to identify if $a/b = c/d$ but I can not say this because I do not have meaning for these quantities. But you see i said the same thing using the permitted vocabulary.
Let Q be the set of equivalence classes. This is our set of rationals. Define

$$(a, b) + (c, d) = (ad + bc, bd), \quad (a, b) \cdot (c, d) = (ac, bd).$$

First observe that bd is nonzero and hence the definition makes sense, the resulting pairs are in our collection. Also observe that these definitions respect the equivalence relation. That is

$$(a, b) \sim (a_1, b_1), (c, d) \sim (c_1, d_1) \Rightarrow (ad + bc, bd) \sim (a_1d_1 + b_1c_1, b_1d_1).$$

$$(a, b) \sim (a_1, b_1), (c, d) \sim (c_1, d_1) \Rightarrow (ac, bd) \sim (a_1c_1, b_1d_1).$$

This observation, that the operations respect the equivalence relation, help us to define the operations on the space of equivalence classes, that is, on Q. These operations make Q into a field.
We define order by

$$(a, b) \leq (c, d) \leftrightarrow ad \leq bc$$

Again, this definition respects the equivalence relation and hence defines an order on Q. This makes Q into a loset and the order is compatible with addition and multiplication.
These operations and order extend the operations and order we already have on Z. Of course, for this sentence to make sense you should first be able to see Z inside Q. The equivalence class containing the element $(a, 1)$ is identified with $a \in Z$. If you want to be formal, you can define map φ on Z into Q and say that this map preserves the 'structure'.
In other words Q is an ordered field. It has the following property, called Archimedian property.

$$x \in Q \Rightarrow (\exists \ k \in N_0) \ x < k.$$

Every number is smaller than some integer.
From now on we think of rationals as we did last year, equivalence classes etc goes out!

From Rationals to Reals, from Q to R:

Why does Q not satisfy all properties required of R? Because there are subsets bounded above which have no upper bound or equivalently there are Cauchy sequences which do not converge. Apply the same trick as earlier. So there are at least two methods to rectify the situation.

Method1: (Dedikind Method)
Let us say that a subset $x \subset Q$ is a cut if it satisfies the following: (i) $\emptyset \neq x \neq Q$; (ii) $p \in x$, $q \leq p$ imply that $q \in x$; (iii) $p \in x$ implies there exists r with $p < r$ and $r \in x$.
Thus a cut is a non-empty proper subset of Q such that if you take an element in it then everything smaller than that is also there and somethings larger than that are also there. (We can not say everything larger than that is also there).

The collection of all cuts is denoted by R. This is our real number system.

When I am thinking of a cut I am actually thinking of the real number: upper bound of that cut in Q. Why did I put condition (iii)? Because I do not want to talk about equivalence classes. If I did not put that condition both sets $x =$ (all rationals ≤ 1) as well as $y =$ (all rationals < 1) are cuts and will have same supremum and so represent same number.
With the above definition this x tis not a cut, there is no question of equivalence classes.

Method 2: (Cantor)

Every real number is a limit of a sequence of rational numbers. This is what we use. Thus consider R^* the space of all Cauchy sequences in Q. Remember a sequence (q_j) of rationals is Cauchy if given any $n \geq 1$ there is a stage after which $|q_i - q_j| < 1/n$. But unfortunately the Cauchy sequences $(1, 1/2, 1/3, \cdots)$ and $(1, 1/4, 1/6, 1/8, \cdots)$ and $(0, 0, 0, 0, \cdots)$ all represent the same number zero. So we need to identify two sequences if they 'seem' to represent the same number. define an equivalence relation in R^*:

$$(a_n) \sim (b_n) \quad \leftrightarrow \quad a_n - b_n \to 0$$

The space of equivalence of equivalence classes, denoted, R is our real number system

Cantor's method is, in a sense, more general: it can be applied later when you learn metric spaces. We discussed Cantor's method in detail, and Dedikind method briefly. I shall review both again quickly and clearly. Then we close this chapter of construction of R. Then you can have a sigh of relief and feel comfortable returning to last year's frame of mind. But please meditate on what all this thing indeed is.

I have not given any references. All of you should first believe, you are capable of understanding. There is nothing deep or difficult. You should first think: what are we doing, why are we doing, how are we doing without much technical details. Then DO.

For set theory You can consult any book on set theory, best to start is Halmos, Naive Set Theory.
For Analysis: One of the best sources is the book Real Analysis by Hewitt and Stromberg. But you can consult Rudin or any other. Generally most of the books contain much more (or much less) than what I intend to do. That is why I have not given any references so far. But please do consult any book you like from the library. In fact, you must consult so that you are exposed to different styles and presentations.

You are free. if you find something that you read from books (related to what we are doing) to be confusing, please do ask me.

Evaluation: Quiz 1 (10 points) , Midsem (30 points), Quiz 2 (10 points), Semestral (50 points).
Quiz dates will be decided in class and may not be announced in notes. If you were absent, it is your responsibility to find out.

We are now in the process of constructing real number system using Q, the set of rational numbers.

Dedikind:
A cut x is a non-empty proper subset of Q such that for any of its elements, everything below is also there and somethings a little above are also there. R is the collection of all cuts. Let me repeat the basic philosophy. The set of all rationals smaller than $+\sqrt{2}$ has no sup in Q. (Theoretically) We put this set itself as its sup!! But then the set of rationals < 2 and the set of rational ≤ 2 have the same sup. To avoid duplication, (and to avoid equivalence classes) we made up our mind to consider sets of the first kind only.

order: Say $x \leq y$ if $x \subset y$. Say $x < y$ is $x \leq y$ and $x \neq y$; that is $x \subset y$ and $x \neq y$.
with this definition R is a loset. The most difficult axiom is now trivial to prove: least upper bound property. Take any (non-empty) collection of cuts C bounded above, thus there is z such that $x \in C$ implies $x \subset z$. Take x^* to be union of all the cuts in x. is it non empty? yes, because there is at least one cut in C and it is non-empty. Is it all of Q? No, because z is not all of Q and anything not in z is also not in x^*. If $p \in x^*$ and $q < p$; then there is a cut $x \in C$ such that $p \in x$ and clearly q is in that x and hence is in x^*. Similar argument shows that if $p \in x^*$ then a little larger than p is also in x^*. Thus x^* is a cut. is it upper bound of C, yes because $x \in C$ implies $x \subset x^*$ and hence $x \leq x^*$. is it least upper bound? yes because if y is an upper bound, then for all $x \in C$, we have $x \leq y$, that is, $x \subset y$ and hence $x^* \subset y$ and so $x^* \leq y$.

addition: Let $x \in R$ and $y \in R$. Define

$$x + y \ = \ \{p + q : p \in x; q \in y\}.$$

This set on right side is non-empty because, x and y are so and hence any $p + q$ with $p \in x$ and $q \in y$ would do. There are also points which are not here. Indeed if $p_1 \notin x$ and $q_1 \notin y$ then for any $p \in x$ we have $p < p_1$ and for any $q \in y$ we have $q < q_1$. Thus $p + q < p_1 + q_1$ showing that $p_1 + q_1 \notin x + y$. Finally, if $p \in x$ and $q \in y$ then x, y being cuts, things r, s a little larger than p, q are also in x, y (respectively). thus $r + s > p + q$ is also in $x + y$. Thus $x + y$ as defined above is indeed a cut.

The equality $x+y=y+x$ follows from the fact that the set of $p+q$ is same as the set of $q+p$. Similarly $(x+y)+z=x+(y+z)$ follows simply because the set of numbers $(p+q)+r$ is same as the set of numbers $p+(q+r)$. See how we are using the corresponding rule for Q.

Let

$$0^* = \{p \in Q : p < 0\}.$$

This is clearly a cut. Note that zero itself is not in the set.

We claim this is zero element. Take any $x \in R$. We show $x + 0^* = x$. If we have $p \in x$ and $q \in 0^*$ then $q < 0$ so that $p + q < p \in x$. Since x is a cut we conclude $p + q \in x$. Thus $x + 0^* \subset x$. Conversely, let $p \in x$. Let $r > p$ and $r \in x$. Note $p - r < 0$ so that $p = r + (p - r) \in x + 0^*$. Thus $x \subset x + 0^*$.

We shall show additive inverse. Let $x \in R$. Define

$$y = \{-q : q \in x^c; q \text{ not least element of } x^c\}.$$

Here x^c, as usual, is the complement of x, that is, all rational numbers q such that $q \notin x$. For example x is all rationals smaller than 4, then x^c is set of all rationals ≥ 4, however when we consider the corresponding y as above, we would not put (-4) in y.

If $p \in x$ and $-q \in y$ then $q \in x^c$ tells $p < q$ so that $p + (-q) < 0$. Thus $x + y \subset 0^*$. Conversely, if we show that every negative rational is in $x + y$ it follows that $x + y = 0^*$ and completes the proof that y is inverse of x. For this it is enough — because $x+y$ is a cut — to exhibit a sequence of elements $p_n - q_n$ in $x+y$ such that every negative rational is smaller than one of these numbers.

We construct the promised p_n, q_n. Take any $p \in x$ and $q \in x^c$. Set $p_0 = p$ and $q_0 = q$. Take mid-point $(p + q)/2$ which is a rational number. If this is in x then this is p_1 and $q_1 = q$. If this mid-point is in x^c then it is q_1 and $p_1 = p$. Now take mid-point of p_1 and q_1. Thus in any case $p_1 \in x$ and $q_1 \in x^c$. Now consider $(p_1 + q_1)/2$ continue and convince yourself you can write the inductive step to construct the sequence (p_n, q_n) so that each p_n is in x and each q_n is in x^c and $q_{n+1} - p_{n+1} = (q_n - p_n)/2$. Thus, by induction, we have

$$q_n - p_n = \frac{q - p}{2^n}.$$

You can easily show $q_n - p_n \to 0$. So $p_n - q_n \to 0$.
As long as q_n is not the leat element of x^c, we know $-q_n \in y$ so that

$p_n - q_n \in x + y$ as promised. As Soham pointed out, for some n, what if $(p_n + q_n)/2$ is the least element of x^c? Then $q_{n+1} = (p_n + q_n)/2 \in x^c$ but $-q_{n+1} \notin y$ (see definition of y.) Then we name that least point of x^c as a, we stop the entire process and simply take $p_n = a - (1/n)$ and $q_n = a + (1/n)$. This satisfies our requirement. Thus $x + y = 0^*$.

We proved that R with addition so defined is a group (abelian). Let us see its relation with order. Let $x < y$ and $z \in R$. Wish to show $x+z < y+z$.

Of course $x \subset y$ tells that if $p \in x$ then $p \in y$ so that for any $q \in z$ we have $p+q \in y+z$ showing that $x+z \subset y+z$, that is, $x+z \leq y+z$. The only question is whether equality can hold here. If so, we would have $x+z = y+z$; add $-z$ to both sides, use associativity (already proved), replace $z+(-z)$ by 0^*, use $x + 0^* = x$ etc to see $x = y$. But this is false.

We need to define multiplication. This is extremely tricky. You can not blindly put, just like addition, $x \cdot y$ to consist of all pq with $p \in x$ and $q \in y$. This will never be a cut, simply because all negative numbers below some stage will be in both x and y showing that all sufficiently large positive numbers are in this suggested set.(Note we are using dot for the multiplication, to be defined, in R, but for product of rational numbers we do not use dot.)

The root cause of the problem is that multiplication of two negative numbers is positive number. This suggests us to define product of positive numbers first. This is what we do now.

multiplication of positive numbers:
let $x > 0^*$ and $y > 0^*$. We define

$$x \cdot y \;=\; \{q \in Q : q \leq 0 \quad \text{or } q = p_1 p_2, \text{ for some } \; 0 < p_1 \in x, \; 0 < p_2 \in y\}.$$

Thus all negative rationals are put in this set right away. This is a cut. In fact, this is clearly non-empty. If $r \notin x$ and $s \notin y$ then you can easily argue that $rs \notin x \cdot y$ so that the suggested set is not all of Q. Let $r \in x \cdot y$ so that there is $p_1 \in x$, $p_2 \in y$ with $0 < p_1$ and $0 < p_2$ and $r = p_1 p_2$. Let $s < r$ be rational. We need to show $s \in x \cdot y$.

If $s \leq 0$ then by definition $s \in x \cdot y$. So let $s > 0$. Then $r > 0$ too. Thus $0 < s/p_2 < r/p_2 = p_1$ and hence $s/p_2 \in x$. Thus

$$s = \frac{s}{p_2} p_2 \in x \cdot y.$$

Finally let $r \in x \cdot y$. Need to show some thing larger than that in $x \cdot y$. If this $r \leq 0$ then any $p_1 p_2$ will do. So let $r > 0$. Then $r = p_1 p_2$. Take a little larger than p_1 in x (remember p_2 is positive) etc to complete the proof.

Thus the suggested set is a cut and hence it is in R. Need to show $x \cdot y = y \cdot x$ and $(x \cdot y) \cdot z = x \cdot (y \cdot z)$. These follow as in the case of addition using the corresponding properties in Q. Shall now exhibit multiplicative unit.

$$1^* = \{q \in Q,\ q < 1\}.$$

This is clearly a cut — need to use between any two rationals there is another one etc to show things a little larger are available.

Take any $x > 0^*$. We show $x \cdot 1^* = x$. All rationals $r \leq 0$ are in both: left side by definition of product and right side because $x > 0^*$. Need to show both sides have the same positive elements. If you take a positive element $p_1 p_2$ on left side then $0 < p_2 < 1$ implies that $p_1 p_2 < p_1$ and hence is in x – remembr $p_1 \in x$. conversely, if we take $p \in x$, then a little larger, say, $r > p$ is also in x and hence

$$p = r\,\frac{p}{r} \in x \cdot 1^*.$$

We now show inverse. Let $x > 0^*$. Let

$$y = \{q \in Q : q \leq 0\} \bigcup \left\{\frac{1}{q} : q \in x^c, q \text{ not least element of } x^c\right\}$$

All negative rationals are in y. If $q \in x^c$ then anything larger than that is in x^c so that any positive smaller than $1/q$ is also in y. If $1/q \in y$ then q being not the least element of x^c take something smaller than that etc; to see some thing larger than $1/q$ is also in y. Thus y is a cut and $y > 0$.
Need to show now that $x \cdot y = 1^*$. All rationals $r \leq 0$ are in both sides. So let us show they have the same positive things too. Let $r \in x \cdot y$. Thus $r = p_1 p_2$ with $0 < p_1 \in x$ and $0 < p_2 \in y$. But definition of y tells $p_2 = 1/q$ with $q \in x^c$. But then $q > p_2$ showing that $p_1 p_2 < 1$ and hence is in 1^*.
Finally we need to show that $0 < r < 1$ implies $r \in x \cdot y$. Repeat the argument given for addition. Argue, using the same construction, to get $p_n/q_n \in x \cdot y$ and $p_n/q_n \to 1$, so that anything smaller than one is smaller than one of these numbers etc. Do not forget to consider the case when q_n is least element of x^c etc. This completes the proof that y is multiplicative inverse of x.

We need to verify $x > 0$ and $y > 0$ implies $x \cdot y > 0$. This is easy. $x > 0$ tells we have $0 < p_1 \in x$ and similarly $0 < p_2 \in y$ and hence $0 < p_1 p_2 \in x \cdot y$

to complete the proof.

completion of the definition of multiplication:

$$x \cdot y = \begin{cases} - & [(-x) \cdot y] & \text{if} & x < 0, & y > 0 \\ - & [(x) \cdot (-y)] & \text{if} & x > 0, & y < 0 \\ & [(-x) \cdot (-y)] & \text{if} & x < 0, & y > 0 \\ & 0 & \text{if} & x = 0, & \text{or} \quad y = 0 \end{cases}$$

These look complicated, but just implementation of the idea: (positive)×(negative) is negative and (negative)× (negative) is positive.

This completes the definition of multiplication. That this satisfies all requirements is routine, but patient, verification — remember need to show + and · are friends; also multiplicative inverse for negative numbers; also for negative numbers too 1^* is multiplicative identity.
You can verify the map $p \mapsto p^* = \{q \in Q : q < p\}$ takes you from Q to R (cuts) which preserves everything: addition, multiplication, order.
This completes construction of R.

Cantor:
As mentioned Q is not a model for the axioms of real numbers because there are Cauchy sequences which are not converging. So using the basic philosophy stated earlier, make the Cauchy sequence itself a point of your space and say that your original Cauchy sequence converges to this point. But make sure, if you feel two Cauchy sequences are converging to the same point, identify them. Thus we need to take equivalence classes, unlike in Dedikind method.

R^* is the space of all Cauchy sequences in Q. Remember a sequence (p_n) of rationals is Cauchy if for every $j \geq 1$ there is N such that $|p_n - p_m| \leq 1/j$ for all $n \geq N$. (we are allowed $\epsilon > 0$, but then say ϵ rational — because we have no reals yet).
We define $(p_n) \sim (q_n)$ if $p_n - q_n \to 0$, that is given any $j \geq 1$ there is N such that $-1/j \leq p_n - q_n \leq 1/j$ for all $n \geq N$. R is the space of equivalence classes of Cauchy sequences.
Here are some facts that you know already but must to prove without using real numbers and using only rational numbers.
(1) Any Cauchcy sequence (p_n) is bounded, that is there is an integer k such that $-k < p_n < k$ for all n.
(2) if $p_n \not\to 0$, then there is N such that $p_n \neq 0$ for all $n \geq N$.

Define

$$[(p_n)] + [(q_n)] = [(p_n + q_n)]; \qquad [(p_n)] \times [(q_n)] = [(p_n \times q_n)]$$

$$[(p_n)] \leq [(q_n)] \longleftrightarrow (\forall j \geq 1)\ \ (\exists N)\ \ (\forall n \geq N)\ \ p_n \leq q_n + (1/j)$$

Why this complication? You can not say $[(p_n)] < [(q_n)]$ iff $p_n < q_n$ after some stage etc. After all the two sequences $(1, 1/2, 1/3, 1/4, \cdots)$ and $(1/2, 1/3, 1/4, \cdots)$ are Cauchy and are 'same'. The above definitions should be accepted only after verifying that those definitions do not depend on whatever specific sequences you take and apply. In other words if $(p_n) \sim (p'_n), \quad (q_n) \sim (q'_n)$, then

$$(p_n + q_n) \sim (p'_n + q'_n); \qquad (p_n q_n) \sim (p'_n q'_n)$$

and

$$(\forall j \geq 1)\ \ (\exists N)\ \ (\forall n \geq N)\ \ p_n \leq q_n + (1/j) \Longleftrightarrow$$
$$(\forall j \geq 1)\ \ (\exists N)\ \ (\forall n \geq N)\ \ p'_n \leq q'_n + (1/j)]$$

This is easy.
Each rational q is identified with the constant sequence where all terms are q (the equivalence containing this constant sequence). This preserves addition, multiplication and order. Additive identity is the equivalence class containing $[(0)]$ and multiplicative identity is the equivalence class containing the constant sequence 1. Additive inverse of $[(p_n)]$ is $[(-p_n)]$. For multiplicative inverse, first note that if $[(p_n)]$ is nonzero, there is N such that $p_n \neq 0$ for all $n \geq N$. Take the sequence $q_n = 1/p_n$ for $n \geq N$ and $q_n = 1$ for $n < N$. Verify $[(p_n)][(q_n)] = [(1)]$.
Thus everything appears smooth. now comes a confusing point. You finally need to show a Cauchy sequence in this space converges. Since points of this space R itself are Cauchy sequences (well, equivalence classes) we need to take a Cauchy sequence of these Cauchy sequences and produce a Cauchy sequence which is its limit. This is simple, just needs maturity; ability to pass through confusion and come out successfully. We shall postpone to a later time when we discuss completion of metric spaces.
OK we have shown that there is a system R that satisfies all the axioms we listed last year.

Now you can comfortably forget the construction and return to your good old picture of real numbers — line ordered with a zero, one etc and 3 coming before 8 on the line and with names for real numbers like $\sqrt{2}$, $\sqrt[39]{55}, e, \pi$ etc as you know. And not space of cuts or space of Cauchy sequences.

Then did I waste your time? No. Firstly, recall that we have never seen any real number and you can even wonder if they exist at all and if all of Analysis is a hoax. After all 2^3 means 2 multiplied by itself three times: $2 \times 2 \times 2$. What is $2^{1/2}$; two multiplied by itself half times? You may show a right-angled triangle and justify its meaning. What is $\sqrt[5]{2}$ raised to the power of $\sqrt[7]{3}$? Our axioms allow meaning to all these. Secondly, you can identify colours, but at some stage your curiosity should make you ask 'what exactly is red colour' and find answers. Same with numbers too. Thirdly, more important, these techniques are useful in diverse contexts – embedding some semigroups in groups; some commutative rings in fields, completing metric spaces; to compactify some spaces and so on.

R, R^n, metric spaces:
We shall proceed to the next topic of our discussion, namely, metric spaces. These are nothing but sets where there is a notion of distance between pairs of points. You can say how far is one point from other. We shall not study the general theory of metic spaces, but things that revolve around Euclidean spaces R and R^n.

Once you have the notion of distance, you can understand the concept of closeness, after all two points are close if the distance between them is not much. Once you can feel closeness of points, you can feel if a sequence of points are getting closer and closer to a given point. In other words, convergence of sequences makes sense. Once you can do this, you can talk about continuous functions. After all continuous functions are just those that preserve convergence. So you see you can build an excellent story, imitate on your set what you have done on R. Of course, you can not add or multiply unless these are possible in your set.
This is not just for the sake of generalization, it has applications far beyond expectations and is essential for all of science — maths, physics, computer science and so on.
For instance the concept of convergence in R helped us to discover new numbers like e and π and γ (Euler constant). In the same way if we have concept of distance among functions and concept of convergence then starting from known functions, by taking limits we can discover new functions. This is helpful when you are searching for functions satisfying certain properties. Of course ew already used limits to discover — via power series — new functions (like sine, cosine, exponential) starting from high school polynomials. For that just convergence of numbers was enough, but we will see problems where convergence in the space of functions is needed.

Let X be a nonempty set. For each pair of points x, y in the set we want to have the concept of distance between the points $d(x, y)$. One thing you feel immediately is that $d(x, x) = 0$. Also if x and y are different points then naturally they are separate and must be at a distance no matter how small; thus $d(x, y) > 0$.
One thing experience tells us is that distance is symmetric. the distance from x to y must be same as the distance from y to x. That is, $d(x, y) = d(y, x)$. Of course, I depended on your experience.

[It is also quite possible to think that this need not be true. For example, suppose you try to measure distance by the energy you need to spend from going from one place to other. For points far apart you need to spend more energy in travelling whereas if the points are close you need not spend too much energy. Such a method appears very reasonable. But imagine x is the point at the base of a hill and y is the point on the top of the hill. You will surely agree that it takes more energy to go from x to y than from y to x. Thus it is also conceivable that distance is not symmetric. However We adapt symmetry.]

Finally again from practice, we see that distance is in a sense 'direct distance' whatever it may mean. Thus going from x to z directly should be not more than going from x to z via y. That is $d(x, z) \leq d(x, y) + d(y, z)$. This is called triangle inequality. This is true for any three points you take in the plane. This we knew from elementary geometry.
We shall adapt just these three as the things to be satisfied by distance and see what can be done and what is its use.

Definition: Let X be a (non-empty) set and $d : X \times X \to [0, \infty)$ satisfying the three conditions. (i) $d(x, y) = 0$ iff $x = y$. (ii) $d(x, y) = d(y, x)$ for any two points x, y. (iii) $d(x, z) \leq d(x, y) + d(y, z)$ for any three points x, y, z. Then we say that d is a distance function on X and we also say (X, d) is a metric space.

If we have a metric space (X, d) and $a \in X$ and $r > 0$ then open ball of radius r with centre a is defined to be the following set: $B(x, r) = \{x \in X : d(x, a) < r\}$.

Example 1: X is real number system R and $d(x, y) = |x - y|$.
In this case $B(a, r)$ is the interval $(a - r, a + r)$. Thus open balls are just open intervals.

Example 2: X is the plane R^2 and for $x = (x_1, x_2)$ and $y = (y_1, y_2)$ we put

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

This is called Euclidean distance. In this case $B(a, r)$ where $a = (a_1, a_2)$ is the set $\{(x_1, x_2) : (x_1 - a_1)^2 + (x_2 - a_2)^2 < r\}$ which is the usual disc centered at a and radius r.
Example 3: X is R^n and $d(x, y) = \sqrt{\sum_1^n (x_i - y_i)^2}$.
Example 4: X is again R, $d(x, y)$ is zero if $x = y$ and one if $x \neq y$. Here $B(a, r)$ consists of just the point $\{a\}$ as long as $r \leq 1$, but it equals all of R as soon as $r > 1$.
Example 5: X is R^2.

$$d(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|\}; \quad x = (x_1, x_2) \;\; y = (y_1, y_2).$$

Here ball of radius one around $(0, 0)$ consists of the square with vertices $(\pm 1, \pm 1)$.
Example 6: X is R^2.

$$d(x, y) = |x_1 - y_1| + |x_2 - y_2|, \quad x = (x_1, x_2) \;\; y = (y_1, y_2).$$

Here ball of radius one around $(0, 0)$ consists of the region bounded by the four lines $(\pm x_1) + (\pm x_2) = 1$.
Example 7: Fix $p \geq 1$. X is again R^2.

$$d(x, y) = (|x_1 - y_1|^p + |x_2 - y_2|^p)^{1/p}, \quad x = (x_1, x_2) \;\; y = (y_1, y_2).$$

Triangle inequality here is not obvious. We shall prove. Observe $p = 2$ corresponds to Euclidean distance.
Example 8: X is $C[0, 1]$ the space of real valued continuous functions on the unit interval $[0, 1]$. And $d(f, g) = \sup\{|f(x) - g(x)| : 0 \leq x \leq 1\}$.
Since continuous functions on $[0, 1]$ are bounded this makes sense. Since for any three functions f, g, h and any point x we have

$$|f(x) - h(x)| \leq |f(x) - g(x)| + |g(x) - h(x)| \leq d(f, g) + d(g, h).$$

This is true for every x and so by taking sup over x on left side we get the triangle inequality.

Here given f and $r > 0$ the ball $B(f, r)$ consists of all continuous functions whose graph lies strictly in the band: graph of $f(x) - r$ and graph of $f(x) + r$. Since we have a closed bounded interval $[0, 1]$, if each $f(x) - g(x)| < r$ then so is the sup.

Holder, Minkowski:
Let us first show that the d_p example above does indeed satisfy the triangle inequality. In what follows we have two numbers $p, q > 0$ such that

$$\frac{1}{p} + \frac{1}{q} = 1. \qquad (\spadesuit)$$

Of course this already implies $p > 1$ and $q > 1$ and given any $p > 1$ there is exactly one q as above.

We start with the simple observation. Let $a \geq 0$ and $b \geq 0$. Then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Since ($\spadesuit$) tells us that the right side is convex combination of a^p and b^q we are advised to look for a convex function and use the definition of convexity. if either a or b is zero, then left side is zero and the inequality holds. So assume $a > 0$ and also $b > 0$. Take numbers x and y such that

$$\exp\{x/p\} = a; \quad \exp\{y/q\} = b$$

so that

$$e^x = a^p; \quad e^y = b^q$$

Now convexity of exponential function completes the proof.

$$ab = \exp\left\{\frac{1}{p}x + \frac{1}{q}y\right\} \leq \frac{1}{p}e^x + \frac{1}{q}e^y = \frac{a^p}{p} + \frac{b^q}{q}.$$

This simple inequality leads to

Holder's inequality: $p > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$.
Let $a_1, \cdots, a_n$ and $b_1, \cdots, b_n$ be real numbers. Then

$$|\sum a_i b_i| \leq \left(\sum |a_i|^p\right)^{1/p} \left(\sum |b_i|^q\right)^{1/q}.$$

If f, g are continuous functions (more generally Riemann integrable) on $[0, 1]$, then

$$\int_0^1 |f(x)g(x)|dx \leq \left(\int_0^1 |f(x)|^p\right)^{1/p} \left(\int_0^1 |g(x)|^q\right)^{1/q}.$$

Sums: If any one of the right side quantities is zero then the corresponding a's or b's are zero and hence left side is also zero and so both sides are zero.

So let us assume that the two quantities on right side are non-zero, say, c and d respectively. Take $\alpha_i = |a_i|/c$ and $\beta_i = |b_i|/d$ to see

$$\sum \frac{|a_i b_i|}{cd} \leq \sum \frac{1}{p} \frac{|a_i|^p}{c^p} + \sum \frac{1}{q} \frac{|b_i|^q}{d^q} = \frac{1}{p} + \frac{1}{q} = 1.$$

gives the stated inequality since $|\sum a_i b_i| \leq \sum |a_i b_i|$.

Integrals: Shall do for continuous functions. Again If any one of the right side quantities is zero then the corresponding f or g is zero and hence left side is also zero and so both sides are zero. So let us assume that the two quantities on right side are non-zero, say, c and d respectively. Take the functions $\alpha = |f|/c$ and $\beta = |g|/d$. Observe $\int \alpha^p(x)dx = 1$ and $\int \beta(x)^p dx = 1$. apply holder point wise to get

$$\alpha(x)\beta(x) \leq \frac{1}{p}[\alpha(x)]^p + \frac{1}{q}[\beta(x)]^q \qquad 0 \leq x \leq 1$$

Now integrate to get the stated inequality. Use $|\int h| \leq \int |h|$ when the functions are not necessarily positive. ∎

Holder leads to

Minkowski Inequality: $p > 1$. With same notation as in Holder,

$$(\sum |a_i + b_i|^p)^{1/p} \leq (\sum |a_i|^p)^{1/p} + (\sum |b_i|^p)^{1/p}.$$

$$\left(\int_0^1 |f(x) + g(x)|^p \, dx\right)^{1/p} \leq \left(\int_0^1 |f(x))|^p \, dx\right)^{1/p} + \left(\int_0^1 |g(x)|^p \, dx\right)^{1/p}.$$

Numbers: To prove this, assume that a's and b's are positive.

$$\sum (a_i + b_i)^p = \sum a_i (a_i + b_i)^{p-1} + \sum b_i (a_i + b_i)^{p-1}$$

use Holder on right side for each sum; note $(p-1)q = p$,

$$\sum (a_i + b_i)^p \leq (\sum a_i^p)^{1/p} \left[\sum (a_i + b_i)^p\right]^{1/q}$$

$$+ (\sum b_i^p)^{1/p} \left[\sum (a_i + b_i)^p\right]^{1/q}$$

Bring the common factor of right side to the left side to complete the proof (if that common factor is zero to start with, nothing need to be proved). If a_i, b_i not necessarily positive replace by their modulus etc.

Proof for integrals is just imitation. ∎
This leads to the triangle inequality in the example 7 above.
Example 9: consider $C[0,1]$. Fix $p \geq 1$. Here is a metric.

$$d_p(f,g) = \left(\int_0^1 |f(x) - g(x)|^p \right)^{1/p}$$

Of course when $p = 1$ this metric is just $d(f,g) = \int |f - g|$.

open/closed/convergence:
Next couple of lectures, the agenda is to imitate whatever we did in R and R^n, namely to define convergence, open sets etc. In what follows (X,d) is a metric space. Sequences are sequences in X and subsets are subsets of X.

Definition: Recall for $a \in X$ and $r > 0$, we defined ball $B(a,r)$ to be the set $\{x \in X : d(a,x) < r\}$. Note strict inequality. This is also called open ball.
A sequence x_n converges to a point x iff $d(x_n, x) \to 0$, equivalently, given $\epsilon > 0$, there is N such that $d(x_n, x) < \epsilon$ for all $n > N$; that is $x_n \in B(x, \epsilon)$.
A subset U is open if $x \in U$ implies there is $\epsilon > 0$ such that $B(x,\epsilon) \subset U$. that is, whenever there is a point in the set, some ball around that point is contained in U. A subset C is closed if its complement is open.

Theorem:
(1) The empty set $\emptyset$ and whole space X are both open and closed.
(2) Union of any collection of open sets is open.
(3) intersection of any collection of closed sets is closed.
(4) Given any $S \subset X$, there is a smallest closed set $\overline{S}$ which contains S. This is called closure of S.
(5) Given any $S \subset X$, there is a largest open set S^o contained in S. This is called interior of S.
(6) A set C is closed iff the following happens: $x_n \in C$ for each n and $x_n \to x$ implies $x \in C$.
(7) emptyset and R are the only subsets of R which are both open and closed.

Proof is simple.
(1) Open:The definition is satisfied for $\emptyset$ because there is no x for which we need to verify. For X, given a any $B(a,r)$ would do. To show closed use definition.
(2) If x is in the union then it is in one of the open sets and a ball is already in that opne set and so contained in the union.
(3) DeMorgan.

(4) intersection of al closed sets containing S would do. There is at least one, namely X.
(5) Union of all open sets contained in S would do; there is at least one $\emptyset$.
(6) Let C be closed. Let each $x_n \in C$ and $x_n \to x$. Need to show $x \in C$. Suppose not, then $x \in C^c$ which is open and so there is $r > o$ so that $x \in B(x,r) \subset C^c$. In particular since each $x_n \in C$ we conclude $d(x_n, x) \geq r > 0$. Thus the hypothesis $x_n \to x$ is violated. Conversely, suppose the stated condition holds. Need to show C is closed, That is C^c is open. Take any $x \in C^c$ if we can not find an $r > 0$ such that $B(x,r) \subset C^c$; then for each n we can find a point $x_n \in B(x, 1/n) \cap C$. Clearly $x_n \to x$, each $x_n \in C$ and $x \notin C$. Contradiction.
(7) Suppose there is an other subset F which is both open and closed. That is, both F and F^c are nonempty and both are open and both are closed. Pick $x \in F$ and $y \in F^c$. Assume $x < y$ (the case $y < x$ is argued similarly). Let $z = \sup\{a : x \leq a \leq y : a \in F\}$.
Is sup well defined? Yes, because the set has at least one element x and is bounded above by y.
Can $z = x$? No because F being open there is some $(x - \epsilon, x + \epsilon) \subset F$ showing $z \geq x + \epsilon$ for this $\epsilon > 0$.
Can $z = y$? No If $z = y \in F^c$ then F^c being open there is $(z - \epsilon, z + \epsilon) \subset F^c$. Also by definition of z, points above z in $[x, y]$ are in F^c. Thus points above $z - \epsilon$ are in F^c and hence z can not be sup.
Thus $x < z < y$.
Can $z \in F$? No because of the same argument above that refuted $z = x$.
Can $z \in F^c$? No because of the same argument above that refuted $z = y$.
This But z must be either in F or F^c. This completes the proof. ∎

Definition: Boundary ∂S of a set S is the set $\overline{S} \setminus S^o$.

Clearly (?) an open interval is an open set and a closed interval is a closed set. Every nonempty open subset of R is a countable disjoint union of open intervals. This is proved as follows. Let U be the open set. For $a \in U$ let I_a be the union of all intervals that include a and are contained in U. There is at least one because U is open and $a \in U$. Union of intervals which have a common point is again an interval (?) and union of open sets is open. Thus I_a is open interval. Consider the collection of all these open intervals I_a for $a \in U$. Clearly their union is all of U because for each $a \in U$, we have $a \in I_a \subset U$. We claim that if you take I_a and I_b then they are either same are they are disjoint. In fact, if they have a point in common then $I_a \cup I_b$ is again an interval and open and includes both a and b and hence (see definitions of I_a and I_b) must equal I_a and I_b. Thus $I_a = I_b$. Now consider only the distinct

intervals I_a. Their union is U. These are countable in number because each contains a rational and different intervals, being disjoint, must have distinct rationals. ∎

Cantor set in R is closed, because each of $[0,1]$ and $[0,1/3] \cup [2/3,1]$ and $[0,1/9] \cup [2/9,3/9] \cup [6/9,7/9] \cup [8/9,9/9]$ $\cdots\cdots$ are all closed and C is their intersection. Hence it is closed.

■ About using ∂ for boundary: Wikipedia + internet, hopefully true:

The symbol ∂ was introduced originally in 1770 by Nicolas de Condorcet, who used it for a partial differential, and adopted for the partial derivative by Adrien-Marie Legendre in 1786. It represents a specialized cursive type of the letter d, just as the integral sign originates as a specialized type of a long s (first used in print by Leibniz in 1686). Use of the symbol was discontinued by Legendre, but it was taken up again by Carl Gustav Jacob Jacobi in 1841, whose usage became widely adopted.
In the early 20th century, as mathematicians connected the boundary of a domain to differentiation via generalized Stokes' theorems and homology in algebraic topology, the letter d naturally evolved into the curly ∂ to denote the boundary operator The use of the partial differential symbol ∂ for the boundary of a set or a boundary operator was popularized by Hermann Weyl and Élie Cartan, and later established in algebraic topology by W. V. D. Hodge, Samuel Eilenberg, Norman Steenrod $\cdots$, originating from the analogy between Stokes' theorem and generalized boundary operators. ■

returning to metic spaces, There is one subtle point. Suppose (X, d) is a metric space. We defined open balls $B(a, r)$; we defined open set as U with the property $a \in U$ implies there is a ball $B(a, r) \subset U$. Thus every open set is union of balls. The subtle point is: is open ball an open set? Yes. Let $B(a, r) = \{x \in X : d(a, x) < r\}$. Let $x_0 \in B(a, r)$; we need to show an s such that $B(x_0, s) \subset B(a, r)$. First note $r - d(a, x_0) > 0$. take s with $0 < s < r - d(a, x_0)$. we claim $B(x_0, s) \subset B(a, r)$. Indeed if $z \in B(x_0, s)$, that is, $d(x_0, z) < s$, then $d(a, z) \leq d(a, x_0) + d(x_0, z) < d(a, x_0) + s < d(a, x_0) + r - d(a, x_0) = r$; that is $z \in B(a, r)$.

subspaces:
If we have a metric space (X, d) and a non-empty sub set $Y \subset X$, then we can regard (Y, d_Y) as a metric space, where d_Y is nothing but d but restricted only to pairs of points of Y. This Y is called a subspace of X. If $a \in Y$ and $r > 0$, then ball $B_Y(a, r)$ in Y, namely $\{y \in Y : d_Y(a, y) < r\}$ is nothing but the corresponding ball in X intersected with Y, that is,

$$\{y \in Y : d_Y(a, y) < r\} = \{x \in X : d(a, x) < r\} \cap Y.$$

Also open sets in Y are nothing but open sets in X intersected with Y. More

precisely $V \subset Y$ is open in Y iff there is $U \subset X$, open in X such that $V = U \cap Y$. This is immediate from the corresponding statement about balls made above and the fact that open sets are nothing but union of balls (remember our definition of ball is open ball).
As a result, $F \subset Y$ is closed in Y iff there is a $C \subset X$ closed set in X such that $F = C \cap Y$.
If y_n and y are all in Y, then $y_n \to y$ in Y iff $y_n \to y$ in X (metric is same?).

complete spaces:
One concept we want to imitate is that of a Cauchy sequence.
A sequence (x_n) in a metric space (X, d) is Cauchy sequence if given any $\epsilon > 0$, there is an N such that $d(x_n, x_m) < \epsilon$ for every $m, n > N$. A metric space is complete if very Cauchy sequence converges.

Thus a Cauchy sequence is a sequence of points which are coming closer and closer. After some stage distance between any two points is at most one; after a later stage distance between any two points is at most $1/2$ and so on. Thus completeness means that any sequence of points which are coming closer and closer are *actually* coming closer to a point. Every Cauchy sequence is heading somewhere (not falling out of a hole!).

This property of completeness is very important, it helps us to discover points in the space which may not be visible to the naked eye. For example in the real line we found that the sequence of points

$$1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}; \quad n = 1, 2, 3, 4 \cdots$$

is a Cauchy sequence and hence converges. it was not an already known point. So we named it e. similarly the sequence of points

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} - \log n; \quad n = 1, 2, 3, 4 \cdots$$

is a Cauchy sequence, and its limit is a number we did not see earlier and named it γ (Euler's constant).

Now you have a panorama of metric spaces, you can discover points in the spaces, that you are not able to see outright. For example we have the space $C[0, 1]$ with sup metric. You can discover that there are elements in this space which are nowhere differentiable; that is, there are nowhere differentiable continuous functions. In fact you will see later that such functions

are far more than differentiable functions.

Let us ask another question which you feel is unrelated to our present discussion. We showed that R can not be expressed as a union of two disjoint non-empty closed sets (equivalently, $\emptyset$ and R are the only subsets of R which are both open and closed) Can you express R as a union of three disjoint non-empty closed sets? That is $R = A \cup B \cup C; \quad A, B, C \neq \emptyset;$ closed, disjoint. No. because $R = A \cup (B \cup C)$. same argument shows that R can not be expressed as a union of finitely many disjoint nonempty closed sets.

Very good, let us change it to countable union. Can you express R as a countable union of disjoint non-empty closed sets? That is,

$$R = \cup A_n; \quad (\forall n) A_n \neq \emptyset; \quad (\forall n) A_n \text{ closed }; \quad (\forall n \neq m) A_n \cap A_m = \emptyset.$$

The answer is not immediate. There is an interesting story behind this discussion. The beauty unfolds slowly. we shall answer this. First a new concept.

Cantor intersection theorem:
In the real line we have shown that a decreasing sequence of closed non-empty intervals with diameter converging to zero have a point in common. We shall now generalize this result to metric spaces. But before doing so, let us understand that the intervals should be as stated in the theorem. A sequence of sets which are decreasing are called nested.

The intervals $\{[n, \infty) : n \geq 1\}$ are closed and nested. But they have no point in common because their diameter is not converging to zero.
The intervals $\{(0,\ 1/n); n \geq 1\}$ have diameter decreasing to zero, they are nested but have no point in common because the sets are not closed.
The intervals $\{[n,\ n + 1/n]; n \geq 1\}$ are closed and have diameter decreasing to zero, but they have no point in common because they are not nested.

Theorem: Let (X, d) be a complete metric space, let

$$C_1 \supset C_2 \supset C_3 \supset \cdots$$

be nested sequence of nonempty closed sets with diameter converging to zero. Then they have exactly one point in common.

The essential thing is that there is a common point. If there are two different points $x,\ y$ in common then the diameter of each of the sets must be at least $d(x, y) > 0$ and hence can not go to zero.

If we take one point $x_n \in C_n$, then the sequence $\{x_n\}$ is Cauchy. Indeed $\{x_n : n \geq N\} \subset C_N$ so that as soon as the diameter of C_N is smaller than ϵ, we can conclude that $d(x_m, x_n) < \epsilon$ for all $m, n \geq N$. By completeness of X this sequence must converge to a point x. Since each C_N is closed and x is also limit of the sequence $(x_n : n \geq N)$ which is contained in C_N we see that $x \in C_N$. This is true for every N. Thus the intersection is exactly one point.∎

Actually the above property characterises completeness. More precisely, let (X, d) be a metric space. Suppose every nested sequence of closed sets with diameter converging to zero have a common point. Then the space is complete. This is seen as follows. Let (x_n) be a Cauchy sequence. Set $C_n = \overline{\{x_i : i \geq n\}}$. Here the overline denotes closure. This is a nested sequence of closed sets. The Cauchy property of the given sequence shows that their diameter is converging to zero. Thus there is a point common to all, say, x. We claim $x_n \to x$.

Baire's theorem:

Let (X, d) be a metric space. A closed set $C \subset X$ is small if it does not contain any non-empty open set. In other words, it has no interior points. More generally, we say that a set $A \subset X$ is small if its closure does not contain any non-empty open set. Here then is an extremely powerful theorem.

Theorem (Baire Category theorem):
A complete metric space can not be written as a countable union of small sets.

In other words, if $C_1, C_2, \cdots$ is a sequence of closed sets each of which has no interior point, then there is a point of X which is not in any one of these sets. Of course, there will be, not one but plenty of points outside all the sets C_n, as we will see.

Proof of this theorem is very simple, but the theorem is itself very powerful. It will help you to see objects which you can not see with ordinary eyes! Let us see two applications before proving this theorem. You will learn later many many applications.

nowhere differentiable functions:

This application is due to Banach. Consider the set $X = C[0, 1]$, space of real valued continuous functions on the interval $[0, 1]$. We show that there are plenty of functions which do not have finite derivative at any point what-

so-ever. In fact, the set of such functions is much larger than differentiable functions.

We know that X is a complete space. For integers $n, m > 1$ consider the following set.

$$E_{n,m} =$$
$$\left\{ x \in C[0,1] : \left(\exists t; 0 \le t \le 1 - \frac{1}{n}\right) \left(\forall h; 0 < h < \frac{1}{n}\right) \left| \frac{x(t+h) - x(t)}{h} \right| \le m \right\}.$$

Observe that if a function is differentiable at some point $t < 1$, then it is in one of the above sets for some $n, m > 1$. Indeed let x be a function having a finite derivative at a point $t_0 < 1$. Get n so that $t_0 < 1 - 1/n$. Consider the fhe function $\varphi(h) = [x(t_0 + h) - x(t_0)]/h$ on $[0, 1/n]$. Of course at $h = 0$, the value is its limit which is the finite right derivative of x at t_0. This is continuous and hence bounded, say, by m. Then clearly $x \in E_{n,m}$ for this n, m.

For each n, m, The set $E_{n,m}$ is closed. To see this first recall that if $x_n \to x$, then x_n converges uniformly to x. If $t_n \to t$ then $x_n(t_n) \to x(t)$. This is because, given $\epsilon > 0$, we can fix N so that $d(x_N, x) < \epsilon/2$ for $n \ge N$. Since x is continuous, we fix N_1 so that $|x(t_n) - x(t)| < \epsilon/2$ for $n \ge N_1$. if now $n \ge N \vee N_1$, then

$$|x_n(t_n) - x(t)| \le |x_n(t_n) - x(t_n)| + |x(t_n) - x(t)| < \epsilon.$$

Now to see that $E_{n,m}$ is closed, take $x_i \in E_{n,m}$ with $x_i \to x$. get t_i for x_i. If necessary take subsequence and assume that $t_i \to t^* \in [0, 1/n]$. Now take any h with $0 < h < 1/n$. Then the facts $t_i \to t^*$ and $t_i + h \to t^* + h$ combined with the observation of the para above show that x also satisfies the required inequality for difference quotients.

Finally we show that the set $E_{n,m}$ does not contain any non-empty open set. This will then show that each $E_{n,m}$ is a small set. We start with an observation. (♠): Let P be a polynomial and $\epsilon > 0$, then we claim that there is a function x such that $d(x, P) < \epsilon$ and $x \notin E_{n,m}$.
How does this help us? If $E_{n,m}$ contains a non-empty open set, then by Weierstrass theorem this open set must contain a polynomial and hence a ball around this polynomial is contained in the open set. In other words, if you assume that interior of $E_{n,m}$ is non-empty, then there is a polynomial P and $\epsilon > 0$ such that $B(P, \epsilon) \subset E_{n,m}$. But the observation (♠) refutes precisely such a statement, it tells that there is a x in this ball which is not

in $E_{n,m}$.
To prove the observation (♠) stated above, fix a bound c for the derivative of P on $[0,1]$. Consider the following function z on $[0,1]$; it is made up of straight line segments; it starts with $z(0) = 0$; increases with slope $s = c + 2m$ till it reaches $\epsilon/2$; then decreases with slope $-s$ till it reaches $-\epsilon/2$; then increases with slope s till it reaches $\epsilon/2$ and then decreases etc; all this continues till you reach $t = 1$ and then stops. Convince yourself that you do reach $t = 1$ after finite number of these ups and downs. You then see that this defines a continuous function on $[0,1]$.

Let $x = P + z$. Then clearly $d(P, x) = \sup |z(t)| < \epsilon$. Now take any $t < 1$. Take any $h > 0$ so that $t + h$ is also before the next corner of z. Thus you have plenty of $h > 0$ at your disposal. Then

$$\frac{x(t+h) - x(t)}{h} = \frac{P(t+h) - P(t)}{h} + \frac{z(t+h) - z(t)}{h}$$

By mean value theorem the first term on right side is between $-c$ and $+c$. By construction the second term is either $c + 2m$ or $-c - 2m$. Keep in mind that $t, t + h$ are in the same 'line segment' of z. As a result the difference quotient for x is at least $2m > m$.

This shows that each $E_{n,m}$ is small. Similarly, the following sets are also small.

$$F_{n,m} =$$

$$\left\{ x \in C[0,1] : \left(\exists t; \frac{1}{n} \leq t \leq 1\right) \left(\forall h; -\frac{1}{n} < h < 0\right) \left|\frac{x(t+h) - x(t)}{h}\right| \leq m \right\}.$$

Thus we have countably many small sets $E_{n,m}$ and $F_{n,m}$ for $n, m \geq 1$. If a function has a derivative at any point of $[0,1]$ then it must be in one of these sets. actually we can be more precise as follows. If a function has a finite right derivative at any point of $[0,1)$ then it must be in one of the $E_{n,m}$. If a function has a finite left derivative at any point of $(0,1]$ then it must be in one of the $F_{n,m}$.

Baire's theorem tells that there are functions outside all these sets. If we take such a function then it can not have a finite right derivative or finite left derivative at any point what-so-ever in $[0,1]$. In particular it is not differentiable at any point.

Just to make you understand the right and left derivatives, let us consider the function $x(t) = |t - 1/2|$. Then this is not differentiable at the

point $t_0 = 1/2$. Convince yourself of this. However this has left derivative equal to -1; and right derivative equal to $+1$ at the point $t_0 = 1/2$. Thus the functions whose existence we asserted above can not even be like this. Think about it.

R as union of closed sets:

Our second application of Baire's theorem is to show the following which answers a question that we raised earlier.

Theorem: R can not be expressed as union of of infinitely many non-empty disjoint closed sets.

Let, if possible

$$R = C_1 \cup C_2 \cup C_3 \cup \cdots\cdots .$$

Let the interior of C_i be denoted by J_i. Denote

$$U = J_1 \cup J_2 \cup J_3 \cup \cdots\cdots .$$

This U is an open set and its complement, denoted by H is therefore a non-empty closed set. Thus H is a complete metric space in its own right, metric is same $d(x, y) = |x - y|$ for $x, y \in H$. Since none of the C_i can be open we see each J_i is a proper subset of C_i, possibly empty. Thus each $C_i \setminus J_i$ is nonempty closed set.

$$H = (C_1 \cap H) \cup (C_2 \cap H) \cup (C_3 \cap H) \cup \cdots\cdots$$

$$= (C_1 \setminus J_1) \cup (C_2 \setminus J_2) \cup (C_3 \setminus J_3) \cup \cdots\cdots .$$

The plan of the proof now is the following. We shall now forget R for a minute and concentrate on the complete metric space H. We show that the sets on the right side above are small in H. This contradicts Baire's theorem for the complete metric space H.

Let us start with an observation. If an interval (a, b) contains a point, say x, of $C_1 \cap H$, then it contains points from other $C_i \cap H$ $(i \neq 1)$ as well. This is easy. If the entire interval (a, b) is contained in C_1 then it would have been removed as part of J_1, So this interval must have points from, say, C_5. Let us say $y \in C_5 \cap (a, b)$. Either $y < x$ or $y > x$ because C_i are disjoint. Suppose $y < x$; similar argument applies in the other case. Consider all points in $[y, x]$ which are in C_5 and take its sup, name it z. This is sensible because the set we are considering includes y and is hence non-empty; moreover it is

bounded by x. Also C_5 being closed we conclude that $z \in C_5$. In particular $z < x$ and points in between these are not in C_5. in other words there is no interval around z contained in C_5 which means $z \in C_5 - J_5$ and thus $z \in C_5 \cap H$ and $z \in (a, b)$ as claimed.

To complete executing our plan, first notice that each $C_i \cap H$ is a non-empty closed subset of H. That it is non-empty is already noted earlier. It is closed in H because whenever a sequence of points from here converge to a limit then the convergence is 'usual convergence' and hence the limit is in both the closed sets C_i as well as H.

finally suppose $C_i \cap H$ contains a non-empty set open in H. That is, there is a point $x \in C_i \cap H$ and $\epsilon > 0$ such that $\{y \in H : d(x, y) < \epsilon\} \subset C_i \cap H$. In other words $(x - \epsilon, x + \epsilon) \cap H \subset C_i \cap H$. But this is not possible as observed above. This completes the proof that each $C_i \cap H$ is a small set in the complete metric space H. And completes proof of the theorem. ∎

Just to impress upon you the phrase 'small in H', let us consider R and Z the set of integers. Each $\{n\}$ is small in R. However none of these singleton sets are small in Z. In fact each of these are both closed and open in Z.

Thus smallness depends on the background. In the background of R, each $\{n\}$ is small. In the background of Z, each $\{n\}$ is not small. You should understand this point.

Proof of Baire:
The execution is made possible by Cantor intersection theorem for complete metric spaces.

So let (X, d) be a complete metric space and let $C_1, C_2, C_3, \cdots$ be small closed sets. We exhibit a point of X which is not in any of the C_i.

In what follows a ball means ball of positive radius. Even though for us balls are, by convention, open balls; we sue open and closed balls in this proof. A closed ball means a set of the form $F = \{x : d(x, a) \leq r\}$ and open ball as usual $B = \{x : d(x, a) < r\}$. Every open ball contains a closed ball with same center (say, any strictly smaller radius) and every closed ball contains an open ball with same centre (same radius). In what follows we ask you to choose an open ball inside a closed ball, Then you should select with the same center as mentioned above (not that such a thing matters, but we should know what we are doing). This is our agreement.

Take any open ball B_1 of your choice. We promise to get open balls $(B_i : i \geq 1)$, closed balls $(F_i : i \geq 1)$ such that the following holds.

$$B_1 \supset F_1 \supset B_2 \supset F_2 \supset B_3 \supset F_3 \supset \cdots \supset B_{n-1} \supset F_{n-1} \supset \cdots \quad (*)$$

$$F_i \subset C_i^c; \quad i = 1, 2, 3, 4, \cdots \qquad (**)$$

$$\text{diameter}(F_i) \leq 1/2^i. \qquad (***)$$

Let us see what happens then. Condition $(*)$ says that the sets F_i are nested closed sets; condition $(***)$ with Cantor intersection theorem then gives a point x common to all F_n; condition $(**)$ says that this point is not in any of the sets C_i. as promised. Actually condition $(*)$ tells that this point is in the open set B_i you gave.
Here is how we construct the sets. Since C_1 is small surely $B_1 \cap C_1^c \neq \emptyset$ and is open. So take a closed ball

$$F_1 \subset B_1 \cap C_1^c.$$

If you have possibility of choosing large ball, restrain, choose ball of radius at most $1/2$.
Take open ball $B_2 \subset F_1$ as per our agreement above. Since C_2 is small $B_2 \cap C_2^c \neq \emptyset$. So select closed ball

$$F_2 \subset B_2 \cap C_2^c.$$

Again make sure diameter of F_2 is at most $1/2^2$. Take open ball $B_3 \subset F_2$ as per our agreement. Then select a closed ball

$$F_3 \subset B_3 \cap C_3^c.$$

Make sure its diameter is at most $1/2^3$.
Here is the inductive step. Suppose we got the balls (B_i) and (F_i) for $i = 1, 2, \cdots, n-1$ as satisfying the three conditions. here is how we construct B_n and F_n. Of course B_n is the open ball contained in F_n as per our agreement earlier. Since C_n is small take closed ball $F_n \subset B_n \cap C_n^c$. If you have a possibility of choosing large F_n, cut it down to have diameter $\leq 1/2^n$. This complete the construction and proof of the theorem. ∎

I would like to stress once again two points which I made earlier.
• First is this. After getting the first three sets you could say, 'continue like this'. But when you write a proof, you must make sure that such a continuation 'for ever' is indeed possible. So it is important for you to show this.
•• Secondly, you see, what we want is just that diameter$(F_n) \to 0$ to get a

point common to all of them; it does not matter how it converges to zero, it does not need to be smaller than $1/2^n$. Thus in stating my conditions suppose I carelessly stated $(***)$ as: diameter $(F_n) \to 0$. You will *not be able to construct* sets by induction simply because it does not make sense to say that we have constructed sets up to n satisfying the three conditions.

Thus whenever you need to make an unending construction you must be able to write the conditions in a way that they make sense inductively; and you should be able to explain that having done the construction up to a stage it can be continued to the next stage.
• • • Please pay attention and do think about it.
There are very many interesting and useful consequences of Baire's theorem. For instance you can show that there is no function on R to R such that its set of discontinuities are exactly the set of irrationals. But we need to continue our journey.

Banach's contraction mapping principle:
We discuss one more important theorem in Complete metric spaces. This is due to Banach.
Let X be a metric space. A map $T : X \to X$ is called a contraction if there is a number c such that $0 \leq c < 1$ and for all points $d(Tx, Ty) \leq cd(x, y)$. Thus application T reduces distances in a strong sense. It is not simply distances are reduced, they are reduced by an assured proportion c.
Contraction is a continuous map. Indeed if $x_n \to x$ then

$$d(Tx_n, Tx) \leq cd(x_n, x) \to 0.$$

Theorem (Banach fixed point theorem, Banach contraction mapping principle).
If T is a contraction of a complete metric space X to itself then there is unique point x^* such $Tx^* = x^*$.

Such a point $Ta = a$ is called a fixed point, because T fixes it. That fixed point is unique is not surprising. If a and b are fixed points then

$$d(a, b) = d(Ta, Tb) \leq cd(a, b); \quad i.e. \quad d(a, b) = 0.$$

Existence of fixed point is also easy, we show that whatever point x you start with, the sequence $x, Tx, T^2x, T^3x, \cdots$ converges to x^*. The space being complete, it is enough to show that this is Cauchy. Because then $T^n x \to z$ (say). By continuity of T, we should have $T(T^n x) \to Tz$ whereas $T(T^n x) = T^{n+1}x \to z$. Thus $Tz = z$.
To show that our sequence is Cauchy, it suffices to show that the series

$\sum_n d(T^n x, T^{n+1}x)$ converges. Because, then given $\epsilon > 0$, you can fix N so that $\sum_{n>N} d(T^n x, T^{n+1}x) < \epsilon$. if $n > m > N$ then by triangle inequality applied several times,

$$d(T^m x, T^n x) \leq d(T^m x, T^{m+1}x) + d(T^{m+1}x, T^{m+2}x) + \cdots d(T^{n-1}x, T^n x)$$

$$\leq \sum_{k>N} d(T^k x, T^{k+1}x) < \epsilon.$$

Finally, to show that the series $\sum_n d(T^n x, T^{n+1}x)$ converges.Just note that

$$d(T^n x, T^{n+1}x) \leq cd(T^{n-1}x, T^n x) \leq c^2 d(T^{n-2}x, T^{n-1}x) \leq \cdots$$

$$\leq c^n d(x, Tx)$$

so that

$$\sum_n d(T^n x, T^{n+1}x) \leq \sum_n c^n d(x, Tx) = \frac{1}{1-c} d(x, Tx)$$

and hence finite, completing proof. ∎

This theorem has several applications and several generalizations. We are not concerned with any generalization. There are several ways contraction maps arise and are useful.

suppose that f is a C^1 map of $[a, b]$ to itself. Since derivative of f is continuous and the interval is closed bounded, there is a number c such that $|f'| \leq c$. Suppose that (by chance) $c < 1$. Then the map f is a contraction map. Indeed by mean value theorem, if you take $x \neq y$ then

$$\frac{|f(x) - f(y)|}{|x - y|} = |f'(\theta)| \leq c.$$

In other words, $|f(x) - f(y)| \leq c|x - y|$. Or equivalently, we have $d(f(x), f(y)) \leq cd(x, y)$.

Before going to next applications, let us remember a little history about the hero of this application: Banach.

There was a time when hell descended on earth: in the form of second world war. Poland suffered very heavily. As far as Maths is concerned here is a brief (incomplete) view.

Some could emigrate early on: Alfred Tarski (logician), Antony Zygmund (analyst), Jerzy Neyman (statistician), Samuel Eilenberg (topologist), Stanislaw Ulam (set theory, computation etc) and several others.

Some stayed on and survived the war, either showing that they have pure blood, whatever it may mean or by going underground: W. Sieprinski, K. Kuratowski, H. Steinhaus and others.

Some committed suicide, like, F. Hausdorff.

Some were put to death in camps: S. Saks, J. Marcinkiewicz, J. Schauder, A. Rajchman, A. Lindenbaum and many many many many others.

Some were saved by Director of a lab (in Lwow, a city in Poland) that makes Typhus vaccine. Since Germans needed it, they allowed him to choose volunteers. This vaccine needs lice. Growing lice is done by carefully packing them, attaching to the calf or thigh of human so that they suck the blood and grow. These volunteers are called lice-feeders. Our hero Banach was one such lice feeder. He survived the war but died soon after due to failed health (and lung cancer).

contraction on R:
Consider the function

$$f(x) = \frac{1}{2}\sin(\cos x) + 239$$

Then you see $|f'| \leq 1/2$ and hence it is a contraction on R. Thus there is a unique fixed point. Of course, this is nothing exciting because you see that the values of the function lie in the interval $239 \pm 1/2$. So you can regard f as a map of this interval to itself.
Note that every continuous map of an interval $[a, b]$ to itself has a fixed point whether it is a contraction or not and so you do not need fixed point theorem for the above map.
The reason that a continuous map of a closed bounded interval $[a, b]$ to itself has a fixed point is the following. $f(x) - x$ is a continuous function. Since values of f are in $[a, b]$, we see $f(a) - a \geq a - a = 0$ and $f(b) - b \leq b - b = 0$. The intermediate value theorem completes the proof. But this proof does not give any algorithm of getting fixed point, whereas for contraction we have an algorithm. Further, for general continuous functions fixed point may not be unique. f is a C^1 function (continuously differentiable) then the derivative is bounded. If $|f'| \leq c < 1$ then f is a contraction and so if you start from any point in this interval and keep applying f successively, you will be heading to the fixed point. Not only that, you get an idea of how close you are to the fixed point. Indeed the estimate obtained above tells you (letting $n \to \infty$) that

$$d(x_m, x^*) \leq c^m \, d(x_0, x_1)/(1 - c).$$

contractions on R^n:
When is a linear map of R^n a contraction? More generally, let us consider an affine map $Tx = Ax + b$ where $A = (a_{ij})$ is an $n \times n$ matrix and b is a vector in R^n. Obviously whether a map is a contraction or not depends on the metric and here we have several metrics on R^n and under each of them it is complete — all the metrics are equivalent. But let us consider, for illustration, only two of those.
Let us consider the Euclidean distance $d(x, y) = \sqrt{\sum (x_i - y_i)^2}$. Thus, using Cauchy-Schwarz,

$$d(Tx, Ty) = \sqrt{\sum_i \left\{ \sum_j a_{ij}(x_j - y_j) \right\}^2} \leq \sqrt{\sum_i \left\{ \sum_j a_{ij}^2 \;\; \sum_j (x_j - y_j)^2 \right\}}.$$

$$= \sqrt{\sum_{i,j} a_{ij}^2 \;\; d^2(x,y)} = \sqrt{\sum_{i,j} a_{ij}^2} \;\; d(x,y).$$

Thus if $\sum_{i,j} a_{ij}^2 < 1$ then the map T is a contraction (in the Euclidean metric) and hence there is a fixed point.
Let us take the distance $d(x,y) = \sum_i |x_i - y_i|$. Then denoting the column sum $\sum_i |a_{ij}| = c_j$ and $c = \max c_j$ we have

$$d(Tx,Ty) = \sum_i |\sum_j a_{ij}(x_j - y_j)| \leq \sum_i \sum_j |a_{ij}| \; |x_j - y_j|$$

$$= \sum_j c_j |x_j - y_j| \leq c d(x,y).$$

Thus if $\max_j \sum_i |a_{ij}| < 1$ then T is a contraction in the d_1 metric and hence has a fixed point.
Let us take the distance $d(x,y) = \max |x_i - y_i|$. Then denoting the row sum $r_i = \sum_j |a_{ij}|$ and $r = \max r_i$,

$$d(Tx,Ty) = \max_i |\sum_j a_{ij}(x_j - y_j)| \leq \max_i \{\sum_j |a_{ij}| d(x,y)\} \leq r d(x,y)$$

Thus if $\max_i \sum_j |a_{ij}| < 1$ then T is a contraction in the d_∞ metric and hence has a fixed points.

Thus the main point is that if any one of the above conditions holds then there is a unique fixed point. Of course you can consider the d_p metrics too.

differential equations:

We shall now discuss another important application. This application is for solving differential equations. In high school you discussed theory of equations. We have a number, do not know what it is, but we know that it satisfies $x^2 - 5x + 6 = 0$. We need to find the number. In this case there is explicit formula, you can 'factorize' this polynomial. Sometimes there was no explicit formula but still you were able to deduce existence of solution using some other rules (Descarte's rule of signs or intermediate value theorem or whatever).
In the study of differential equations, you have a function; you do not know what it is; but you know that the function and its derivatives obey some equation. You need to find the function. Just as in the case of theory of equations it is in general difficult to find explicit formula for the function.

Sometimes you can solve explicitly and get a formula for the function. But such a luxury is very rare. We should be satisfied if we know that there is a function at all satisfying the differential equation. We should be more than happy if we know all possible solutions. If we succeed, we shall try to understand properties of the solution more.

If someone asked you to find a function satisfying: $x'' - 5x' + 6 \equiv 0$, you can proceed like solving equations. Denoting D for differentiation once, $D^2 = DD$, differentiation twice, we can rewrite our equation as $(D^2 - 5D + 6)x \equiv 0$. That is (•) $(D-3)(D-2)x \equiv 0$ or (••) $(D-2)(D-3)x \equiv 0$. Observe that if $(D-2)x \equiv 0$ then (•) holds. Similarly, if $(D-3)x \equiv 0$ then (••) holds. Thus in either case we have a solution to our problem. We easily see $x(t) = e^{2t}$ satisfies $(D-2)x \equiv 0$ and $x(t) = e^{3t}$ satisfies $(D-3)x \equiv 0$. Thus $x(t) = e^{2t}$ and $x(t) = e^{3t}$ are solutions of our original problem. One can see easily that linear combinations $x(t) = c\ e^{2t} + d\ e^{3t}$ are also solutions of our problem. The interesting point is that these are the only solutions, there are no more. Of course, the entire argument using symbol D looks mysterious, but can be justified. Thus we solved our problem using just high school factorization of $(D^2 - 5D + 6)$.

If some one asked you to solve $x'(t) = \sin t$ then it is very easy again. because the function is $x(t) = -\cos t\ +\ c$ for some number c. This was simple because of two reasons. firstly there is no appearance of the function x on the right side. Secondly we could integrate the function. You can integrate whenever right side does not involve the function x (or its derivatives etc). Whether you can get a formula or not depends on the right side. For example $x'(t) = \exp\{-t^2\}$ will not allow you to come up with a formula. But in these cases you are sure that the integral of the right side is indeed a solution. Thus some problems are solved by integration.

Serious problems arise if the right side also involves the function x. For example solve: $x'(t) = \sin x(t)$. It does not make sense to write $x(t) = \int_0^t \sin x(s)ds$ and say we solved the problem. We do not even know if there is a solution at all (yes, there is, you will see). There is no need for us to define what is a differential equation and what is meant by a solution. This is reserved for a later course. We consider a specific problem, so you will have no trouble following.

Differential equations arise in several contexts. More important are they in physics. Imagine that in the space in this room at every point there is a force. Of course, when I say force you will ask; on what is it acting. No, it is

not yet acting — it has the potential to act. For example, we place magnets at several points. It creates a field. We place a particle (iron dust) at the point P, the force at that point will push it in a particular direction. suppose it is pushed to Q, then the force at that point Q will push it to R and the force at the point R will push it to etc and so on.
There is just one subtle point which is very important. I simply said the force at P will act on my particle and push it to Q. Actually when the particle is on its way to Q it will pass through intermediate points, but there are forces at these points and they do their bit as well to push. Thus the particle may actually not reach Q and it may be pushed to Q'. Pause and think about the situation our particle is in. since there are forces at every point, our particle is pushed at every time instant. In other words it is continuously pushed around. Life for our particle is not as discrete as I made it out in the earlier para.

So what is the problem. Well, I now place a particle at this point P. Tell me how it travels. Tell me the trajectory or path the particle takes. so you should tell me at every time instant t the position of the particle. We shall not consider in this generality and in three dimensions.

Another situation where these arise is in geometry. Basically, I want to draw a curve and I have been instructed 'how the curve should curve'. More precisely, at every point of the plane, R^2, a vector (two dimensional) is given. I was given a starting point P. I should draw my curve starting at the given point P and at any point on the curve, the tangent is as prescribed at the point, remember there are placed vectors at every pint of R^2 and if your curve passes through a point $(x(t), y(t))$ then its slope $(x'(t), y'(t))$ should be the given vector at the point $(x(t), y(t))$. Thus $t \mapsto (x(t), y(t))$ is a R^2 valued function. However we consider a simpler version: R valued function as explained below.

Now let us make matters precise. We are given an open set $U \subset R^2$ and a point $(t_0, x_0) \in U$. We are given a function $F : U \to R$. We are asked to locate an interval $(t_0 - \delta, t_0 + \delta)$ and a differentiable real valued function x on this interval satisfying two conditions: $x(t_0) = x_0$ (initial condition) and for every t in this interval $x'(t) = F(t, x(t))$ (differential equation).

You see the change of attitude. Our task is 'local'. In a small interval around the given point t_0 we should solve the problem. The idea is that once you do this you can 'continue' from where you 'reached'. Try to make sense of it. It does not concern us now because we are not going to carry it out and

find out what is the largest interval on which the solution can be defined. We shall show locally and stop. Observe that $(t, x(t))$ is a point on the curve and we are wanting at that point on the curve its tangent should have slope $F(t, x(t))$. We solve this problem assuming certain conditions. Here is the precise theorem.

■ Theorem (Picard): Given

(i) Open set $U \subset R^2$ and a point $(t_0, x_0) \in U$.

(ii) $F : U \to R$ which is continuous and there is a number M such that $|F(t, x) - F(t, y)| \leq M|x - y|$ for all points (t, x) and (t, y) in U.

Then there exists

an interval $(t_0 - \delta, t_0 + \delta)$ and a real valued differentiable function x on this interval such that its graph lies in U and at every point t in this interval $x'(t) = F(t, x(t))$. ■

Proof is very simple. fix a closed ball B with centre (t_0, x_0) such that $B \subset U$. Let K be a bound for $|F|$ on the compact set B. Choose $\delta > 0$ so that

(i) $M\delta < 1$ and

(ii) $[t_0 - \delta, t_0 + \delta] \times [x_0 - K\delta, x_0 + K\delta] \subset B$.

Before we produce the function, let us make an observation that motivates the later considerations.

Solving $x'(t) = F(t, x(t))$ and $x(t_0) = x_0$ (♠)

is same as

solving $x(t) = x_0 + \int_{t_0}^{t} F(s, x(s))ds$. (♣)

Indeed, suppose (♠) holds. Then x being differentiable, must be continuous and F being continuous $F(t, x(t))$ is continuous. In other words x' is continuous. Then rules of integration tell us that

$$x(t) - x(t_0) = \int_{t_0}^{t} x'(t)dt = \int_{t_0}^{t} F(s, x(s))ds$$

which is (♣). Conversely suppose (♣) holds. Then the fundamental theorem of integration gives (♠).

Consider the space X consisting of functions $x \in C[t_0 - \delta, t_0 + \delta]$ satisfying the following two conditions.

(i) $x(t_0) = x_0$ and

(ii) values of x lie in the interval $[x_0 - K\delta, x_0 + K\delta]$.

In other words the space of continuous functions on the said interval whose value at t_0 is as required and $|x(t) - x_0| \leq K\delta$ for every t in this interval

that is, graph of x lies in our rectangle $[t_0-\delta, t_0+\delta]\times[x_0-K\delta, x_0+K\delta] \subset B$.

The space $C[t_0-\delta, t_0+\delta]$ is a complete metric space — exactly like $C[0,1]$ is complete. Its subset X we are considering is a closed subset. If a sequence of continuous functions satisfying the two conditions converge to a function in the space (so uniform convergence) then the limit function also satisfies the two conditions. Hence X is a complete metric space.
Let us define a map T of X to itself by

$$Tx(t) = x_0 + \int_{t_0}^{t} F(s, x(s)ds.$$

The above analysis suggests that if we find a fixed point of this map then it satisfies our requirements. Thus the proof is completed by showing firstly, that the map T we defined takes X to itself and secondly, it is a contraction.

Since F is continuous and indefinite integral is continuous we see that Tx is continuous function. Actually the indefinite integral is continuous even if integrand is not continuous, but we never proved such a theorem. We however proved that for a continuous function the indefinite integral is again a continuous function. This is all what we need to see Tx is a continuous function.

Also from definition of Tx we see that its value at t_0 is indeed x_0. Further, since x takes values in the interval $[x_0 - K\delta, x_0 + K\delta]$ the graph of x lies in our rectangle $[t_0 - \delta, t_0 + \delta] \times [x_0 - K\delta, x_0 + K\delta]$ and value of F at points here are bounded by K. Thus

$$|Tx(t) - x_0| = \left| \int_{t_0}^{t} F(s, x(s)ds \right| \leq K|t - t_0| \leq K\delta$$

Thus $Tx(t)$ is also between $x_0 \pm K\delta$. Hence $Tx \in X$ for each $x \in X$.
Finally take any $x, y \in X$ and $t \in [t_0-\delta, t_0+\delta]$. Let us assume $t > t_0$; similar argument applies if $t < t_0$.

$$|Tx(t) - Ty(t)| = \left| \int_{t_0}^{t} \{\ F(s, x(s)) - F(s, y(s))\ \} ds \right|$$

$$\leq \int_{t_0}^{t} |F(s, x(s) - F(s, y(s)|ds \leq \int_{t_0}^{t} M|x(s) - y(s)|ds$$

$$\leq \int_{t_0}^{t} Md(x, y)\ ds \leq Md(x, y)(t - t_0) \leq M\delta d(x, y).$$

This shows $d(Tx, Ty) \leq (M\delta)d(x, y)$ and since $M\delta < 1$ we have proved that T is indeed a contraction. This complete the proof. ∎

Recall that a function g of one variable is said to be Lipschitz if there is a number M such that $|g(x) - g(y)| \leq M|x - y|$ for all x, y in its domain. This number is called the Lipschitz constant. Thus what we demanded, apart from continuity of F, is that it be Lipschitz in the second variable x for each fixed value for the first variable t. Further the Lipschitz constant does not depend on t, thus the same number M works for all t. This condition is expressed by saying that F is Lipschitz in the second variable, uniformly in the first variable.

You might be wondering what has such a theorem got to do with physics, because, the equations you come across there involve second derivatives. Remember, the simplest situation is $F = ma$ or $mx''(t) = F$. Here $x(t)$ is position of particle at time t and $x'(t)$ will give you velocity and $x''(t)$ gives the acceleration. so to know where the particle is you should solve a more complicated equation that we considered. I shall explain the trick.
Suppose you want to solve $x''(t) = -x(t)$. Of course, you know the solutions already. Solving for one function which involves second derivative is transformed to a problem of solving for two functions but involves only first derivative as follows. Solve for two functions x and y with

$$x'(t) = y(t); \qquad y'(t) = -x(t)$$

If you could solve the first problem and get x, take $y = x'$ to see you have a solution for the later problem. Conversely, if you have solution x, y for the second problem then x solves the first problem.
The method for solving for two functions is similar to the above, involves no new ideas but we shall not get into.

Compactness:
We have been discussing complete metric space and what can be done there: Baire category, Banach Contraction etc. Let us pass to another topic which is equally important.

A metric space (X, d) is compact if the following is true: Given any collection of open sets whose union is X, we can pick finitely many of those open sets whose union equals X. In other words, given a collection of open sets that cover X, there are finitely many of those which also cover X.

Theorem: A space (X, d) is compact iff given any family of closed sets whose intersection is empty, there are finitely many of those closed sets whose intersection is empty.
Equivalently, if a family of closed sets is such that intersection of any finitely many of them is nonempty, then their total intersection is non-empty, thus there is a point common to all the closed set in the family.
Proof is just DeMorgan. A collection of closed sets has empty intersection iff their complements (which are open) makeup all of X. ∎

Theorem: A space (X, d) is compact iff every sequence has a limit point.

Definition: Let (x_n) be a sequence in a metric space (X, d). A point $z \in X$ is said to be a limit point of the sequence if for any $\epsilon > 0$, the ball $B(z, \epsilon)$ contains x_n for infinitely many n, i.e. $(\forall \epsilon > 0)(\forall N)(\exists n > N) x_n \in B(z, \epsilon)$.

Proof of the theorem: Suppose that we have a sequence (x_n) which has no limit point. We show space is not compact, by exhibiting a collection of open sets whose union is X but no finite union is X.
Since (x_n) has no limit point, for every $p \in X$, you can fix a ball $B(p, \epsilon_p)$ to which no x_n belongs after some stage. Clearly X is union of all these balls. If you take any finitely many of these, there is a stage after which no point of the sequence is in the union. Thus no finite union can equal X.

Conversely, suppose every sequence has a limit point. First we claim that given any $\epsilon > 0$ we can cover X by finitely many balls of radius ϵ. Indeed let $\epsilon > 0$ be given. take any point x_1. If $B(x_1, \epsilon) \neq X$ then take $x_2 \in X \setminus B(x_1, \epsilon)$ If $B(x_1, \epsilon) \cup B(x_2, \epsilon) \neq X$ take $x_3 \in X \setminus [B(x_1, \epsilon) \cup B(x_2, \epsilon)]$. Having got $x_1, \cdots, x_{n-1}$, if $\cup[B(x_i, \epsilon); 1 \leq i \leq n-1] \neq X$ pick $x_n \in X \setminus \cup[B(x_i, \epsilon); 1 \leq i \leq n-1]$. If the process does not stop at a finite stage, you get an infinite sequence $(x_n, n \geq 1)$ such that distance between any two of them is at least ϵ. So it cannot have a limit. Thus the process stops at a finite stage.

Next we claim that given any open cover of X there is an $r > 0$ such that whatever point $x \in X$ you take $B(x, r)$ is contained in some open set in the given cover. If you say this is not true, I can get for each n, a point x_n such that $B(x_n, 1/n)$ is not contained in any open set in the given cover. But then, hypothesis tells you that the sequence (x_n) has a limit point, say z. Get open U in the given cover such that $z \in U$. get $\epsilon > 0$ such that $B(z, \epsilon) \subset U$. Take N so large that $(1/N) < \epsilon/2$. Since z is limit point of (x_n), get $k > N$ such that $x_k \in B(z, \epsilon/2)$. Observe $(\bullet)$: $B(x_k, \epsilon/2) \subset U$. this is because if $y \in B(x_k, \epsilon/2)$, then $d(z, y) \leq d(z, x_k) + d(x_k, y) < (\epsilon/2) + (\epsilon/2) = \epsilon$ and

hence $y \in B(z,\epsilon) \subset U$. Here then is a contradiction. Since $k > N$ we have $(1/k) < (1/N) < \epsilon/2$ so that $B(x_k, 1/k) \subset B(x_k, \epsilon/2) \subset U$ where as by choice $B(x_k, 1/k)$ is not contained in any set in the given cover.

Now to complete proof that the space is compact, take an open cover, take $r > 0$ as in the para above and using the earlier para cover the space by finitely many balls $B(x_i, r)$; $1 \leq i \leq k$; for each i choose an open set U_i in the given cover that contains the ball $B(x_i, r)$. Clearly these finitely many U_i cover X. ∎

Theorem: X is compact iff every sequence has a convergent subsequence.
this follows immediately from the previous theorem and the following observation.

Theorem: z is a limit point of (x_n) iff there is a subsequence of (x_n) which converges to z.
This was already done in Analysis 1. Here is the proof: If z is a limit point choose inductively points $x_{n_k} \in B(z, 1/k)$ with $n_k > n_{k-1}$. Conversely, if there is a subsequence then every ball around z contains infinitely many indices n_k of the subsequence itself such that x_{n_k} is in the ball. ∎

Theorem: (X, d) is compact iff (i) it is complete and (ii) given $\epsilon > 0$, finitely many balls of radius ϵ cover X.

Proof is straightforward. Suppose X is compact. Firstly, by previous theorem every sequenc has a convergent subsequence. But if a subsequence of a Cauchy sequence converges, then the Cauchy sequence itself convergence. Thus completeness follows. Also ϵ balls around each point cover the space and you can pick finitely many of them by compactness.
Conversely, suppose (i) and (ii) hold. We show every sequence has a convergent subsequence. Let (x_n) be a sequence. Cover the space by finitely many balls of radius 1 (use (ii)). there must be one which contains infinitely many points of the sequence, Say B_1. Cover B_1 by finitely many balls of radius $1/2$ — covers X by using (ii) and consider their intersection with B_1. Since B_1 contains x_n for infinitely many n, One of these intersections must contain x_n for infinitely many n, say $B_1 \cap B_2$ contains infinitely many points of the sequence. Having got balls B_i, for $1 \leq i \leq n-1$ such that B_i has radius $1/i$ and $\cap[B_i : 1 \leq i \leq n-1]$ contains infinitely many points of the given sequence, cover this balls of radius $1/n$ etc and pick B_n so that $\cap[B_i : 1 \leq i \leq n]$ contains infinitely many points of the sequence. Now pick for each k a point $x_{n_k} \in \cap[B_i : 1 \leq i \leq k]$ with $n_k > n_{k-1}$. This subsequence (x_{n_k}) of the given sequence is Cauchy; by (i) must converge to a point. this completes the proof. ∎

We shall now discuss compactness of subsets. Let (X, d) be a metric space. Let $Y \subset X$. Recall that we can regard Y itself as a metric space; metric being the restriction of d to $Y \times Y$.

Theorem: The following two statements are equivalent.

(i) Given any family of subsets of Y which are open in Y and cover Y, there are finitely many whose union equals Y.

(ii) Given any family of open subsets of X whose union includes Y, there are finitely many of those whose union includes Y.

The importance of this theorem is the following. When you re talking about compact subsets of a pace X, any of the above two statements can be used as definition. The statement (i) forgets the background and says that the metric space (Y, d) is compact. On the other hand statement (ii) does not forget the background and states in terms of the open sets of X, does not refer to open subsets of Y at all.
To prove the theorem we only need to observe the following which we noted earlier.

(*) A set $U \subset Y$ is open in Y iff there is a set $V \subset X$ which is open in X such that $U = V \cap Y$.

continuity:
So far we have been taking about sets — closed, open, compact, first category, second category etc. We shall now study functions between metric spaces.

Remember our main idea in taking up metric spaces is the feeling that if we have a concept of 'how close things are' we can do at least a part of the real analysis that deals with convergence and continuous functions. After all $x_n \to a$ meant that x_n is getting closer and closer to a. Further, our idea of continuous function on R to R is that: when x is close to a then $f(x)$ should be close to $f(a)$. This was made precise by saying that whenever $x_n \to a$ then $f(x_n) \to f(a)$. We shall now explore continuous functions and their properties.

Let (X, d) and (Y, ρ) be metric spaces and $f : X \to Y$ be a function and $a \in X$. We say that f is continuous at a if the following is true: $x_n \to a$ in X implies $f(x_n) \to f(a)$ in Y. We say f is continuous function if it is continuous at every point $a \in X$.
Thus f is continuous if it preserves convergence. That is, $x_n \to x$ in X

implies that $f(x_n) \to f(x)$ in Y. All the results that we did in calculus concerning functions on R are all true with nearly the same proofs.

Theorem: f is continuous at a iff the following holds: for every $\epsilon > 0$ there is a $\delta > 0$ such that $\rho(f(x), f(a)) < \epsilon$ whenever $d(x, a) < \delta$.
Suppose the condition holds. To show f is continuous at a take any sequence $x_n \to a$. We need to show that $f(x_n) \to f(a)$. So fix $\epsilon > 0$. We shall show a stage after which $\rho(f(x_n), f(a)) < \epsilon$. With this $\epsilon > 0$ find $\delta > 0$ as assured. Since $x_n \to a$ we have $d(x_n, a) < \delta$ after some stage and clearly then $\rho(f(x_n), f(a)) < \epsilon$.
To prove the converse, assume that the condition fails. Fix an $\epsilon > 0$ for which we can not find $\delta > 0$ satisfying the condition. Since $\delta = 1/n$ does not satisfy the requirement we can pick x_n so that $d(x_n, a) < 1/n$ and yet $d(f(x_n), f(a)) \geq \epsilon$. Clearly $x_n \to a$ but $\rho(f(x_n), f(a)) \geq \epsilon$ for all n. So f is not continuous at a.

Theorem: f is continuous at a iff following holds: For any set $V \subset Y$ which includes a ball around $f(a)$ the inverse image $f^{-1}(V)$ includes a ball around a.
suppose the condition holds. Then we verify the statement of the previous theorem. So let $\epsilon > 0$ be given for which we need to find $\delta > 0$. Take V to be the ball of radius ϵ around $f(a)$ and use condition to get a ball $B(a, \delta) \subset f^{-1}(V)$. This δ will do.
To prove the converse, suppose that the statement of the previous theorem holds. We show that condition of the present theorem holds. Take V as stated, suppose it includes ball of radius $\epsilon > 0$. get a $\delta > 0$ as in the previous theorem. Observe that $f^{-1}(V)$ includes $B(a, \delta)$.

Theorem: f is continuous iff for every open set $V \subset Y$, the set $f^{-1}(V)$ is open in X.
This is immediate from previous theorem. Indeed, suppose f is continuous. Let V be open in Y. To show $f^{-1}(V)$ is open in X take any point $a \in f^{-1}(V)$. Then $f(a) \in V$ and V being open, contains a ball around $f(a)$ so by previous theorem $f^{-1}(V)$ includes a ball around a. Thus $f^{-1}(V)$ includes a ball around each of its points. that is, it is open.
to prove the converse, suppose that condition of the theorem holds. To show f is continuous at say $a \in X$, take $x_n \to a$. Need to show $f(x_n) \to f(a)$. Take open ball of radius $\epsilon > 0$ with centre $f(a)$. Inverse image of this ball is open and also contains a and so from hypothesis, includes a ball around a, say, $B(a, \delta)$. Since $x_n \to a$ we conclude that $x_n \in B(a, \delta)$ after some stage. But then after that stage $f(x_n)$ is in the ϵ ball around $f(a)$. Thus $f(x_n) \to f(a)$.

Theorem: f is continuous iff for every closed set $C \subset Y$, the set $f^{-1}(C)$ is closed in X.

f is continuous and C^c is open implies $f^{-1}(C^c)$ is open and hence $f^{-1}(C)$ is closed. Conversely, Given condition implies that for any open set $f^{-1}(U^c)$ is closed and hence $f^{-1}(U)$ is open. Here we used properties of inverse images

Theorem: If $f : X \to Y$ is continuous and X is compact, then $f(X)$ is compact.

If you have a collection $\mathcal{U}$ of open sets in Y covering $f(X)$ then the family of open sets $\{f^{-1}(U) : U \in \mathcal{U}\}$ covers X and take a finite family that covers X, say, $\{f^{-1}(U_i) : 1 \leq i \leq k\}$ then the finite family $\{U_i : 1 \leq i \leq k\}$ covers $f(X)$. (You should verify this and not simply reproduce this sentence).

In particular, since we know that compact sets are precisely closed and bounded sets we conclude as a special case of the above theorem: if f is a continuous function defined on a closed bounded subset of R, then the range of the function is again closed and bounded. In particular, it attains its bounds too, as they are part of the range.

Connected space:
A metric space (X, d) is connected if the only subsets which are both open and closed are $\emptyset$ and X. Equivalently $X = A \cup B$, A, B disjoint, both closed implies that one of them is empty.
We saw that $[0, 1]$ and R are connected.

If we have a metric space (X, d) and $Y \subset X$, then Y is said to be connected if (Y, d) is connected. Thus background does not matter, you restrict everything to Y.

Theorem 7: If $f : X \to Y$ is continuous and X is connected, then $f(X)$ is connected.
That is, image of a connected set is connected. Observe that $f : X \to f(X)$ is a continuous function. If $f(X)$ is not connected, then there is a non-empty proper subset $A \subset f(X)$ which is both closed and open. Clearly its inverse image is both closed and open and non-empty and proper subset of X.

In particular, since we know that the only connected subsets of R are intervals we conclude the following: Range of a real valued continuous function defined on a connected metric space is again an interval. In particular range of a continuous function defined on an interval in R is again an interval. as

a result the function has intermediate value property, that is, if $f(x) = a$ and $f(y) = b$ and $a < c < b$ then there is a point z such that $f(z) = c$ and $x < z < y$.

If $Y \subset X$ is connected then its closure $\overline{Y}$ is also connected. Indeed if $\overline{Y} = A \cup B$ where A, B are nonempty disjoint closed (in $\overline{Y}$) then $(\bullet)$ $Y = (A \cap Y) \cup (B \cap Y)$. Since A is closed in $\overline{Y}$, there is a closed set $C \subset X$ such that $A = C \cap \overline{Y}$. But $\overline{Y}$ being a closed set we conclude that A (intersection of two closed sets) is itself closed in X. Thus $A \cap Y$ is closed in Y. Similarly $B \cap Y$ is closed in Y. Thus the two sets in the decomposition $(\bullet)$ are closed in Y. Since A is closed in X, A^c is open in X, since A is non-empty and has points of closure of Y it must have points of Y as well. Similarly $B \cap Y$ is nonempty too. Thus in the decomposition $(\bullet)$ the two sets are non-empty. This shows $(\bullet)$ is a disconnection of Y, in. case, there is a disconnection of $\overline{Y}$.

Pathconnectedness:
Let (X, d) be a metric space and $Y \subset X$. We say Y is path connected if given any two points in Y there is a path in Y joining the two points. A path in a space Y is a continuous function $f : [0, 1] \to Y$.
You can think of path as just the function or its image in Y – this later picture is what we do most of the time. This path said to join the two points $f(0)$ and $f(1)$. Note that path in Y has to live in Y, that is, for all $0 \leq t \leq 1$ we should have $f(t) \in Y$.

Clearly if Y is pathconnected then it is connected. Indeed if $Y = A \cup B$ disjoint non-empty sets, then pick $a \in A$, $b \in B$ and let f be a path joining a to b. Let $s = \sup\{t : 0 \leq t \leq 1 : f(t) \in A\}$. This set is non-empty because it contains 0 and is not all of $[0, 1]$ because 1 is not here. By definition of sup, there is a sequence t_n in the set above such that $t_n \to s$ and f being continuous $f(t_n) \to f(s)$. So if A were closed, then $f(s) \in A$. Since $f(1) \notin A$ we conclude $s < 1$. But then take $s < u_n < 1$; $u_n \to s$ to see $f(u_n) \in B$ and $f(u_n) \to f(s)$, showing B can not be closed.

Consider the following set in R^2.

$$S = \{(x, \sin(1/x)) : 0 < x \leq 1\} \cup \{(0, y) : -1 \leq y \leq 1\}$$

This is a compact space being a closed bounded subset of R^2. This is also connected. Here is how to see. The graph of $\sin(1/x)$ on $(0, 1)$ is connected being image of $(0, 1)$ under the continuous map $x \mapsto (x, \sin(1/x))$. It is easy to see that S is nothing but the closure of this graph.

S being a closed bounded subset of R^2 it is compact. To show that S is connected, you can, for example, say the interval $(0, 1]$ is connected and its continuous image $x \mapsto (x, \sin(1/x))$ is connected and hence its closure, namely S, is connected.

Of course, you can show S is connected with bare hands as follows. Let $A \subset S$ be non-empty subset which is both closed and open. Suppose it has a point $(a, \sin(1/a))$ then the entire curve should be in A. In fact, if some $(b, \sin(1/b))$ is not in A then consider all t between a and b such that $(t, \sin(1/t)) \in A$ and arrive at a usual (?) contradiction by considering the sup or inf of this set of points t. If the entire curve is in A then A being closed, it must include the other points of S from the Y-axis too. Note that each point $(0, y)$ in S is limit of a sequence of points on the curve.

However S is not path connected. For example there is no path joining $P = (0, 0)$ to $Q = (1, \sin 1)$. Pause and Think. Looks obvious but you can not think of a simple argument!.
Suppose there is a continuous $f : [0, 1] \to S$, $f(0) = (0, 0)$ and $f(1) = (1, \sin 1)$. The map t going to first coordinate of $f(t)$ includes in its range the points zero and one and it is a continuous map on $[0, 1]$ and hence its range being connected must include all of $[0, 1]$. But the only point in S with first coordinate $u \neq 0$ is $(u, \sin(1/u))$; hence the entire graph must be in range of f. Now use continuity of f at 0. get $\delta > 0$ such that $t \in [0, \delta)$ implies $d(f(0), f(t)) < 1/4$. The oscillations of the $\sin 1/x$ graph should rigorously lead you to a contradiction.

Here is a useful theorem:
Theorem: Any open connected subset of R^2 is pathconnected.
Let U be open connected (nonempty). Fix $a \in U$ Let S be the set of all points $b \in U$ such that there is a path in U from a to b. Since U is open, there is a ball $B(a, r) \subset U$ and clearly $B(a, r) \subset S$. Thus S is nonempty. In fact the same argument shows that S is open: if $b \in S$, then there is an r such that $B(b, r) \subset U$ and you can take a path from a to b and extend it to a path to any point in this ball. Thus $B(b, r) \subset S$. Also S is closed in U. indeed if $b_n \in S$ and $b_n \to b \in U$, then first get $r > 0$ such that $B(b, r) \subset U$ and then an N such that $x_N \in B(b, r)$. Now a path from a to b_N can be found (because $b_N \in S$) and can be extended to a path to b within this ball. Thus S is both open and closed in U and U being connected we conclude $S = U$. ∎

One can define local concepts too like locally connected, locally compact, locally path connected etc but those are reserved for a later course in topol-

ogy. We have miles to go.

separability:
Let (X, d) be a metric space. A subset $D \subset X$ is said to be dense in X if every nonempty open set contains a point of D. equivalently $\overline{D} = X$.
If $X = R$ is real line with usual topology, then Q is dense, Q^c is also dense, complement of Cantor set is dense, R itself is dense.

A space is separable if it has a countable dense set. Complete separable metric space is called Polish space (respecting work of Polish mathematicians on such spaces).

R (rationals), R^n (points with rational coordinates), $C[0, 1]$ (polynomials with rational coefficients) are separable spaces.
R with metric $d(x, y) = 0$ or 1 according as $x = y$ or $x \neq y$ is not separable. In fact no subset other than R itself can be dense!
$B[0, 1]$ the collection of all bounded real valued functions defined on the unit interval $[0, 1]$ with $d(x, y) = \sup |x(t) - y(t)|$ is metric space. It is not separable because for each number $0 \leq u \leq 1$ define the function $x_u(t) = 1$ when $t = u$ and zero when $t \neq u$ and observe that $d(x_u, x_v) = 1$ for $u \neq v$. thus if you take ball of radius $1/4$ around each of these you get disjoint non-empty open sets and any dense set should have a point in each of these.

Homeomorphism/isometry:

You are familiar with isomorphism of groups, rings, fields, vector spaces and so on. Basically these are functions between algebraic structures that help you recognize when two things are same, provided you look at them the right way. In topology isomorphism is called homeomorphism. Let X and Y be metric spaces.

A function $f : X \to Y$ is called a homeomorphism if it is one-one, onto, continuous and inverse map is also continuous from Y to X.
Thus open sets in X are carried to open sets in Y, all open sets in Y are images of open sets in X; same with closed sets. A sequence converges to a point in X iff the image sequence converges to the image point in Y. Thus homeomorphism preserves open sets/closed sets/convergence.
Thus, if two spaces are homeomorphic then by renaming, one looks like the other. Renaming x as $f(x)$ the space X looks like Y. But please do remember, this appearance is only for convergence; not for actual distances.
$[0, 1]$ and $[55, 10000]$ are homeomorphic (usual metric). $[0, 1]$ and $(0, 1)$ are not homeomorphic.

Homeomorphism is an equivalence relation among metric spaces. Aha, the collection of metric spaces is NOT a set! (Do not waste time on this point.) What we mean is the following. If you take any specific collection of metric spaces, then among the collection homeomorphism is an equivalence relation. We can state the same property without talking about collections at all as follows.
X is homeomorphic to X (identity map).
If X is homeomorphic to Y then Y is homeomorphic to X (take inverse map).
If X is homeomorphic to Y and Y is homeomorphic to Z, then X is homeomorphic yo Z (take composition of maps).

But homeomorphism need not preserve the metric. for example the function $f(x) = 1/x$ is a homeomorphism of $(0, 1)$ onto $(1, \infty)$ that does not preserve the distance. Then why is it called homeomorphism? Generally when you say isomorphism or homomorphism etc they preserve the structure that we have. Homomorphisms preserve 'forward' and isomorphisms preserve both ways (for this to make sense isomorphism has to be bijections and etc).
Homeomorphism is an important concept in understanding spaces. For example if you have understood a group, then you have understood all groups isomorphic to it. If you have understood a vector space, then you have understood all vector spaces that are isomorphic to it. similarly, if you have understood open sets/convergence/ in a metric space, then you have understood in all spaces homeomorphic to it. Remember homeomorphism may not preserve distance. Thus understanding here means convergence and continuous functions. You can demand that metric is also preserved, this has a different name.
$\varphi : (X, d) \to (Y, \rho)$ is called an isometry if it is distance preserving bijection, that is, one-one, onto and $d(x_1, x_2) = \rho(f(x_1), f(x_2))$ for all pairs of points.
Obviously isometry is a homeomorphism and much more.

Cantor's completion of metric spaces:
Having seen how important it is to have a complete space, the natural question is the following. If the metric space is not complete, is there anything we can do complete it?

Why is the space not complete. There are Cauchy sequences which are not converging. So either we should make sure such sequences are not Cauchy or provide a point (of convergence) for each such sequence. The first alternative works if we can provide suitable metric without changing the notion

of convergence. This is possible if the space is already an open subset of a complete metric space. This is possible even if the space is a set which is countable intersection of open sets in a complete space. It stops there and does not work for any metric space.
Besides, in the procedure described above, there are two problems. firstly, I said if your space is an open subset of a complete space you can change the metric. But how do you recognize that there is indeed a bigger space which is complete and our space is indeed an open subset of it? Second point is that one may not like to change the distance. After all, if a particular metric is natural then changing it, just to achieve some other desired property, makes the new metric artificial and devoid of meaning. such a thing should be avoided.
So we look for the second alternative. This is what Cantor did. We do the same thing that we did in Cantor's construction of real numbers (we did not carry out all the details then!). Exactly the same procedure works, not only for the set of rational numbers, but for any space. This is what we do now.

So let (X, d) be a metric space. Consider the space C_X of all cauchy sequences in X. Thus each element of C_X is a Cauchy sequence (x_n). We identify if two Cauchy sequences if they 'appear' to be converging to the same point.

Define $(x_n) \sim (y_n)$ if $d(x_n, y_n) \to 0$. This is an equivalence relation on C_X; you use triangle inequality for transitivity of this relation. Denote the space of equivalence classes by X^*. Thus elements of X^* are bags. Each bag contains Cauchy sequences which are equivalent. If there is one Cauchy sequence (x_n) in a bag then every Cauchy sequence (y_n) equivalent to (x_n) is also in that bag — and nothing else is there. Elements of X^* are denoted $[(x_n)]$ — sometimes when there is no confusion by $[x]$ there by meaning $[(x_n)]$.

Let us observe that if (x_n) and (y_n) are Cauchy sequences in X then the limit $\lim d(x_n, y_n)$ exists. Indeed, to show this, it is enough to show that the sequence of real numbers $\{d(x_n, y_n)\}$ is a Cauchy sequence of real numbers. But this is easy, triangle inequality

$$d(x_n, y_n) \leq d(x_n, x_m) + d(x_m, y_m) + d(y_m, y_n).$$

so that

$$d(x_n, y_n) - d(x_m, y_m) \leq d(x_n, x_m) + d(y_n, y_m). \qquad (\bullet)$$

similarly,

$$d(x_m, y_m) \leq d(x_m, x_n) + d(x_n, y_n) + d(y_n, y_m).$$

so that

$$d(x_m, y_m) - d(x_n, y_n) \leq d(x_n, x_m) + d(y_n, y_m). \qquad (\bullet)$$

Thus by the two (•) above,

$$|d(x_n, y_n) - d(x_m, y_m)| \leq d(x_n, x_m) + d(y_n, y_m).$$

and hence can be made as small as we please for all large values of m and n. You only need to see that both terms on the right side can be made small.

Further if $(a_n) \sim (x_n)$ and $(b_n) \sim (y_n)$ then $\lim d(a_n, b_n) = \lim d(x_n, y_n)$. This is again because

$$|d(a_n, b_n) - d(x_n, y_n)| \leq d(a_n, x_n) + d(b_n, y_n) \to 0.$$

Let us define for $[(x_n)]$ and $[(y_n)]$ in X^*

$$d^*([x], [y]) = \lim d(x_n, y_n).$$

The above analysis shows that this limit exists and is well-defined on X^*.

We show that d^* is a metric on X^*. Clearly $d^*([x], [x]) = 0$. Also $d^*([x], [y]) = 0$ implies, by definition of the equivalence relation, $(x_n) \sim (y_n)$, that is $[x] = [y]$. Symmetry, $d^*([x], [y]) = d^*([y], [x])$ is also clear. the triangle inequality of d leads to the triangle inequality for d^*.

We shall now identify X as a subset of X^*. For $p \in X$, let $\varphi(p)$ be the constant sequence $x_n = p$ for all n, more precisely, $\varphi(p)$ is the bag containing this sequence. Then this map is one-one. This preserves distance too. That is $d(p, q) = d^*([p], [q])$.
We shall now show this space is complete.

■ Some history about our hero Georg Cantor. He constructed the Palace of Set Theory single-handed, formulated Continuum Hypothesis and so on.

Unfortunately even in scientific investigations we have resistance to accept new ideas. — after all we are human beings. Just as the famous Mach and other scientists opposed Boltzman's ideas; jut as the Church opposed Galileo's ideas; here too we have Georg Cantor being opposed from several quarters. the famous Kronecker and Poincare opposed rather very very vehemently. So did the Church. After all God is infinite and hence (!) infinity is God. How can the infinity that represents integers be different from the

infinity that represents real numbers? There can not be many Gods, there is only one God! Worse, you are saying given any God there is a bigger God! Definitely not acceptable.

Cantor passed through bouts of severe depression. It was left to Hilbert to say: No one can dislodge us from the Paradise created by Cantor. Paul Erdos used to say: Keep your brains open. Obviously brains of Kronecker and Poincare were not open during that period.
One is reminded of Max Planck who said (I have not verified): A new scientific truth does not triumph by convincing its opponents and making them see the light, but rather because its opponents eventually die, and a new generation grows that is familiar with it.

We have a metric space (X, d). We considered the space C_X of Cauchy sequences (x_n). Defined an equivalence relation $(x_n) \sim (y_n)$ if they appear to be converging to the same point; more precisely, if $d(x_n, y_n) \to 0$. X^* is the space of equivalence classes or bags as we called them. Define $d^*([x], [y]) = \lim d(x_n, y_n)$. We showed that this limit exists and does not depend on the sequences you have taken from the two bags.

Defined the map $\varphi : X \to X^*$ as follows. For $p \in X$, $\varphi(p)$ is the bag containing the constant sequence all of whose terms equal p. We showed that this is distance preserving map.

Recall that, for a metric space (Z, ρ), a subset $D \subset Z$ is dense in Z if every non-empty open set contains a point of D. equivalently, given $z \in Z$ and $\epsilon > 0$, there is $p \in D$ such that $\rho(z, p) < \epsilon$..

We now show that range $\varphi(X)$ is dense in X^*. To do this, take $[x] \in X^*$ say the bag containing the Cauchy sequence (x_n). Let $\epsilon > 0$. We shall show $p \in X$ such that $d^*(\varphi(p), [x]) < \epsilon$. Choose N such that $n, m \geq N$ implies $d(x_n, x_m) \leq \epsilon/2$. Take $p = x_N$. We show that this does. Since $\varphi(p)$ is the constant sequence $\{p, p, p \cdots\}$ we see, by calculation,

$$d^*(\varphi(p), [x]) = \lim d(x_N, x_n) \leq \epsilon/2 < \epsilon.$$

We now show that X^* is a complete metric space. But before we do this let us make an observation which will avoid some notational confusion later.

Theorem: Let (S, d) be a metric space and $T \subset S$ be a dense subset. Assume the following: (x_n) is a Cauchy sequence; $x_n \in T$ for all n then $\exists x \in S$ such that $x_n \to x$. Then the metric space S is complete.
In other words, if you know Cauchy sequences with terms coming from T converge (in S, we are not saying that they converge to points in T) then every Cauchy sequence in S converges.

Proof is simple and as follows. let (x_n) be a Cauchy sequence in S. for each n, using denseness of T pick $y_n \in T$ so that $d(x_n, y_n) < 1/2^n$. Now we see that (y_n) are all points in T. The plan is to show that (y_n) is a Cauchy sequence and get its limit guaranteed by hypothesis, and show that original sequence (x_n) also converges to this limit. Towards showing (y_n) is Cauchy,

fix $\epsilon > 0$. Choose N so that $m, n \geq N$ implies $d(x_n, x_m) < \epsilon/2$ and also $\sum_{m \geq N} 2^{-m} < \epsilon/2$. if now, $m, n \geq N$ then $d(y_m, x_m) + d(x_n, y_n) \leq \sum_{k \geq N} 2^{-k} < \epsilon/2$ so that

$$d(y_m, y_n) \leq d(y_m, x_m) + d(x_m, x_n) + d(x_n, y_n) \leq (\epsilon/2) + (\epsilon/2).$$

So let $y_n \to y$. Towards showing that $x_n \to y$, you only need to note that $d(x_n, y_n) \leq 1/2^n \to 0$. Or, explicitly, given $\epsilon > 0$, choose N so that $|y_n - y| < \epsilon/2$ for $n \geq N$ and also $1/2^N < \epsilon/2$. Then for $n \geq N$,

$$d(x_n, y) \leq d(x_n, y_n) + d(y_n, y) \leq (\epsilon/2) + (\epsilon/2).$$

This completes the observation. Let us now return to showing that X^* is complete. From the observation, it suffices to show that Cauchy sequences whose points come from the dense set $\varphi(X)$ converge. Accordingly, take a Cauchy sequence $\varphi(p_n)$ where $p_n \in X$ for each n. Since φ is distance preserving map, we conclude that $x = (p_n)$ is a Cauchy sequence in X and is an element of C_X, the space of Cauchy sequences. We show $\varphi(p_n) \to [x]$. Take $\epsilon > 0$. Choose N so that $d(p_n, p_m) < \epsilon/2$ for $n, m \geq N$. We show now $d^*(\varphi(p_i), [x]) < \epsilon$, for $i \geq N$. Keep in mind that $\varphi(p_i)$ is the constant sequence

$$p_i,\ p_i,\ p_i,\ p_i, \cdots$$

and x is the sequence

$$p_1,\ p_2,\ p_3,\ p_4, \cdots .$$

Use definition of d^* to see

$$d^*(\varphi(p_i), [x]) = \lim_n d(p_i, p_n) \leq \epsilon/2.$$

The last inequality is from the fact that $i \geq N$ and as soon as $n \geq N$ we know $d(p_i, p_n) \leq \epsilon/2$.

definition of Completion:

Let (X, d) be a metric space. By completion of (X, d) we mean a metric space (Z, ρ) and a map $\varphi : X \to Z$ such that the following hold. (i) Z is a complete metric space. (ii) φ is distance preserving map. (iii) Range $\varphi(X)$ is dense in Z.

Condition (ii) tells you that $\rho(\varphi(x), \varphi(y)) = d(x, y)$. In particular if $x \neq y$ then $\varphi(x) \neq \varphi(y)$. Thus φ is a bijection from X to $\varphi(X)$. Since the distance is also preserved, when you see $\varphi(X)$ it looks exactly like X. In other words

you see a replica of X in this new space. To put it differently, if you rename the point $\varphi(x)$ as x, then you see X in Z. Thus many times we regard $X \subset Z$. Just like, while constructing real number system starting from rationals, we considered either cuts or Cauchy sequences; but no matter what, we regarded Q as a subset of the R we constructed.
Such a view leads to the nice feeling that the extra points, that is points of $Z \setminus X$, are the new ones needed to show as limits of Cauchy sequences. Thus the new space is not as abstract as it appears, but original set of points with new things thrown in where ever necessary.
Condition (iii) tells you that you have not added unnecessary points. More precisely, if $z \in Z \setminus \varphi(X)$, then denseness of $\varphi(X)$ tells you that there is a sequence $(x_n) \subset X$ such that $\varphi(x_n) \to z$. But then $\varphi(x_n)$ is a Cauchy sequence (recall any convergent sequence is a Cauchy sequence, in any metric space). But φ being distance preserving, we conclude (x_n) itself is a Cauchy sequence in X. It does not converge in X [if it did, say to x then $\varphi(x_n)$ converges to $\varphi(x)$ a point of $\varphi(X)$ contradicting that it converges to a point of $Z \setminus \varphi(X)$.] Thus the point z is essential to show as limit of the Cauchy sequence (x_n).
Condition (i) tells you that your new space is complete. Thus we 'embedded' X in a complete space without bringing in un-necessary points in the process.

Thus what we have shown is that
Theorem: every metric space has a completion.
(X^*, d^*) satisfies all the three conditions.

We shall now show that a completion is unique. More precisely,
Theorem: if there are two completions (Z_1, ρ_1, φ_1) and (Z_2, ρ_2, φ_2) then there is distance preserving bijection between them that keeps X fixed. This means there is a bijective isometry$f : Z_1 \to Z_2$ such that $f(\varphi_1(x)) = \varphi_2(x)$.

Remember $x \in X$ looks like $\varphi_1(x)$ in Z_1 whereas it looks like $\varphi_2(x)$ in Z_2. Sometimes statement of a claim itself includes how to start its proof. Thus let there be two completions with notation as above. We define f as follows. For $x \in X$ we define $f(\varphi_1(x)) = \varphi_2(x)$. Now take any $z \in Z_1$. Since $\varphi_1(X)$ is dense in Z_1 take a sequence $\varphi_1(x_n) \to z$ in Z_1. In particular, $\{\varphi_1(x_n)\}$ is a Cauchy sequence in Z_1. Since φ_1 is distance preserving we conclude that (x_n) is Cauchy in X. But now φ_2 is distance preserving tells us that $\{\varphi_2(x_n)\}$ is Cauchy in Z_2. since Z_2 is complete the limit $\lim \varphi_2(x_n)$ exists in Z_2 and this limit is defined as $f(z)$.
This is a good definition because if some one takes a different sequence

$\varphi_1(y_n) \to z$ then we see

$$\rho_2(\varphi_2(x_n), \varphi_2(y_n)) = d(x_n, y_n) = \rho_1(\varphi_1(x_n), \varphi_1(y_n)) \to 0$$

so that $\varphi_2(y_n)$ also converges to the same limit as $\varphi_2(x_n)$.
This completes the definition of f on Z_1 to Z_2. It is distance preserving because for two points $z, w \in Z_1$ take $\varphi_1(x_n) \to z$ and $\varphi_1(y_n) \to w$ and see

$$\rho_2(f(z), f(w)) \;=\; \lim \rho_2(\varphi_2(x_n), \varphi_2(y_n)) \;=\; \lim d(x_n, y_n)$$

$$=\; \lim \rho_1(\varphi_1(x_n), \varphi_1(y_n)) \;=\; \rho(z, w).$$

It follows that f is one-one too. Indeed if $z \neq w$, then $\rho_2(f(z), f(w)) = \rho_1(z, w) \neq 0$, showing that $f(z) \neq f(w)$.
Shall show that f is onto Z_2. Indeed, if $w \in Z_2$, then using $\varphi_2(X)$ is dense, take $\varphi_2(x_n) \to w$. Repeating the earlier argument conclude $\{\varphi_2(x_n)\}$ and hence $\{x_n\}$ and hence $\{\varphi_1(x_n)\}$ are Cauchy (in their respective spaces) Using completeness of Z_1 get limit z of this last sequence and argue $f(z) = w$.

This completes the proof of the fact that every metric space has a completion and it is unique in the sense that between any two completions there is a distance preserving bijection that is identity on X, which is inside both. More precise statement is as mentioned earlier.

Some times, completion has a concrete representation. For example consider $(0, 1)$ with usual $d(x, y) = |x - y|$. The space is not complete. To complete it you should consider space of Cauchy sequences. But in this case completion is just $[0, 1]$. This is obvious if you consider the map φ to be the map $\varphi(x) = x$ on $(0, 1)$. Here $X = (0, 1)$, $Z = [0, 1]$. We know Z is complete, φ preserves distance and its range $\varphi(X)$ is dense in Z.
But then what happens if you repeated the construction? Well, you see apart from the bags containing constant sequences $\{p, p, p \cdots\}$ for $p \in (0, 1)$; you will get only two new bags. They are: bag containing the sequence $\{1/2, 1/3, 1/4, 1/5, \cdots\}$ and bag containing $\{1/2, 2/3, 3/4, 4/5, \cdots\}$. If you name these two bags as zero and one you have $[0, 1]$.

You should also note that $([91, 92], |x - y|, \varphi)$ where $\varphi(x) = 91 + x$ for $x \in (0, 1)$ is also completion of $(\ (0, 1), |x - y|\ )$. But we do not consider such simply because when there is a 'natural' representation of the completion, we take it. For example the space $C[0, 1]$ with the metric $d(f, g) = \int_0^1 |f(x) - g(x)|dx$ is not complete and its completion also has a natural representation; we shall probably discuss later.

This process is due to Cantor. He used this method to construct real numbers starting from rationals, R is precisely this completion of Q, no more and no less.
This analysis shows some general facts. here is one.

Theorem: Suppose (Z_1, ρ_1) and (Z_2, ρ_2) are two complete metric spaces. suppose D is a dense subset of Z_1 and f is a distance preserving map defined on D into Z_2. Then the map can be extended to a distance preserving map on Z_1 into Z_2. Further, if $f(D)$ is dense in Z_2 then the extension is distance preserving bijection.
Proof is already contained in what we have done above. Take any $z \in Z_1$; use D is dense, get $x_n \in D$ such that $x_n \to z$, conclude (x_n) is Cauchy, conclude $f(x_n)$ is Cauchy, use Z_2 is complete, get its limit and declare that as $f(z)$. One shows that this is well defined; does not depend on the sequence (x_n) you have taken from D.
Also this is distance preserving just as above. Such an extension is unique too, because any distance preserving map has the property that when $x_n \to x$, then $f(x_n) \to f(x)$. This shows that for z, its value $f(z)$ must equal to what we defined.
Of course this extension may not be onto. One can show that it is onto iff $f(D)$ is dense in Z_2.

Here is another fact that comes out of the analysis above.
Definition: Let X^1 be a set and $d^1 : X \times X \to [0, \infty)$ satisfying the following: (i) $d^1(x, x) = 0$. (ii) $d^1(x, y) = d^1(y, x)$. (iii) $d^1(x, z) \leq d^1(x, y) + d^1(y, z)$.. Then d^1 is called a pseudometric and (X^1, d^1) is called a pseudometric space.

Thus d^1 is nearly a metric. It falls short of being metric only because it may not satisfy: $d^1(x, y) = 0$ implies $x = y$. If this condition is also satisfied then this is indeed a metric.
Let us start with (X^1, d^1) as above. then define $x \sim y$ if $d^1(x, y) = 0$. This is an equivalence relation. Indeed, $x \sim x$ because $d^1(x, x) = 0$. If $x \sim y$ then $y \sim x$ by symmetry of d^1. Triangle inequality and non-negativity of d^1, shows that $x \sim y$ and $y \sim z$ implies $x \sim z$.
Let us consider the space of equivalence classes, denote, X. Define $d([x], [y]) = d^1(x, y)$. This is well defined. Indeed, if $u \in [x]$ and $v \in [y]$, then

$$d^1(u, v) \leq d^1(u, x) + d^1(x, y) + d^1(y, v) = 0 + d^1(x, y) + 0 = d^1(x, y).$$

and

$$d^1(x, y) \leq d^1(x, u) + d^1(u, v) + d^1(v, y) = 0 + d^1(u, v) + 0 = d^1(u, v).$$

Thus $d^1(x,y) = d^1(u,v)$.
It is not difficult to verify that (X,d) is a metric space. Thus d satisfies all the three rules above and the missing rule too.
If you carefully see the completion process this is precisely what we did. The space X^1 is the space of Cauchy sequences and $d^1((x_n),(y_n)) = \lim d(x_n,y_n)$. The above equivalence relation is precisely what we used to make bags of Cauchy sequences and the new metric was d^*.
It is also easy to see that if (X^1,d^1) is complete, then so is (X,d).
For example, you can consider R^2 and define $d^1(x,y) = |x_1 - y_1|$ where $x = (x_1,x_2), y = (y_1,y_2)$. This is a pseudo metric but not a metric. If you look at the equivalence classes, you get R (in a precise sense)

Thus the process of completion due to Cantor not only outlined how to complete a metric space, producing a construction of real numbers; it has also thrown out certain general techniques like the above two. [Of course, historically, matters are different. Cantor did for real line construction. But what he did was so powerful that it applied for metric spaces when they were discovered].

Miscellaneous:
As mentioned at the beginning of our excursion into metric spaces, the plan is to execute convergence, continuity, and imitate some of our calculus concepts in a more general setting. We discussed sequences; Cauchy sequences; convergence; continuous functions between metric spaces and several routine results from calculus could be imitated with proofs almost same as in Calculus. We had new concepts like compact, complete, connected, path connected I have resisted the temptation of freely using technical words. This is because many times students use such a technical word but unfortunately can not explain what it means. You should know that technical word is only an agreement to use a compact word instead of long expression. The most important thing is to know and understand what the word stands for.

(i) Distance preserving maps are called isometries. Thus an isometry from a metric space (Z_1,d_1) to a metric space (Z_2,d_2) is a map T satisfying $d_1(x,y) = d_2(Tx,Ty)$. note that isometry does not require it to be surjective. Clearly they are continuous maps.

(ii) In the context of compactness we came across he following for a metric space (X,d): Given any $\epsilon > 0$ finitely many balls of radius ϵ cover the space. When this happens the space is called totally bounded. Thus a metric space is totally bounded if for every $\epsilon > 0$, the space can be covered by finitely

many balls of radius ϵ.. There is another concept: a set $F \subset X$ is called an ϵ-net if every point of X is at a distance at most ϵ from some point of F – equivalently, ϵ balls around points of F cover X. Thus totally bounded can equivalently be stated as: (X, d) is totally bounded iff there is a finite ϵ-net, for every $\epsilon > 0$.

(iii) We talked about compactness. Every compact set is closed. Indeed suppose $A \subset X$ and A is not closed. Take $x \in \overline{A} \setminus A$. Note that every non-empty open set U which contains x must intersect A; otherwise $A \subset U^c$ and hence $\overline{A} \subset U^c$. Easy to see that if B_n is the closed ball $\{z : d(x, z) \leq 1/n\}$ then B_n^c is open and $\{B_n^c : n \geq 1\}$ cover A and finitely many of these do not cover A.
Sets A such that their closure is compact are called precompact. Thus A is precompact iff $\overline{A}$ is compact.

(iv) A set D is dense if every non-empty set contains a point of D. Such sets are also called everywhere dense. Thus Q is everywhere dense in R.
However, if I take $Z \cup (Q \cap [0, 1])$ then it is not everywhere dense, but it is dense in the open interval $(0, 1)$.
On the other hand if I take just the set of integers Z then it is not dense in any (always now non-empty) open interval. In other words given any open interval I, I can find an open subinterval $J \subset I$ such that J has no point of Z. Such sets are called nowhere dense.
A set A in (X, d) is called nowhere dense if given any (always non-empty now) open set $U \subset X$, we can find an open $V \subset U$ such that V has no point of A. Note that when this happens A has empty interior; in fact $\overline{A}$ has empty interior. Because if $\overline{A}$ has. non-empty interior U, then for this U you can not find any V as required above. These sets we called small sets. Countable union of such nowhere dense sets are called first category sets or meagre sets. A set which is not first category is called a second category set. Complements of first category sets are called Residual set.
Q is a first category set in R; $(0, 1)$ is a second category set in R but not a residual set. Q^c is a residual set. The set of nowhere differentiable functions in $C[0, 1]$ is a residual set because its complement was shown to be first category. Thus there are so many such functions, the rest is a countable union of small sets.

(v) The facts that, under continuous map, image of compact set is compact and images of connected set is connected are very useful. For example $[0, 1]$ and $(0, 1)$ are not homeomorphic as one is compact and other is not. Also $[0, 1)$ and $(0, 1)$ are not homeomorphic. Now compactness fails

because both are non-compact. However, if there is a homeomorphism: $f : [0,1) \to (0,1)$ then removing zero keeps the first space connected, but removing $f(0)$ makes the second space disconnected where as restriction of f is still a homeomorphism. Same kind of argument shows that $[0,1]$ and the circle $\{(x,y) : x^2+y^2 = 1\}$ are not homeomorphic. (now you argue removing two points carefully). Note both of these are compact.
What about R and R^2? They are not homeomorphic; you can easily show using earlier arguments (remove a point). But what about R^2 and R^3? More generally R^m and R^n with $m \neq n$? Well, they are not homeomorphic and this is. a famous theorem which we shall not prove. This is one place where topology meets algebra: you associate a group with each R^m and show homeomorphism leads to isomorphism of the associated groups. Finally compute the groups explicitly and show they are not isomorphic.

(vi) You should be careful and remember that Schroder-Bernstein is not true for homeomorphisms! For example $[0,1]$ is homeomorpghc to a subset of $(0,1)$; and $(0,1)$ is homeomorphic to a subset of $[0,1]$. But obviously $(0,1)$ and $[0,1]$ are not homeomorphic.

(vii) A set A in a metric space is bounded if there is a number M such that $d(x,y) \leq M$ for all $x, y \in A$. You can state the same thing as follows: Take any $z \in X$. A set A is bounded iff there ia a number c such that $d(z,x) \leq c$ for all $x \in A$. This is immediate from triangle inequality and does not depend on the point z. You must not be under the impression that closed bounded sets are compact!

(viii) Let (X,d) be e metric space. Distance between points leads to concept of distance between points and regions and distance between regions as follows. For $u \in X$ and $A \subset X$ (non-empty) we define

$$d(x,A) = \inf\{d(x,y) : y \in A\}$$

and for two non empty sets A, B contained in X

$$d(A,B) = \inf\{d(x,y) : x \in A, y \in B\}$$

They are useful in several contexts. Just as we defined balls around points, we can define balls around sets. Let $S \subset X$ (non-empty). The ϵ ball (open is our convention; you can replace our $<$ be $\leq$ below too but we DO NOT)

$$B(S,\epsilon) = \{x : d(x,S) < \epsilon\} = \bigcup_{u \in S} B(u,\epsilon)$$

(ix) We discussed the notion of limit point for a sequence (x_n) — z such that $B(z,\epsilon)$ contains x_n for infinitely many values of n; whatever be $\epsilon > 0$.

Here is notion of limit point for a set A. A point z is limit point of a set A if whatever be $\epsilon > 0$, the ball $B(z, \epsilon)$ contains a point of A other than z. Of course if $z \notin A$, then this last phrase 'other than z' is unnecessary. This is just imitation of real line notion, remember we proved that every infinite bounded set has at least one limit point (Bolzano-Weierstrass Theorem). In a compact metric space every infinite set has a limit point — we have proved it in disguise. Also closure of a set A is nothing but A along with the set of all limit points of A.

(x) There is a very important and useful notion of distance between compact sets defined by Hausdorff. For simplicity, let us take a compact metric space (X, d), you keep in mind the set $[0, 1]$. Let 2^X be the collection of all non-empty compact subset of X. Define

$$d_H(K, L) = \inf\{\epsilon > 0 : L \subset B(K, \epsilon); K \subset B(L, \epsilon)\}$$

It is not difficult to show that d_H is a metric on 2^X. It is also true that $(2^X, d_H)$ is a compact metric space.
Here is an interesting result. consider the two functions $f_1(x) = x/3$ and $f_2(x) = (2/3) + (x/3)$. Let $K_0 = \{0, 1\}$ two point set. Let, in general, $K_n = f_1(K_{n-1}) \cup f_2(K_{n-1})$. Thus

$$K_1 = \{0, 1/3, 2/3, 1\}$$

and

$$K_2 = \{0, 1/9, 2/9, 3/9, 6/9, 7/9, 8/9, 9/9\}.$$

each of these is compact, being finite. In fact these are nothing but the end points of remaining intervals at the successive stages.One can show that the sequence (K_n) converges, in 2^X to C, the Cantor set.
Instead of $[0, 1]$ you keep $[0, 1] \times [0, 1]$ in mind and two nice continuous functions f, g on this set to itself and a $0 < p < 1$, then you can play a nice game. Start with an arbitrary point z in this square, pick one of these functions with probabilities p or $(1 - p)$, say φ_1 (this is either f or g but we do not know, random) and move to $\varphi_1(z)$ and repeat; again independently select f or g with the same probabilities, say φ_2 and move to $\varphi_2(\varphi_1(z))$ and continue. You get a trajectory $\{\varphi_n(\varphi_{n-1}(\cdots(\varphi_1(z))\cdots)); \ n \geq 1\}$ It so happens that even though this is random trajectory, there is a density for the trajectory. Thinking of the square as computer screen, the computer can be instructed to colour the screen depending on the density of the trajectory in each region. This leads to nice pictures, may even end up with your face. These are called random iterations and appear in the context of fractals.
After these miscellaneous remarks concluding our general discussion of metric spaces, we take up an interesting and useful theorem in Analysis.

Arzela-Ascoli:

Let us consider the space $C[0,1]$. How do you know if a given subset of this space is compact? Answer to this question is provided by a theorem that was proved (independently) by two Italian mathematicians. Arzela and Ascoli. This is what we discuss now. Why are we interested in compact sets? Because we want to locate functions satisfying certain given conditions; say satisfying a differential equation. If too many conditions are to be satisfied, may be we can find functions satisfying some conditions, and then functions satisfying some more and then functions satisfying some more etc and hopefully in the limit — assuming that there is a limit function — all the conditions would be satisfied.

But first observe the following. For each $n = 2, 3, 4, \cdots$ define a continuous function x_n on $[0,1]$ as follows. $x_n(t)$ starts at zero when $t = 0$ and increases with slope $2n$ till it reaches 1 at $t = 1/2n$ and then decreases the same way reaching zero at $t = 1/n$ and it stays at that value from then on. Let C be the set consisting these functions $\{x_n; n \geq 2\}$.

All these functions are non-negative and are bounded by one. Thus this is a bounded set. Remember a set is bounded if there is a number M such that distance between any two points of the set si at most M. Also this is a closed set. Indeed, if there is any limit point, equivalently, if there is a subsequence that converges to some x then it must also converge pointwise too and hence $x \equiv 0$. But all these functions are far from zero, indeed, $d(x_n, 0) = 1$ for each n. So there is no limit point.

Thus this set is closed and bounded but not compact. We need more than 'bounded' for a set to be compact. This is what we discover now.

Consider $C[0,1]$ with sup metric. Let K be a compact subset. Let us see some of its properties.

(i) K is closed.

We knew it, but let us recall. take any sequence in K, then it contains a subsequence converging to a point of K, so if the sequence itself converges to a then $a \in K$. Thus K is closed.

Since $[0,1]$ is a compact subset of R, every continuous function on it is bounded. That is, given $x \in C[0,1]$ there is a number M such that $|x(t)| \leq M$ for all t. Even if you take finitely many continuous functions then there is one bound for all of them. In general, if you take infinitely many functions then there may not be a common bound for all these functions. But if you have a compact set of functions then there is a common bound.

(ii) There is a number M such that $|x(t)| \leq M$ for all $x \in K$ and all $0 \leq t \leq 1$.
Indeed we know that the function $\varphi(x) = d(x, 0) = \max |x(t)|$ is a continuous function on $C[0, 1]$, in particular, on K. Thus K being compact this function must be bounded on K, that is, there is a number M such that $\varphi(x) \leq M$ for all $x \in K$. This is what we wanted.

Since $[0, 1]$ is a compact subset of R, every continuous function on it is uniformly continuous. In other words, let $x \in C[0, 1]$. Given $\epsilon > 0$, there is a number $\delta > 0$ such that $|x(t) - x(s)| < \epsilon$ for all t, s with $|t - s| < \delta$. Even if you take finitely many continuous functions then there is one $\delta > 0$ which works for all of them. In general, if you take infinitely many functions then there may not be a common $\delta > 0$ which works for all them. But if you have a compact set of functions then there is one $\delta > 0$ that works for all of them..

(iii) Given $\epsilon > 0$, there is a $\delta > 0$ such that $|x(t) - x(s)| < \epsilon$ whenever $x \in K$ and $|s - t| < \delta$. (Of course, through out it is assumed that s and t are in $[0, 1]$).

Proof is simple. Fix $\epsilon > 0$. Remember, K being compact, there are finitely many balls of radius $\epsilon/3$ that cover K; say $B(x_i, \epsilon/3); \quad 1 \leq i \leq n$. Fix $\delta > 0$ such that $|s - t| < \delta \Rightarrow (\forall i;\ 1 \leq i \leq n)|x_i(s) - x_i(t)| < \epsilon/3$. This does. For, given now any $x \in K$ and any s, t with $|s - t| < \delta$; let us pick i so that $x \in B(x_i, \epsilon/3)$. remember, this means $|x(u) - x_i(u)| < \epsilon/3$ for all u. Thus

$$|x(s) - x(t)| \leq |x(s) - x_i(s)| + |x_i(s) - x_i(t)| + |x_i(t) - x(t)| \leq 3 \times \epsilon/3 = \epsilon.$$

Arzela-Ascoli Theorem Let $K \subset C[0, 1]$. Then K is compact iff the followng three conditions hold.
(i) K is closed.
(ii) There is a number M such that $|x(t)| \leq M$ for all $x \in K$ and all $0 \leq t \leq 1$.
(iii) Given $\epsilon > 0$, there is a $\delta > 0$ such that $|x(t) - x(s)| < \epsilon$ whenever $x \in K$ and $|s - t| < \delta$. (Of course, through out it is assumed that s and t are in $[0, 1]$).
Moreover, if conditions (ii) and (iii) hold for a set K, then they hold for its closure and hence its closure is compact.

Last sentence is obvious. Indeed if a sequence of functions converges in the space, then they converge point-wise too. Thus the same bound M works

for closure too and the δ using $\epsilon/2$ works for the closure, for a given $\epsilon > 0$.

We have already observed that if K is compact, then the three conditions hold. Let us now assume that K satisfies the three conditions and show that it is compact. So take a sequence (x_n) in K. We need to exhibit a subsequence converging to a point of K. Of course if we just exhibit a converging subsequence then it must converge to a point of K because of condition (i).

Since the space $C[0,1]$ is complete, it is enough to exhibit a subsequence which is Cauchy. This is achieved in two steps.

(A) there is a subsequence $x_{n_1}, \quad x_{n_2}, \quad x_{n_3}, \cdots\cdots$ such that for every rational number $r \in [0,1]$, the sequence of numbers $x_{n_1}(r), \quad x_{n_2}(r), \quad x_{n_3}(r), \cdots\cdots$ converges.
(B) the sequence in (A) is Cauchy.

Let us prove (B) first. To avoid ugly notation, let us rename $y_1 = x_{n_1}$, $y_2 = x_{n_2}$, and so on. Thus (y_n) is a sequence of elements in K and for every rational number r, the sequence of numbers $\{y_n(r)\}$ converges. Let us show (y_n) is Cauchy. Fix $\epsilon > 0$. Use condition (iii) with $\epsilon/3$ and get a $\delta > 0$. Take partition $0 = r_0 < r_1 < r_2 < \cdots < r_k = 1$ where each r_i is rational and $r_{i+1} - r_i < \delta$. Such a partition can easily be got. For example, you can, by taking smaller δ if necessary, assume your δ is rational. Take now multiples of $\delta/2$ and stop when you go out of $[0,1]$ taking the last point to be one.

Now choose N such that

$$n, m \geq N \Rightarrow (\forall i \; 1 \leq i \leq k) \; |y_n(r_i) - y_m(r_i)| < \epsilon/3.$$

This is possible because each of the sequences $(y_n(r_i) : n \geq 1)$ is convergent and hence Cauchy. And we have finitely many of them for $1 \leq i \leq k$. We claim this N will do.
To see this, fix any $n, m > N$ and any $t \in [0,1]$. we need to show that $|y_n(t) - y_m(t)| < \epsilon$. Let us say, $r_i \leq t \leq r_{i+1}$

$$|y_n(t) - y_m(t)| \leq |y_n(t) - y_n(r_i)| + |y_n(r_i) - y_m(r_i)| + |y_m(r_i) - y_m(t)|$$

$$\leq 3 \times \epsilon/3 = \epsilon.$$

The first and last terms are smaller than $\epsilon/3$ by choice of δ and the middle term by choice of N. This proves (B)

Let us now prove (A). This is routine 'diagonal argument' we have done once long ago. Let us do it again. Again to avoid ugly notation let us reformulate our problem as follows.

(AA) For each $i = 1, 2, 3, \cdots$ we have a bounded sequence of reals, $\{x^i(n) : n \geq 1\}$. The claim is that there is one subsequence of integers $\{n_1 < n_2 < n_3 < \cdots\}$ so that for each i the subsequence $\{x^i(n_1), x^i(n_2), x^i(n_3), \cdots\}$ converges.

How does this show (A)? We are given a sequence of functions (x_n) from K. So for each rational r we can evaluate our functions at this point to obtain a sequence of numbers $\{x_1(r), x_2(r), x_3(r), \cdots\}$. This sequence is bounded by condition (ii) of the theorem. Thus we have countably many sequences as in the above para. the only difference is that in the earlier para I have one sequence for each i, here we have one sequence for each r. But it does not matter because the set of rationals is countable and we can enumerate it as one sequence. Thus (A) is a consequence of (AA)

We shall now proceed to prove (AA). The idea is simple but execution needs vocabulary. Here is the idea. List your sequences as follows.

$$x^1(1),\ x^1(2),\ x^1(3),\ x^1(4),\ \cdots$$
$$x^2(1),\ x^2(2),\ x^2(3),\ x^2(4),\ \cdots$$
$$x^3(1),\ x^3(2),\ x^3(3),\ x^3(4),\ \cdots$$
$$\vdots$$

Since the first row is bounded, select a subsequence which converges. In other words, put cross marks on some terms so that if you read the sequence along the cross marks then it converges. Now read the second row only along the cross marks. Since it is bounded, there is a subsequence which converges. In other words if you make some cross marks into double crosses, then the second row read along the double crosses converges. Read the third row only along the double crosses and make some of these into triple crosses so that when you read third row along the triple crosses it converges. Continue. The required subsequence is the first crossed place, second double crossed place, third triple crossed place etc. This does. Too much english!

Here is the execution. Choose

$$n_{11} < n_{12} < n_{13} < \cdots \tag{1}$$

so that the sequence

$$x^1(n_{11}), \quad x^1(n_{12}), \quad x^1(n_{13}), \quad \cdots$$

converges. Possible because we started with a bounded sequence. Now consider the sequence of numbers

$$x^2(n_{11}), \quad x^2(n_{12}), \quad x^2(n_{13}), \quad \cdots$$

and take a subsequence that converges. That is,

$$n_{21} < n_{22} < n_{23} < \cdots \tag{2}$$

from among (1) so that

$$x^2(n_{21}), \quad x^2(n_{22}), \quad x^2(n_{23}), \quad \cdots$$

converges. Now consider

$$x^3(n_{21}), \quad x^3(n_{22}), \quad x^3(n_{23}), \quad \cdots$$

and take a subsequence that converges. That is

$$n_{31} < n_{32} < n_{33} < \cdots \tag{3}$$

from among (2) so that

$$x^3(n_{31}), \quad x^3(n_{32}), \quad x^3(n_{33}), \quad \cdots .$$

In general choose

$$n_{k1} < n_{k2} < n_{k3} < \cdots \tag{k}$$

from among

$$n_{k-1,1} < n_{k-1,2} < n_{k-1,3} < \cdots \tag{$k-1$}$$

so that

$$x^k(n_{k1}), \quad x^k(n_{k2}), \quad x^k(n_{k3}), \cdots$$

converges. Put

$$n_1 = n_{11}, \quad n_2 = n_{22}, \quad n_3 = n_{33}, \quad \cdots \tag{$*$}$$

We claim that for each i,

$$x^i(n_1), \quad x^i(n_2), \quad x^i(n_3), \quad \cdots$$

converges. We first argue that $n_k > n_{k-1}$. This is because the k-th selection is a subset of $(k-1)$-th selection. In other words, at the worst we might have

$$n_{k1} = n_{k-1,1}, \quad n_{k2} = n_{k-1,2}, \quad \cdots n_{k,k-1} = n_{k-1,k-1}$$

showing $n_k = n_{kk} > n_{k-1}$. Thus we have a strictly increasing sequence of numbers (n_k).

Since (*) is a subsequence of (1) we see that the first sequence converges along this subsequence. Since (*) is a subsequence of (2) except possibly for the first term, we see second sequence converges along the subsequence. Since (*) is a subsequence of (3) except possibly for the first two terms we see that the third sequence converges along this subsequence. In general (*) is a subsequence of (k) except possibly for the first $(k-1)$ terms so that the k-th sequence converges along this subsequence.
This completes proof of (AA) and thus of (A) and thus of the theorem

There are just two points that need to be mentioned.

Firstly, I used a phrase like 'the first sequence converges along this subsequence'. we have never precisely defined what it means. I hope it is clear, but here it is. suppose we have a sequence $a = (a_n)$ of real numbers and a strictly increasing sequence of integers $S = (m_i)$. We say that the sequence a converges along the subsequence S, if the sequence $\{a_{m_1}, \ a_{m_2}, \ a_{m_3}, \ \cdots\}$ converges.
Second point is the following. we did some construction by induction. I cautioned once that when ever you do such a construction, you should precisely write down what you are going to do and then carry out. Carry out first step and assuming that you constructed up to $(k-1)$-th step, explain how you would do next step. do not say 'like this' 'like that' 'so on' etc. I may also add that subtle point is that when you *before hand* list properties, they should be interpretable inductively. if you are not clear you should refer too an earlier notes where this was discussed.

I carried out an inductive construction without listing before hand what I am proposing to do. That was done in order not to interrupt the thought process set in motion by the motivation. Here is the claim. We construct for each k a strictly increasing sequence

$$S_k = \{n_{k1}, \quad n_{k2}, \quad n_{k3}, \quad \cdots\}$$

of natural numbers so that (i) elements of S_k are among S_{k-1} and (ii) the k-th sequence (x^k) converges along S_k.

Returning to Arzela-Ascoli, property (ii) says that there is one bound for all functions in A. This property is called equi-bounded or uniformly bounded. Thus a subset A is equi-bounded or uniformly bounded if there is a

number M so that $|f(x)| \leq M$ for all $f \in A$ and all x.

Similarly we had the property that given $\epsilon > 0$ there is one $\delta > 0$ such that $|f(s) - f(t)| < \epsilon$ for each $f \in A$ and each s, t with $|s - t| < \delta$ This property is called *equi-uniformly continuous* (or uniformly uniformly continuous, but this sounds awkward).

Thus we can state AA theorem as:
A subset of $C[0,1]$ is compact iff it is closed, equi-bounded and equi-uniformly continuous.
a subset of $C[0,1]$ is pre-compact iff it is equi-bounded and equi-uniformly continuous.

AA for compact metric:
Let X be a compact metric space and $C(X)$ be the collection of real continuous functions on X. Since X is compact these are bounded functions ans sup metric makes sense; $d(f,g) = \sup\{|f(x) - g(x) : x \in X\}$. Also repeat your usual proof to show that uniform limit of a sequence of continuous functions is continuous to conclude that this is a complete metric space. Also repeat usual proof to show that each continuous function on X is uniformly continuous. Here is the theorem.

Arzel-Ascoli: $K \subset C(X)$ is compact iff
(i) K is closed and
(ii) K is bounded, that is, there is M such that $|f(x) \leq M$ for all $f \in K$ and all $x \in X$ and
(iii) K is equi-uniformly continuous, that is, given $\epsilon > 0$, there is $\delta > 0$ such that $|f(x) - f(y)| < \epsilon$ for all $f \in K$ and for all $x, y \in X$ with $d(x,y) < \delta$.

Proof is exactly as above. Set of rationals we used is replaced by countable dense set. We know every compact metric space is separable; union of finite $1/n$ nets ($n \geq 1$) would do. The only difference is when we said earlier $|s - t| < \delta$ we now say $d(x,y) < \delta$ and so on. It is a good exercise for you to pause and meditate.

Here is a promised application of AA.
Ω is an open region in R^2; $F : \Omega \to R$ continuous; $(t_0, x_0) \in \Omega$.
Wanted: An $a > 0$ and a function x in $(t_0 - a,\ t_0 + a)$ satisfying (i) $x(t_0) = x_0$ (initial condition) and and (ii) $x'(t) = F(t, x(t))$ for all t (the differential equation/law of motion/curving rule for the curve). Here 'all t' means $t \in (t_0 - a,\ t_0 + a)$. Also (ii) implies that for all t, the point $(t, x(t))$ is in Ω.
Such a function is called an integral curve for: $\frac{dx}{dt} = F(t, x(t))$ passing through (t_0, x_0). Usual notation: $x'(t) = F(t, x)$; $x(t_0) = x_0$.

■ Theorem (Peano): There is an integral curve. ■

We discussed importance of this problem while discussing Picard. You can think of it as a problem in solving differential equations. You can think of it as a problem in drawing curves. You can think of it as problem in describing motion of a particle. notice the local nature of the question.

Proof: Picard solves it like an artist. He draws one curve, then changes to a different curve closer to required one, then he changes again etc. Peano solves it like a mason. draws straight lines — chords – of the unknown curve. then he makes it better by taking points closer and drawing smaller chords and getting closer to the curve etc. (Observe how a mason makes round pillar with rectangular bricks). Idea is the following:
(♠) start with (t_0, x_0). Draw a line with slope $F(t_0, x_0)$ and a little later stop, say you stopped at (t_1, x_1). From Now on start line with slope $F(t_1, x_1)$ and a little later stop, say you stopped at (t_2, x_2). From Now on start line with slope $F(t_2, x_2)$. Continue this way till you reach $t_0 + a$ (though I have not told you what is a). In other words we have a function whose graph is 'polygonal line'. Suppose the points of slope-change are called link points and link length is say $1/k$. It is believable that as k becomes large we seem to be meeting demand at more and more points. Note that at the link points the right derivative agrees with data, by construction. Of course, left derivative is probably not that value, however the continuity of F should make us believe that as k increases, the left derivative is also coming close to the required value. Hopefully in the limit we would meet demands at all points. AA theorem assures us a limit to talk about. This will be made precise now.

First let us fix a (closed) rectangle $S = [t_0-\alpha, t_0+\alpha]\times[x_0-\beta, x_0+\beta] \subset \Omega$. Get M such that $|F(t,x)| \leq M$ on this rectangle. Fix a number $a > 0$ such that the rectangle $T = [t_0 - a, t_0 + a] \times [x_0 - aM, x_0 + aM] \subset S$. You only need to take $a = \min\{\alpha, \beta/M\}$. Our drama from now on takes place on this rectangle. We show curve on $[t_0,\ t_0 + a]$. A similar construction gives on $[t_0 - a,\ t_0]$.

Let us make an observation. Consider an interval $[u, v]$. A function f on this interval will be called a polygonal curve if it is continuous and there is a finite partition $u = a_0 < a_1 < \cdots < a_k = v$ such that on each $[a_i, a_{i+1}]$ the graph is a straight line. Norm of the partition is called link length and the points $\{a_i\}$ are link points. We consider partition into intervals of equal length (except the last interval $[a_{k-1}, a_k = v]$). Suppose f is a polygonal curve. Assume that maximum of the slopes is at most M, in modulus. Then for any two points $x < y$ we have $|f(y)-f(x)| \leq M(y-x)$. Indeed if x, y are in one link, say, $[a_i, a_{i+1}]$ then equation of straight line tells you $f(x) = mx+c$ so that $f(y) - f(x) = (\text{slope})(y - x)$ and hence $|f(y) - f(x)| \leq M(y - x)$. If they are in different links $x \in [a_i, a_{i+1}]$ and $y \in [a_j, a_{j+1}]$ with $j > i$, then expressing

$$f(y) - f(x) = [f(y) - f(a_j)] + [f(a_j) - f(a_{j-1})] + \cdots + [f(a_{i+1}) - f(x)]$$

we apply earlier observation and using each slope is at most M in modulus we see

$$|f(y) - f(x)| \leq M[y - a_j] + M[a_j - a_{j-1}] + \cdots + [a_{i+1} - x] = M(y - x)$$

In particular for any $x \in [u, v]$ we have $|f(x) - f(u)| \leq M|x - u|$

$$f(x) \in [f(u) - M(x - u),\ \ f(u) + M(x - u)] \quad (\bullet)$$

Let us also record earlier observation,

$$|f(x) - f(y)| \leq M|x - y| \quad x, y \in [u, v] \qquad (\bullet\bullet)$$

Can do better. Suppose that slope within each link is between m and M $(m < M)$. Then instead of converting to modulus above we can outright see that for $x > u$,

$$m(x - u) \leq f(x) - f(u) \leq M(x - u) \quad (\bigstar)$$

After these observations, let us return to the original problem. Taking link length $1/k$ let us define polygonal curve x_k as stated in (♠). Keeping in

mind that $|F(t,x)| \leq M$ we conclude from ($\bullet$) that at any point $x_k(t) \in (x_0 - aM, x_0 + aM)$. Thus the curves x_k all lie in the rectangle T. In particular $\{x_k, k \geq 1\}$ is a bounded set. ($\bullet\bullet$) shows that the family is equicontinuous in $C[t_0, t_0 + a]$. In fact, given $\epsilon > 0$, we just take $\delta = \epsilon/M$. Get a (uniformly) convergent subsequence, by Arzela-Ascoli. Renaming, we shall assume that $\{x_k\}$ is convergent. Thus $x_k \to x$ uniformly on this interval. but now link length for x_k may not be $1/k$; say it is is η_k. Observe $\eta_k \downarrow 0$. We shall show x is differentiable at every point $u \in [t_0, t_0 + a)$ and $x'(u) = F(u, x(u))$. So fix u. Fix $\epsilon > 0$. Shall show $\eta > 0$ so that

$$0 < |h| < \eta \Rightarrow \left| \frac{x(u+h) - x(u)}{h} - F(u, x(u)) \right| \leq \epsilon$$

Since we have not yet defined below t_0, interpret this as right derivative if your point u is t_0.

Since we understand the functions x_k that we constructed, we transfer the problem regarding unknown x to our known functions. We exhibit a $\eta > 0$. and a K such that

$$0 < |h| < \eta; k > K \Rightarrow \left| \frac{x_k(u+h) - x_k(u)}{h} - F(u, x(u)) \right| \leq \epsilon \qquad (\clubsuit)$$

In all such statements it is understood that both u and $u + h$ are within our interval, now $[t_0, t_0 + a]$. This being true for all $k > K$ by taking limit in ($\clubsuit$) and using $x_k \to x$ we get the previous inequality, which was to be proved.

choose $\eta > 0$ so that

$$|t - t'| < 2\eta; |x - x'| < 4M\eta \Rightarrow |F(t,x) - F(t',x')| < \epsilon \qquad (\blacklozenge)$$

possible by uniform continuity of F on S. Choose K so that

$$k > K \Rightarrow \eta_k < \eta; \quad \sup |x_k - x| < M\eta \qquad (\blacksquare)$$

possible because $\eta_k \downarrow 0$ and $x_k \to x$ uniformly. remember u, ϵ are given to us. Now we made choice of $\eta > 0$ and K. towards showing ($\clubsuit$), let us fix $k > K$ and fix $0 < h < \eta$. Can run similar argument if $-\eta < h < 0$. We can rewrite ($\clubsuit$) as follows:

$0 < |h| < \eta; k > K$ implies $x_k(u+h) - x_k(u)$ is between $[F(u, x(u)) - \epsilon]h$ and $[F(u, x(u)) + \epsilon]h \qquad (\clubsuit)$

But since x_k is made up of line segments, in view of the observation made earlier, namely ($\bigstar$), we only need to show that the slopes of the line segments that enter in our calculation of $x_k(u+h) - x_k(u)$ are within $[F(u, x(u)) \pm \epsilon]$.

Since the link length $\eta_k < \eta$ and $0 < h < \eta$ it is possible that u and $u+h$ are in different link segments of x_k. Let us denote the partition on which x_k is based by $a_0 < a_1 < \cdots < a_n$. Let $a_i < u \leq a_{i+1}$ and $a_{i+m-1} < u+h \leq a_{i+m}$ where $m \geq 1$. Let us see the slopes of each line segment of x_k during this period.

The first stretch has slope $f(a_i, x_k(a_i))$. (0)
But link length tells you $|a_i - u| \leq \eta_k < \eta$ (1)
From (••) which is valid for all our x_k, we have $|x_k(a_i) - x_k(u)| \leq M|a_i - u| \leq M\eta$. Further, $k > K$ implies by (■) that $|x_k(u) - x(u)| \leq M\eta$. Thus, combining with earlier ineuality, we have, $|x_k(a_i) - x(u)| \leq 2M\eta$. (2)
Now using (1) and (2); ♦ leads to $|F(a_i, x_k(a_i)) - f(u, x(u))| < \epsilon$.
Thus (0) implies slope of first stretch is between $F(u, x(u)) \pm \epsilon$ (A)

Regarding slope for other stretch during this period, consider $[a_j, a_{j+1}]$. Slope of this stretch is $F(a_j, x_k(a_j))$. First observe
$0 < a_j - u \leq h < \eta$. (1)
By (••), valid for all our x_k tells $|x_k(a_j) - x_k(u)| \leq M|a_j - u| \leq M\eta$. But $k > K$ implies by (■) that $|x_k(u) - x(u)| \leq M\eta$. Thus
$|x_k(a_j) - x(u)| \leq 2M\eta$. (2)
Now (1), (2), ♦ imply slope of this stretch is between $F(u, x(u)) \pm \epsilon$ (B)

Thus slopes of all stretches during this period are between $F(u, x(u)) \pm \epsilon$. You can argue exactly same way (take care you do not go below t_0!) for $-\eta < h < 0$. It is here that you need to use 2η, because when you consider $u - h$, the slope of the segment where this lies is determined by that a_i which is below $u - h$. Even though $u - h$ is within distance η of u, the same can not be said about a_i. However a_i is within distance 2η of u because link length $\eta_k < \eta$.
We discussed only on $[t_0, t_0 + a]$ and as mentioned earlier similar analysis holds for $[t_0 - a, t_0]$. Draw polygonal curves starting from t_0 (backwards) upto $t_0 - a$.
This completes the proof. ■

Stone-Weierstrass:
We now proceed to another important theorem regarding dense subsets of $C(X)$ for compact metric X. Compact metric spaces arise often in practice. For example the space of infinite sequences of $+1$ and -1 is a very natural phase space for a spin system consisting of very very large number of particles. This is a compact metric space.

This theorem generalizes the Weierstrass theorem, you learnt last year, which says the following. Given any real valued continuous function f on the interval $[0,1]$, and $\epsilon > 0$, there is a polynomial P such that $|P(x) - f(x)| < \epsilon$ for each $x \in [0,1]$. With our present notation we can restate this as follows: given $f \in C[0,1]$ and $\epsilon > 0$, there is a polynomial P such that $d(f,P) < \epsilon$. Or simply put, the set of polynomials is dense in $C[0,1]$.
Given any polynomial P, and $\epsilon > 0$, there is a polynomial Q with rational coefficients such that $d(P,Q) < \epsilon$. Since the set of polynomials with rational coefficients is countable, we can also conclude that the space $C[0,1]$ is separable. Thus it is a separable complete metric space or a Polish space.

Let $\mathcal{P}$ be the collection of all polynomials on $[0,1]$. This collection satisfies three conditions which are easy to verify: (i) This is an algebra, that is linear combinations and products of elements in $\mathcal{P}$ are again in $\mathcal{P}$. (ii) constant functions are also in $\mathcal{P}$. (iii) Given any two points in $[0,1]$ there is an element of $\mathcal{P}$, — that is a polynomial — which takes different values at the two given points.
The theorem that we are going to discuss is a generalisation of the above. Let X be a compact metric space and $C(X)$ be the space of real valued continuous functions on X with sup metric.

Theorem:
Let $D \subset C(X)$ where X is a compact metric space. If D satisfies he following three conditions then it is dense in $C(X)$.
(i) D is an algebra. That is, linear combinations, products of functions in D is again in D. (ii) Constant functions are in D. (iii) D separates points. That is, given two points $p \neq q$ in X, there is an $f \in D$ with $f(p) \neq f(q)$.
We could have stated (ii) to simply say that the constant function $x \equiv 1$ is in D. Then by (i) every constant function would also be in D.
Note that if condition (iii) fails then D can not be dense. This is obvious because if every function in D takes the same value at the two points p and q then so do their limits and hence it can not be dense. For example the function $f(x) = d(p,x)$ will not be in the closure of D.
That D is an algebra simply means that it is a vector space and closed under multiplication.

This theorem gives new information even in the case of $[0,1]$. For example, if you take any integer $k \geq 1$ then the set of polynomials

$$a_0 + a_1 x^k + a_2 x^{2k} + a_3 x^{3k} + \cdots + a_n x^{nk}; \quad n \geq 0, a_i \in R$$

is dense in $C[0,1]$. That is, polynomials in powers of x^k are dense. For ex-

ample, you can consider powers of $x^{1000000}$ and their linear combinations.

Or you can take polynomials in $\sin x$. They are dense too. That is, the set of functions

$$a_0 + a_1 \sin x + a_2 (\sin x)^2 + a_3 (\sin x)^3 + \cdots + a_n (\sin x)^n : \quad n \geq 0; a_i \in R$$

is dense. Of course these you can obtain from a clever change of variable from usual Weierstrass theorem.

Proof of the theorem:
Let $\overline{D}$ be the closure of D. We try to understand this set enough to conclude that it must be all of $C(X)$.

1°. The set $\overline{D}$ is again an algebra separating points and containing constants.

Since D already has the last two properties any larger set, in particular, closure has those properties. We only need to show that it is an algebra. But it is clear from the fact that if $f_n \to f$ and $g_n \to g$ then
$f_n + g_n \to f + g$ and $cf_n \to cf$ and $f_n g_n \to fg$. remember all these are uniform convergences.

2^0. D and hence $\overline{D}$ strongly separates points: given $p \neq q \in X$ and numbers a, b there is $f \in D$ such that $f(p) = a$ and $f(q) = b$.

Indeed take φ which takes different values at p and q and define

$$f(x) = \frac{\varphi(x) - \varphi(q)}{\varphi(p) - \varphi(q)} a + \frac{\varphi(x) - \varphi(p)}{\varphi(q) - \varphi(p)} b$$

Even though φ appears at several places, only numbers calculated from it appear mostly. The constant function $\varphi(q)$ is in D and hence so is $\varphi - \varphi(q)$ and so is the ratio appearing in the first term above and so is the first term itself. Thus it is a function in D. So is the second term. so is their sum. It takes the required values at the given points.

3°. $f \in \overline{D}$ implies $|f| \in \overline{D}$.

Indeed, take $\epsilon > 0$. Let the range of f be contained in the interval $[c, d]$. By usual Weierstrass theorem get a polynomial $P(t)$ such that for all t in this interval $||t| - P(t)| < \epsilon$. In other words the function $|t|$ is approximated by a polynomial on this interval. It is clear that $||f(x)| - P(f(x))| < \epsilon$, for all $x \in X$. In other words the distance between the two functions $|f|$ and $P(f)$ is at most ϵ. Remember that if p is the polynomial,

$$P(t) = a_0 + a_1 t + a_2 t^2 + \cdots + a_k t^k$$

then $P(f)$ is the function

$$a_0 + a_1 f + a_2 f^2 + \cdots + a_k f^k.$$

This is in $\overline{D}$ because f is in it and it is an algebra. Thus there are elements close to $|f|$ which are in $\overline{D}$. The later being closed, we conclude that $|f|$ itself is in it.

4°. $\quad f, g \in \overline{D} \Rightarrow f \vee g, f \wedge g \in \overline{D}$.

Recall $f \vee g(x) = \max\{f(x), g(x)\}$; $\quad f \wedge g(x) = \min\{f(x), g(x)\}$. The claim is immediate from the previous observation and the fact

$$f \vee g = \frac{f + g + |f - g|}{2}; \quad f \wedge g = \frac{f + g - |f - g|}{2}.$$

5°. Fix $f \in C(X)$, fix a point $p \in X$ and fix a number $\epsilon > 0$. Then there is a function $h \in \overline{D}$ such that $h(p) = f(p)$; $\quad h(x) < f(x) + \epsilon$

Of course this function depends on p and should have been denoted by h_p but to lighten the notation we did not do that. Thus we can get a function in our set which is close to a given f. Well, not really close; but does not go too far above f; it may however go too far below f.
Proof is simple. For any $q \neq p$ fix a function $\varphi_q \in \overline{D}$ which takes same values as f at both p and q. This is possible by previous observation. Let

$$U_q = \{x : \varphi(x) < f(x) + \epsilon\}.$$

Then for each q we have an open set conning the point q and of course all these sets contain p as well thus they cover X. Pick finitely many, say

$$U_{q_1};\ U_{q_2};\ U_{q_3};\ \cdots U_{q_k}.$$

Consider the function $h = \varphi_{q_1} \wedge \varphi_{q_2} \wedge \cdots \varphi_{q_k}$.

Since each $\varphi_q(p) = f(p)$ we see $h(p) = f(p)$. Given any point $x \in X$ it is in some U_{q_i} and so $\varphi_{q_i}(x) < f(x) + \epsilon$ and so $h(x) < f(x) + \epsilon$. Since each $\varphi_q \in \overline{D}$ previous observation tells us that $h \in \overline{D}$.

6°. Fix $f \in C(X)$ and $\epsilon > 0$ the there is a $g \in \overline{D}$ such that

$$f(x) - \epsilon < g(x) < f(x) + \epsilon; \quad \forall\, x \in X.$$

In other words $d(f, g) < \epsilon$. Since this is true for every $\epsilon > 0$ and since $\overline{D}$ is closed we conclude that $f \in \overline{D}$. Since $f \in C(X)$ is arbitrary we conclude that $\overline{D} = C(X)$. In other words D is dense in $C(X)$; proving the theorem.

6° is proved as follows. For each $p \in X$, use earlier observation to get h, now let us denote it by h_p so that

$$h_p \in \overline{D}; \quad h_p(p) = f(p); \quad h_p(x) < f(x) + \epsilon \ \ \forall\, x \in X.$$

Set $V_p = \{x : h_p(x) > f(x) - \epsilon.\}$ Then $p \in V_p$. In other words the family of all V_p covers the space X and take a finite sub family that covers, say,

$$V_{p_1};\ V_{p_2};\ \cdots ; V_{p_k}.$$

Set

$$g = h_{p_1} \vee h_{p_2} \vee \cdots \vee h_{p_k}.$$

Given any point x, it is in some V_{p_i} and so $h_{p_i}(x) > f(x) - \epsilon$ and hence $g(x) > f(x) - \epsilon$. Also given $x \in X$ each $h_p(x) < f(x) + \epsilon$ and hence so is the finite max $g(x)$. By earlier observation, we see $g \in \overline{D}$.

This completes the proof.

We have already seen implications of this theorem for continuous functions on the interval $[0, 1]$. We can use this theorem to show that for a compact metric space X, the space $C(X)$ is a separable metric space. Thus it is a Polish space (complete separable metric space).
We can also show that trigonometric polynomials are dense in the space of continuous functions f on $[0, 2\pi]$ with $f(0) = f(2\pi)$. by applying this theorem to the unit circle.

Space Filling Curves or Peano Curves:
Cantor startled the math community by showing that R^2 has same number of points as R. But of course, he could exhibit only some identification and it was unclear whether one could get a continuous function.

Peano startled the math community by showing that there is a continuous function from the unit interval $[0, 1]$ onto the unit square $[0, 1] \times [0, 1]$. After all, a continuous function from the interval to the plane should 'trace a curve' which should be 'one dimensional'. But that mental picture was proved to be wrong. This curve of Peano fills up all of the square. That is why such functions from then on are called Peano curves or space filling curves.
Soon after words Hilbert gave another geometric construction which is more fertile, in the sense, it inspired others and also gave a general method of construction. It also helped to evaluate some interesting properties such maps can possess.

One knew properties such maps can not possess. For example such a map can not be one-one. If f is a continuous map of a compact metric space X onto another compact metric Y space which is one-to one; then its inverse is already continuous and hence such a map is a homeomorphism. In fact, if V is open in X, then V^c is closed and hence compact (because X is so) and hence so is $f(V^c)$ and hence it is closed in Y and hence $f(V) = [f(V^c)]^c$ is open. This shows that f^{-1} is continuous.
In our case $X = [0,1]$ and $Y = [0,1]\times[0,1]$ and we knew that they can not be homeomorphic. It was also known that such maps can not be differentiable.

At that time when they were found, these were curious objects. But now-a-days they are finding applications in computer science. After all, by talking about point t on the line you can mean point $f(t)$ in the square. In other words, you can stay on the line and talk about two dimensional things. This was found useful in data organisation. I do not know this topic.

However, there are some interesting algorithmic questions where these maps are found useful — for example, for the travelling salesman problem (known to be 'hard' in some sense). This problem has a data of finitely many points in the unit square and asks for a tour of these points which is economical; shortest possible. There is no idea how to make a beginning of a tour. If you have a Peano curve you can make a beginning (good or bad). Take those points given to you, take 'some' points $\{t_i\}$ such that your data is $\{f(t_i)\}$ and tour in the increasing order of the points t_i. In other words the problem is linearized, an order has been brought. Apparently Sierpinski also had one which is used in this context, but I do not know.

I shall describe the Hilbert method but do not carry out because it needs some vocabulary. Our interest is only to make it clear to ourselves that such curves exist. We are not interested in either studying their properties or in using them. So we follow a simpler method later and completely work out. But you should know Hilbert method.

Divide the interval $[0,1]$ into four parts

$$I_1 = [0,1/4];\ I_2 = [1/4,1/2];\ I_3 = [1/2,3/4];\ I_4 = [3/4,1].$$

Divide the square also into four parts

$$R_1 = [0,1/2]\times[0,1/2]; \quad R_2 = [0,1/2]\times[1/2,1]$$

$$R_3 = [1/2,1]\times[1/2,1]; \quad R_4 = [1/2,1]\times[0,1/2].$$

Draw a curve joining by straight lines, the mid point of R_1 to mid point of R_2 to R_3 to R_4. Set up a map f that takes I_j to the part of your curve in R_j. This is our first function f_1.
Now divide each I_j into I_{jk} for $k = 1, 2, 3, 4$, and similarly the squares. assign the four parts of the line I_j to the four parts of the square R_1 and so on and draw a curve made up of straight lines. This is our second function f_2. You need to be careful and give an algorithm on how to do it. This can be done.

To do this you need either picture or vocabulary. But the point is that these functions, so obtained, converge to a function which is continuous and its image is all of the square. We shall not pursue this process.

We exhibit our curve (due to Lebesgue) after two observations.
Remember the construction of the Cantor set. We define

$$I_0 = [0, 1/3]; \qquad I_2 = [2/3, 1]$$

$$I_{00} = [0, 1/3^2];\ I_{02} = [2/3^2, 3/3^2];\ I_{20} = [6/3^2, 7/3^2];\ I_{22} = [8/3^2, 9/3^2].$$

and so on.

$$C_0 = [0, 1];\ C_1 = I_0 \cup I_2;\ C_2 = \cup\{I_{ij}; i, j = 0, 2\} \quad C = \cap C_n$$

is the Cantor set.

Observation 1: There is a continuous function f defined on the Cantor set onto the unit interval $[0, 1]$.
This is simple. we know that a point x is in the Cantor set iff it has triadic expansion involving only the digits zero and two. That is,

$$x = \frac{x_1}{3} + \frac{x_2}{3^2} + \frac{x_3}{3^3} + \cdots; \quad x_i = 0, 2.$$

equivalently

$$x = \frac{2x_1}{3} + \frac{2x_2}{3^2} + \frac{2x_3}{3^3} + \cdots; \quad x_i = 0, 1.$$

Keep in mind that such an expansion is unique. It is worth remembering that the point $1/3$ is in C and it has the eligible ternary expansion $.0222222\cdots$.
Also remember given any sequence of zeros and ones, the series $\sum(2x_i)/3^i$ converges and the number so defined is a point in $[0, 1]$ and is in the Cantor set.
Here then is the function

$$f(x) = \sum \frac{x_i}{2^i}; \qquad x = \sum \frac{2x_i}{3^i}.$$

This is good definition because, given a point $x \in C$ it has a unique eligible ternary expansion (that is involving zeros and twos).
Clearly given any number $y \in [0,1]$, it has a binary expansion $\sum y_i/2^i$ with each y_i zero or one and then the point $x = \sum(2y_i)/3^i$ is in C and $f(x) = y$ to show that f is onto $[0,1]$.
Finally continuity of f is seen as follows. let $x_n \to x$, all in C. If $x \in I_0$ then first digits of the x_n after some stage must be zero. Otherwise $x \in I_2$ and after some stage the first digit of the x_n must be two.
Similarly depending to which of the four intervals I_{ij} the point x belongs, we see that the second digit of x_n must be same as that of x after some stage. In other words, given any k, $f(x_n)$ will have the same first k digits as that of $f(x)$ after some stage. This shows continuity.

Observation 2: there is a continuous one-one map g on C onto $C \times C$.
Of course any such map is a homeomorphism, as noted earlier. We are thinking $C \times C \subset [0,1] \times [0,1]$. If $x \in C$ and

$$x = \frac{x_1}{3} + \frac{x_2}{3^2} + \frac{x_3}{3^3} + \cdots; \quad x_i = 0, 2.$$

we put

$$g_1(x) = \frac{x_1}{3} + \frac{x_3}{3^2} + \frac{x_5}{3^3} \cdots; \qquad g_2(x) = \frac{x_2}{3} + \frac{x_4}{3^2} + \frac{x_6}{3^3} + \cdots;$$

$$g(x) = (g_1(x), g_2(x)).$$

Since each x_i is zero or two, we see that both $g_1(x)$ and $g_2(x)$ are in C. We have just separated the even and odd digits. This is one-one. Remember an eligible expansion is unique for points in C. This is also onto $C \times C$ because given two points in C we can interlace the digits of these two points to get a point of C whose image is the given pair.
That it is continuous is immediate because the digits are unchanged. (of course, this is to be interpreted carefully, there is a change in the 'place' of the digit).

Observation 3: Given a continuous function h on C to $S = [0,1] \times [0,1]$ we can extend it to a continuous map of $I = [0,1]$ to S.

The only issue is to define the function on the points of the deleted intervals. Let $x \in (a,b)$ a deleted interval. Thus $a, b \in C$. Hence we know $h(a)$ and $h(b)$. Extend linearly in between. More precisely put

$$h((1-\theta)a + \theta b) = (1-\theta)f(a) + \theta f(b); \qquad 0 \le \theta \le 1$$

Since unit square is convex, h so defined still takes values in S. As the point θ goes from 0 to 1; the point $t = (1-\theta)a + \theta b$ travels from a to b and the h values travel on the straight line from $h(a)$ to $h(b)$. We are stll using the same symbol h for this extended function. Thus now h is defined on all of I and takes values in S.

We need to show that h so defined is continuous at all points. Of course, it is continuous at all points outside the Cantor set. Indeed, let $x \in I_n$, one of the deleted open intervals and $x_i \to x$. Then after some stage each $x_i \in I_n$ and so the definition given above takes over to show continuity.

Let $x \in C$. Let $x_i \to x$. We need to show that $h(x_i) \to h(x)$. Given $\epsilon > 0$ plot (a', b') around x so that for every point $y \in C \cap (a', b')$ we have $d(f(x), f(y)) < \epsilon$. This is possible because f is given to be continuous on C. Let us take $a, b \in C$, $a' < a < x < b < b'$. This is possible because every point of C is a limit point of C.

We now claim that for every point y in this interval (a, b), whether y is in C or not, we have the inequality $d(h(x), h(y)) < \epsilon$. Since this last inequality holds for points of $y \in C$ in this interval, we only need to show that it holds for points not in C. But then any such point is in a deleted interval (a_n, b_n). Since $a, b \in C$ we conclude that

$$d(h(a_n), h(x)) \leq \epsilon; \quad d(h(b_n), h(x)) \leq \epsilon.$$

In other words, both $h(a_n)$ and $h(b_n)$ are in the ϵ-ball around $h(x)$. Remember balls are convex. Since $h(y)$ is on the line joining $h(a_n)$ and $h(b_n)$, we see that $d(h(y), h(x)) \leq \epsilon$. This shows continuity.

We shall now exhibit space filling curve.
Take $f : C \to [0, 1]$ as in observation (1).
Take $g(x) = (g_1(x), g_2(x)) : C \to C \times C$ as in observation (2).
Define $h : C \to S$ by

$$h(x) = (h_1(x), h_2(x)); \quad h_1(x) = f(g_1(x)); \; h_2(x) = f(g_2(x)).$$

The map h is a continuous function on C — it is composition of continuous functions. Given $(a, b) \in S$; pick $s, t \in C$ with $f(s) = a$ and $f(t) = b$ and pick $x \in C$ with $g(x) = (s, t)$. Then verify $h(x) = (a, b)$. This shows that h is onto S. Now use observation (3) to extend the map as a continuous map from all of I to S. Of course, it still remains onto S.

This does.

Cantor sets:

We discussed once that theoretically one could construct similar to the Cantor set, by cutting into more number of pieces. For example, at each stage you divide an interval into five subintervals of equal length; instead of dividing into three. You pick the first, third and fifth. Continue, dividing these three selected intervals in the same manner. The procedure remains same. Take intersection. Such sets are also called Cantor sets. Here is the precise definition.

A compact subset of R is called perfect if every point is a limit point of the set. For example the usual Cantor set, or the interval $[0, 1]$ are such sets. The set $\{0,\ 1,\ 1/2,\ 1/3,\ \cdots\}$ is closed but not perfect. The set Q of all rationals is not closed, but every point of it is a limit point of the set.

A perfect nowhere dense set is called a Cantor set. This means our set should be closed, bounded, every point must be a limit point, should not contain any (non-empty) open interval.

Thus usual Cantor set is an example of a Cantor set, it is called the Cantor set. The construction described above will also lead to a Cantor set. Mathematics, like music, can be improvised with your ideas. For example should we cut the interval into fixed number of pieces? First step cut into three pieces and leave even one, that is, the middle one. You select the other two intervals. Next, cut each selected part into five pieces and leave even ones, that is, the second and fourth part in each of the first level intervals. You have six intervals. Next cut each selected interval into seven parts etc. It is all your pick! If you want you can divide into p_n subintervals at the n-th stage. Here p_n is the $(n+1)$-th prime.

You would probably think that you are getting many many very different sets. But this is only illusion as far as sets are concerned.

Here is a very interesting theorem: Such sets are all the same! More precisely, let P and Q be two such sets. Then there is a homeomorphism h of R to itself which sends P to Q. This last phrase means $h(P) = Q$. This is proved exactly like a similar theorem we proved quite a while ago, namely, given two countable dense sets in R, there is a homeo that sends one to the other. We shall not carry out the proof. The basic observation is this: If

a, b are respectively the minimum and maximum of your Cantor ser then the deleted open intervals form a countable linearly ordered set without first and last elements and between any two there is another one.

You should not come to the erroneous conclusion that all such sets are same. No. We only said that they look same. But different sets have different qualities and expertise from analysis/arithmetic point of view. We shall not pursue this line of thought.

We completed our discussion of metric spaces. We have done some basic results and applications. We shall leave our long excursion to metric spaces and return to real line once again. In fact we leave the topics of set-theoretic flavour, take up important topic of Fourier series which has analytical flavour.

We first discuss two loose ends before taking up Fourier series.

1. $(R, <)$ and Cantor and Souslin:

Recall that the set of rationals $(Q, <)$ is characterized as
a (nonempty) linearly ordered set $(S, <)$ such that (i) it has no first or last element and between any two (distinct) elements there is one more; (ii) it is countable. In other words given any such set satisfying these two conditions we can setup a bijection between S and Q preserving the order.

Cantor has a similar theorem for reals too. Let us say that a linearly ordered set $(S, <)$ is complete if the following holds: Given any (nonempty) bounded set, it has infimum and supremum. A is bounded means $(\exists a, b)$ $(\forall x \in A)$ $(a \leq x \leq b)$. We say c is supremum of A if (i) $(\forall x \in A)$ $(x \leq c)$ and (ii) $[(\forall x \in A)$ $(x \leq b)$ implies $c \leq b]$. The first clause says c is an upper bound (define upper bound, you know it) and second clause says it is least upper bound. Thus supremum is least upper bound. We say d is infimum of A if (i) $(\forall x \in A)$ $(d \leq x)$ and (ii) $[(\forall x \in A)$ $(a \leq x)$ implies $a \leq d]$. The first clause says d is a lower bound (define lower bound, you know it) and second clause says it is greatest lower bound. Thus infimum is the greatest lower bound.

Of course, this implies that (i) any nonempty set bounded above has a supremum. Indeed if $A \subset S$ nonempty and bounded above, then take any $a \in S$ and consider $S_0 = \{x \in S : a \leq x\}$ then this is nonempty (it has a) and bounded (same upper bound as S and lower bound a) and so S_0 has a supremum. It is easy to show that this is supremum of S too.
Similarly, we can show that the above implies (ii) every nonempty set bounded below has infimum.
Further (i) and (ii) themselves are equivalent and any one of them is, in turn, enough to show that every bounded set has infimum and supremum.

Let us say $D \subset S$ is dense if every nonempty open interval contains a point of D. That is, if $(a < b)$ and if there are points x with $a < x < b$ then there is a point $d \in D$ such that $a < d < b$.

$(R, <)$ is characterized as
a (nonempty) linearly ordered set $(S, <)$ such that (i) it has no first or last

element and between any two (distinct) elements there is one more; (ii) it is complete and (iii) it has a countable dense set. In other words given any such set satisfying these three conditions we can setup a bijection between S and R preserving the order.
Proof is simple. Let D be a countable dense set in S, verify that it satisfies the conditions of the previous result and setup a order preserving bijection $h : Q \to D$. Here is the required bijection on R onto S. Take any $a \in R$, consider the set $L_a = \{q \in Q : q \leq a\}$ and observe that the set $h(L_a) = \{h(q) : q \in L_a\}$ is bounded above. Indeed if r is any rational with $r > a$, then $h(r)$ is an upper bound for $h(L_a)$. Let $h(a) = \sup h(L_a)$. There is no confusion in using the same notation h because it is easy to see that when $a \in Q$, then this process indeed gives us original $h(a)$. To show that this is one-one is easy because given $a < b$ there are two rationals $a < q_1 < q_2 < b$ and $h(a) \leq h(q_1)$ where as $h(b) \geq h(q_2)$ and $h(q_1) < h(q_2)$. This is onto S is shown by reversing the argument. More precisely, given any $s \in S$, consider $\{h^{-1}(d) \in Q : d \leq s, d \in D\}$ observe it is bounded above in R and if x is its sup, then $h(x) = s$. You should carry out the details completely. ∎

Now D is dense in S and is countable implies that if we consider any collection of nonempty disjoint open intervals in S then the collection must be countable (because we can show a point of D in each of them and D is countable). The Russian Mathematician Souslin proposed the following:

Souslin Conjecture: $(R, <)$ can be characterized as a (nonempty) linearly ordered set $(S, <)$ such that (i) it has no first or last element and between any two (distinct) elements there is one more; (ii) it is complete and (iii) Any collection of nonempty disjoint open intervals is countable. In other words given any such set satisfying these three conditions we can setup a bijection between S and R preserving the order.
This statement is not decidable. In other words, assuming the above is true will not lead to a contradiction; also assuming above is not true — that is, there is such a set $(S, <)$ not order isomorphic to $(R, <)$ — will not lead to any contradiction.

You can characterize the set of rationals or reals using the algebraic operations. For example R is a complete Archimedian ordered field. (any complete Archimedian ordered field is isomorphic to R, in all respects; addition, multiplication, order). One would like to see R as a line without the algebraic operations but with order; then what characterizes it? (Of course, if you ignore even the order and just consider it as just a set, then only its cardinality characterizes it and not too interesting).

(ii) cardinalities:

In showing that R and R^2 are isomorphic as additive groups, we considered them as vector spaces over the field of rationals and considered Hamel bases B_1 and B_2. Then saying that $|B_1| = |B_2|$ we setup a bijection between these bases and extended. However we have not proved that $|B_1| = |B_2|$. This is what we do now through a series of simple observations.

(i) First recall Every infinite set contains a copy of N, that is, contains a countably infinite set.
Proof: Fix a choice function f for subsets of X, the given infinite set. This means for each non-empty subset $S \subset X$ we have $f(S) \in S$. Define $x_1 = f(X)$; $x_2 = f(X \setminus \{x_1\})$; $x_3 = f(X \setminus \{x_1, x_2\})$ and in general $x_{k+1} = f(X \setminus \{x_1, \cdots, x_k\})$. You can continue forever because X is not finite and so $(X \setminus \{x_1, \cdots, x_k\})$ is never empty. The set $(x_i : i \geq 1)$ does the job. Carefully note that these are distinct elements.

(ii) If X is an infinite set and $F \subset X$ is a finite set, then $|X| = |X - F|$. That is, X and $X - F$ have the same cardinality.
Proof: Denote your finite set by $F = \{x_1, \cdots x_k\}$. Fix a choice function as above. Put $x_{k+1} = f(X \setminus \{x_1, \cdots, x_k\})$ $x_{k+2} = f(X \setminus \{x_1, \cdots, x_k, x_{k+1}\})$ and in general for $n \in N$, $x_{k+n} = f(X \setminus \{x_1, \cdots, x_{k+n-1}\})$.
Here is a bijection on X onto $X - F$. If $x \notin (x_i : i \geq 1)$ put $h(x) = x$. Put $h(x_i) = x_{i+k}$ for $i = 1, 2, \cdots$. note $x_1, \cdots, x_k$ are not in the range of h.
This does the job. You only need to note that X is partitioned into two sets $A_1 = \{x_1, x_2, \cdots\}$ and $A_2 = X - A_1$. Similarly $X - F$ is partitioned into two sets $B_1 = \{x_{k+1}, x_{k+2}, \cdots\}$ and $B_2 = X - A_1$ (watch out, not $X - B_1$). The map h is a bijection of A_i and B_i.

(iii) If X is an infinite set then $|X \times \{0, 1\}| = |X|$. That is, two copies of X has the same potency as one copy of X. Remember $X \times \{0, 1\}$ consists of all pairs $(x, 0)$ for $x \in X$ and also all pairs $(x, 1)$ for $x \in X$.
Proof: let P consist of all pairs (S, f) where $S \subset X$ and f is a bijection on $S \times \{0, 1\}$ to S. This is non-empty. Indeed, by earlier result, you get S which is a copy of N. Since $S \times \{0, 1\}$ is also countably infinite there is a bijection f of $S \times \{0, 1\}$ to S. This (S, f) is in P.
Define $(S, f) \leq (T, g)$ if $S \subset T$ and g extends f. This makes sense because $S \times \{0, 1\} \subset T \times \{0, 1\}$. This makes P a poset.
Every chain has an upper bound. If (S_α, f_α) is a chain then take S as the union of these S_α. If $(x, i) \in S \times \{0, 1\}$ put $h(x, i) = f_\alpha(x, i)$ in case $x \in S_\alpha$.

again the fact that we have a chain tells us there is no conflict in case x is in two of the sets S_α. It is a bijection too. If you take two distinct points (x,i) and (y,j) then there is one S_α in which both x and y are present and so $h(x,i) \neq h(y,j)$. If you do not like my use of the suffix alpha for the collection in the chain, you can avoid, do not name the chain and try to argue.

Let (T,g) be a maximal element, exists by Zorn. Can $X \setminus T$ be infinite? No because of the following reason. If it were infinite, use above and get a copy of N in $X \setminus T$, denote it by A. Since $A \times \{0,1\}$ is a countably infinite set, fix a bijection φ on $A \times \{0,1\}$ onto A. Take $T_1 = T \cup A$. Define g_1 on $T_1 \times \{0,1\}$ as follows. If $x \in T$, then $g_1(x,i) = g(x,i)$ and if $x \in A$, then $g_1(x,i) = \varphi(x,i)$. Note that φ takes you onto A whereas g takes you onto T. This helps you to show that g_1 is bijection. But then $(T,g) < (T_1,g_1)$ contradicting maximality of (T,g).

Thus $X \setminus T$ is finite and hence by earlier result, T has the same cardinality as X. Now $X \times \{0,1\} \sim T \times \{0,1\} \sim T \sim X$.

(iv) if X is infinite then $|X \times N| = X$.
Proof: By above, write $X = X_1 \cup Y_1$, disjoint union, where $|X_1| = |Y_1| = |X|$. Now write $Y_1 = X_2 \cup Y_2$ disjoint union, where $|X_2| = |Y_2| = |Y_1| = |X|$. If you got X_i, Y_i for $i \leq n$ such that (i) $Y_i = X_{i+1} \cup Y_{i+1}$ for $i < n$ disjoint union, where $|X_{i+1}| = |Y_{i+1}| = |Y_i| = |X|$, then get X_{n+1}, Y_{n+1} by decomposing Y_n. Thus we have $(X_i, i \geq 1)$ disjoint subsets of X such that $|X_i| = |X|$ for each i. Clearly if $Z = \cup X_i$ then $|Z| = |X \times N|$. since $Z \subset X$ we conclude $|X \times N| \leq |X|$; but clearly $|X| \leq |X \times N|$ and hence equality holds.

(v) If $|S| < |X|$ and $|T| < |X|$ and S, T are disjoint, all are infinite sets, then $|S \cup T| < |X|$.
Proof: recall $|S| < |X|$ means there is an injection from S to X but there is no bijection. We know (from Axiom of choice) either $|S| \leq |T|$ or $|T| \leq |S|$, there is no loss in assuming that $|S| \leq |T|$. So there is an injection $f : S \to T$. Define g from $S \cup T$ to $T \times \{0,1\}$ by $g(x) = (f(x), 0)$ if $x \in S$ and $g(x) = (x,1)$ if $x \in T$. Note that ranges of S and T under g are disjoint. Easy to see that g is an injection showing

$$|S \cup T| \leq |T \times \{0,1\}| = |T| < |X|.$$

where the equality above follows from earlier result. The first inequality is witnessed by g and the last inequality is hypothesis.

(vi) if X is infinite then $|X \times X| = |X|$.
Proof: this is similar to (ii) but slightly involved. We proceed exactly as in that prrof. Let P be the set of all pairs (S, f) where $S \subset X$ and f is a bijection on $S \times S$ onto S. This is partially ordered by inclusion. That is $(S_1, f_1) \leq (S_2, f_2)$ if $S_1 \subset S_2$ and f_2 is an extension of f_1. This means $f_2(x, y) = f_1(x, y)$ for $x, y \in S_1$. This makes P a poset. P is non-empty because from earlier result you can take a countably infinite set $S \subset X$ and use the fact that $S \times S$ is also countably infinite to get a bijection $f : S \times S \to S$. Then $(S, f) \in P$.
Every chain has upper bound. Indeed if (S_α, f_α) is a chain, here is the upper bound: S is the union of all the S_α. For $x, y \in S$, if they both belong to S_α we put $f(x, y) = f_\alpha(x, y)$. Since we have a chain, there is one α such that both x and y are in S_α. Since we have a chain, this definition does not depend on which α we take. That this is a bijection is routine. if you take two pairs (x_1, y_1) and (x_2, y_2) then there is one S_α in which all these four points x_1, x_2, y_1, y_2 are available and f_α is a one-one map on this $S_\alpha \times S_\alpha$ so that the f_α values of these two pairs are different, but these are f values as well. Also given any $a \in S$, get an α such that $a \in S_\alpha$ and hence there is an $(x, y) \in S_\alpha \times S_\alpha$ such that $f_\alpha(x, y) = a$ so that $f(x, y) = a$. So far it is exactly as earlier. here also, if you do not like the suffix alpha, you can avoid it. You just keep on saying chain, (A, f) is in the chain etc.

Take a maximal element (T, g). Since $T \subset X$ there are only two possibilities, either $|T| = |X|$ or $|T| < |X|$ — the identity map shows that $|T| \leq |X|$. If $|T| = |X|$ we are done because $|X \times X| = |T \times T| = |T| = |X|$. Assume now $|T| < |X|$. we show contradiction for maximality by producing (T_1, g_1) such that $(T, g) < (T_1, g_1)$.
Towards this end we first note that $X \setminus T$ can not be finite. If it were, then X and T differ by finitely many points so that $|X| = |T|$ which is not the case now.
Since $|T \times T| = |T|$ we conclude that T is infinite (we know T is not empty!); we saw that $X \setminus T$ is infinite. We are assuming that $|T| < |X|$. If we also have $|X \setminus T| < |X|$ then earlier result leads to a contradiction: $|X| = |T \cup (X \setminus T)| < |X|$. Thus we must have $X \setminus T \sim X$.
We are now ready to contradict maximality of (T, g). Since $X \setminus T$ and X are of the same cardinality, pick a subset $S \subset X \setminus T$ with $|S| = |T|$. You only need to fix a bijection φ on X onto $X \setminus T$ and take $S = \varphi(T)$. Let $T_1 = T \cup S$. We shall now define a bijection g_1 on $T_1 \times T_1$ onto T_1 which extends g.

Let us make two observations. Firstly, $T_1 \times T_1$ is disjoint union of the four sets $A_0 = T \times T$; $A_1 = T \times S$; $A_2 = S \times T$; $A_3 = S \times S$. And also these

four sets have the same cardinality simply because S and T have the same cardinality. Since $|T \times T| = |T|$ (Remember g) we conclude that all these four sets are equipotent with T and also with S.
Second observation is the following. By earlier result We can express S as disjoint union of three sets S_1, S_2, S_3 all having the same cardinality as S.

To complete the proof define g_1 on $T_1 \times T_1$ as follows. On $A_0 = T \times T$ follow g to take you to T; on the other three sets A_1, A_2, A_3 which makeup the remaining part of $T_1 \times T_1$ use any maps taking you in a bijective way to S_1, S_2, S_3 respectively.
This completes the proof by showing an element larger than the alleged maximal element and thereby establishing that T must have same cardinality as X and thereby $X \times X$ must have a bijection to X.

(vii) Let B be a Hamel basis of R. Let $B_1 = QB = \{qb : q \in Q, b \in B\}$ and $B_2 = B_1 + B_1 = \{x+y : x, y \in B_1\}$ and in general $B_n = \{x_1+x_2+\cdots+x_n : x_i \in B_1\}$. Then $|B_n| = |B|$ for all n.
Proof: Firstly, B has to be infinite. Otherwise, if $= \{b_1, \cdots, b_k\}$ its linear span would be $\{\sum_{j=1}^{k} : q_j b_j\}$ where q_j are rationals and easily seen to be countable, where as R is not.
So B is infinite. but then Q being countable, an earlier result tells us $|Q \times B| = |B|$, Note that $(q, b) \mapsto qb$ is actually one-one, except when $q = 0$. $|B_1| \leq |Q \times B| = |B|$. And clearly $|B| \leq |B_1|$. Thus $|B_1| = |B|$.
The map $(x, y) \mapsto x + y$ takes you from $B_1 \times B_1$ onto B_2. Thus $|B_2| \leq |B_1 \times B_1| = |B_1|$. But of course, $|B_1| \leq |B_2|$. Thus $|B_2| = |B_1| = |B|$.
You can show by induction that $|B_n| = |B|$ for all n.

(vi) If B is a Hamel basis of R then $|B| = |R|$. Similarly if B_2 is Hamel basis of R^2 then $|B_2| = |R^2| = |R|$. Thus $|B| = |B_2|$.
Proof: By definition of Hamel basis, $\cup B_n = R$ and earlier result tells you $|B_n| = |B|$ for all n and also $|B \times N| = |B|$. Thus $|R| = |\cup B_n| = |N \times B| = |B|$, completing proof.
There are several other useful things about sets but we shall not get into.

Wave equation:

Vibration of a string: Consider a rod, pictured as interval $[0, 1]$. Imagine a tight wire tied at both end points. Thus the wire is on the rod initially. It is now stretched and released.

Problem: describe the displacement of the wire at position x on the rod,

at time t.

If this displacement is $u(x,t)$ then (i) $u(x,t)$ is a function for $(x,t) \in [0,1]\times[0,\infty)$. (ii) since the wire is fixed at end points $u(0,t) = u(1,t) = 0$ for all t. (iii) Since initially we stretched, we know $u(x,0) = f(x)$ for $0 \le x \le 1$, a known function (with $f(0) = f(1) = 0$). We now need one more property obeyed by the function: the wave equation: (iv) $u_{tt} - u_{xx} = 0$ on the open region $(0,1)\times(0,\infty)$. There is usually constant floating around $u_{tt} - ku_{xx} = 0$, k depending on the tension of the wire etc. But we are taking $k = 1$. One of the standard techniques is: hope for the best, try a function which is product of two functions, one function of x and other a function of t. This is called separation of variables method. Thus if you pretend $u(x,t) = f(x)g(t)$ then the differential equation takes the form

$$g''(t)f(x) - f''(x)g(t) = 0; \qquad \frac{g''}{g} = \frac{f''}{f}$$

We assumed nonvanishingness in this division. After all, we are searching for non-zero solution, so there is at least one point where u is nonzero and hence locally this is justified. Since g''/g is a function of t alone and f''/f is a function of x alone we conclude that these ratios must be constant, say λ. Thus

$$g''(t) = \lambda g(t) \qquad f''(x) = \lambda f(x)$$

such problems as above are called eigen value problems in analogy with Linear algebra $Av = \lambda v$. After all if D denotes derivative on 'appropriate' function space then it is indeed linear transformation and so is the second derivative $A = D^2$. The above equation is just $Ag = \lambda g$.
Suppose $\lambda \ge 0$ then $f''(x) = \lambda f(x)$ has the following solutions: – If $\lambda = 0$ then we have $f(x) = c_1 x + c_2$; if $\lambda > 0$ we have $f(x) = c_1 e^{\sqrt{\lambda}x} + c_2 e^{-\sqrt{\lambda}x}$. These are uninteresting and do not fit the bill because these are not oscillatory and our string, we know, oscillates. Thus for our problem we should have $\lambda < 0$, say $\lambda = -\mu^2$ where $\mu > 0$. thus for some constants a, b, c, d,

$$f(x) = a\cos\mu x + b\sin\mu x; \qquad g(t) = c\cos\mu t + d\sin\mu t$$

$$u(x,t) = [a\cos\mu x + b\sin\mu x] \times [c\cos\mu t + d\sin\mu t]$$

$u(0,t) = 0$ for all t tells $a[c\cos\mu t + d\sin\mu t] \equiv 0$ identically in t. The second factor can not be zero, if it were then $u \equiv 0$, not acceptable. So $a = 0$. Thus

$$u(x,t) = b\sin\mu x[c\cos\mu t + d\sin\mu t]$$

Again $u(1,t) \equiv 0$ tells

$$b(\sin \mu)\ [c \cos \mu t + d \sin \mu t] \equiv 0$$

If $b = 0$ again $u \equiv 0$, not accepted. If the last factor is zero, then again $u \equiv 0$, not acceptable. Thus $\sin \mu = 0$ or μ is a multiple of π, say $n\pi$ where n is an integer. Thus

$$u(x,t) = \sin n\pi x\ \ [c \cos n\pi t + d \sin n\pi t]$$

We have absorbed the extra constant b, into c, d.
To take stock, Ignoring the divisions we made earlier etc, you can now directly verify that the above function satisfies the wave equation, and also initial conditions at end points (technically these are called boundary conditions, nothing initial there, condition is for all time). The question is: $u(x,0) = f(x)$? Instead of answering this, let us ask for what f could we solve our problem? If $f(x) = \sin n\pi x$ we have solved our problem. By superposition (or linear combination of solutions) we can solve if f is of the form

$$f(x) = \sum_{n=1}^{k} \alpha_n \sin n\pi x; \qquad u(x,t) = \sum_{n=1}^{k} \alpha_n \sin n\pi x\ \ [\cos n\pi t + d \sin n\pi t]$$

whatever be the integer k.
early seventeen hundreds, D'Alembert, Euler, Bernoulli and others discussed the wave equation. The method we discussed, separation of variables, is also called standing wave approach. $f(x) = \sin nx$ is a wave, $u(x,t) = f(t)\ \sin nx$ is a standing wave, at every time its profile is just $\sin nx$, with a multiple. On the other hand you can think of $u(x,t) = \sin(x - 5t)$. This is also a wave for each t. This is called travelling wave, it is sine wave but moving with a certain speed; its profile is not just a multiple of a fixed wave.

Returning to our problem,
How nice if every f on $[0,1]$ with $f(0) = f(1) = 0$ is a – possibly infinite – combination $\sum_n \alpha_n \sin n\pi x$. (♠)
At least we can look at the corresponding infinite sum of solutions (and discuss its convergence, differentiability, being a solution etc).

Heat equation:

I have an infinite rod, R. Initially I have heat distributed over the rod, say $\varphi(x)$ at position x. tell me the heat at time t at position x in the rod.

We did this in our probability course. here is the answer.

$$p(t,x,y) = \frac{1}{\sqrt{2\pi t}} e^{-(x-y)^2/2t}; \qquad u(x,t) = \int_{-\infty}^{\infty} \varphi(y) p(t,x,y) dy$$

We had heat equation, $u_t = \frac{1}{2} u_{xx}$ confirming to some probability formulae. Interpretation is the following: p is the fundamental solution; if unit heat is placed at y at time zero, then at time t at position x, the heat is $p(t,x)$. If you place several amounts at several places it adds up (or integrates up).

The present problem is different. As in the wave equation, we consider a finite rod, $[0,1]$ There is a heat distribution at time zero: $\varphi(x)$. Let us assume that the two ends are connected to a reservoir and maintained at temperature zero. Tell me the heat distribution at time t in the rod at position x. (we are using the terms heat/temperature synonymous, let us accept). This is what Fourier considered in the late seventeen hundreds.
So we need a function $u(x,t)$ on $[0,1] \times [0,\infty)$. Initial condition: $u(x,0) = f(x)$. Boundary conditions: $u(0,t) \equiv 0$ and $u(1,t) \equiv 0$. Finally u satisfies $u_t = u_{xx}$. Again, there is a constant $u_t = k u_{xx}$, but for simplicity we took $k = 1$. again separation of variables leads, to state briefly:

$$u(x,t) = f(x)g(t); \quad fg' = f''g; \qquad f'' = \lambda f; \qquad g' = \lambda g$$

$$f(x) = A\cos\sqrt{-\lambda}x + B\sin\sqrt{-\lambda}x; \qquad g(t) = Ce^{\lambda t}$$

Even though we can talk about complex valued functions, which we shall soon do, let us now restrict to real valued functions. (You can also bring in the realistic condition that at no point heat tends to $\pm\infty$ and so the multiplying factor must decrease to zero, can not be a positive exponent, but do not bother) So $\lambda \leq 0$, say $\lambda = -\mu^2$. thus

$$f(x) = A\cos\mu x + B\sin\mu x; \qquad g(t) = Ce^{-\mu^2 t}$$

$u(0,t) = 0$ tells $A = 0$. And $u(1,t) = 0$ tells $\mu = n\pi$ giving

$$u(x,t) = e^{-n^2\pi^2 t}\sin n\pi x$$

if you ask for what initial f you have now solved the problem, for what more you can solve, you will arrive as earlier to the conclusion:
How nice if every f on $[0,1]$ with $f(0) = f(1) = 0$ is a possibly infinite combination $\sum_n \alpha_n \sin n\pi x$. (♠)

this is precisely what Fourier asserted boldly:
Every 2π-periodic real valued function on R is a superposition of the sine and

cosine functions. That is, for some numbers a_n and b_n; it is of the form $\sum_0^\infty (a_n \sin 2\pi nx + b_n \cos 2\pi nx)$ (♡)

Of course it is not true as stated. However, in understanding this statement, much of the mathematics grew and existing maths was made rigorous: what is meant by convergence of infinite series? In what sense is the above statement true? In fact, origins of set theory also lie in Fourier series; Cantor was understanding problems concerning series of functions of the above type.

complex n-dimensional vector space:

We would quickly recall what we know about $\mathfrak{C}^n$, the complex n dimensional space. Let us recall. $\mathfrak{C}^n$ is the set of all n tuples $z = (z_1, z_2, \cdots, z_n)$ of complex numbers. We define inner product

$$\langle z, w\rangle = \sum_k z_k \overline{w_k}; \quad \langle z, z\rangle = ||z||^2 = \sum |z_k|^2.$$

$$d(z, w) = \sqrt{\langle z - w, z - w\rangle} = \sqrt{\sum |z_k - w_k|^2}.$$

Here $\overline{w}$ is the complex conjugate of w. Recall if $w = a + ib$ where a, b are real then its conjugate is $a - ib$.
A vector z is normal if $\|z\|^2 = 1$. Two vectors z, w are orthogonal if $\langle z, w\rangle = 0$. This space has an orthonormal basis consisting of n vectors. One such choice is the standard basis, but there are many other choices too. Thus there are n vectors $e_1, e_2, \cdots, e_n$ such that

$$\langle e_k, e_l\rangle = 1 \text{ if } k = l; \quad \text{and} \quad = 0 \text{ if } k \neq l.$$

Any vector v can be uniquely written as

$$v = \sum_1^n \langle v, e_k\rangle e_k = \sum \hat{v}_k e_k.$$

This is called expansion of v in terms of the orthonormal basis. Sometimes this is also called Fourier expansion (w.r.t. the basis). Also the coefficients $\hat{v}_k = \langle v, e_k\rangle$ are called Fourier coefficients (w.r.t the basis). It follows from properties of inner product that

$$||v||^2 = \sum_1^n |\hat{v}_k|^2.$$

This is sometimes called Parseval equation. Interesting point, trivial but should be noted, is that this is true whatever orthonormal basis you take. If you have orthonormal vectors $(e_j : 1 \leq j \leq k)$ and you do not know whether they form a basis, you can still say

$$\sum_1^k |\hat{v}_j|^2 \leq \|v\|^2; \qquad \hat{v}_j = \langle v, e_j \rangle$$

Sometimes this is called Bessel's inequality. Of course, in the finite dimensional case which we are now discussing, we know that the system forms a basis iff $k = n$.

Returning to the Orthonormal basis, Let us take an integer $1 \leq m < n$ and consider only the partial sum

$$v^* = \sum_1^m \hat{v}_k e_k.$$

Then clearly v^* is in the subspace spanned by the first m-basis vectors, that is,

$$v^* \in \text{ span } \{e_1, e_2, \cdots, e_m\} = S; \quad \text{say.}$$

The vector $v - v^*$ is orthogonal to every vector in S. This is because

$$\langle v - v^*; \sum_1^m a_k e_k \rangle = \langle v; \sum_1^m a_k e_k \rangle - \langle v^*; \sum_1^m a_k e_k \rangle$$

$$= \sum_1^m \overline{a_k} \langle v, e_k \rangle - \sum_1^m \overline{a_k} \langle v^*, e_k \rangle = 0;$$

because, $\langle \sum_{l=1}^m \hat{v}_l e_l, e_k \rangle = \hat{v}_k = \langle v, e_k \rangle$.

Since $v = (v - v^*) + v^*$ and $v - v^* \perp v^*$ we see

$$||v||^2 = ||v - v^*||^2 + ||v^*||^2.$$

Thus $||v^*||^2 \leq ||v||^2$. We could have got it by direct computation because

$$||v^*||^2 = \sum_1^m |\hat{v}_k|^2 \leq \sum_1^n |\hat{v}_k|^2 = ||v||^2.$$

In a sense v^* captures the entire part of v in S (whatever it may mean) so that what remains is orthogonal to S. Not only this, v^* is the vector closest to v in the subspace S. That is

$$d(v, v^*) \leq d(v, w) \quad \forall \ w \in S.$$

This is seen as follows. $v - w = (v - v^*) + (v^* - w)$ and the two terms on the right are orthogonal we see

$$||v - w||^2 = ||v - v^*||^2 + ||v^* - w||^2 \geq ||v - v^*|^2$$

complex numbers and complex valued functions

Now a little analysis on $\mathfrak{C}$. A sequence $z_n = a_n + ib_n$ converges to a complex number $z = a + ib$ if $d(z_n, z) = \|z_n - z\| \to 0$. This is same as saying $a_n \to a$ and $b_n \to b$. Whenever we write complex number as $a + ib$ it is understood that a is the real part and b is the imaginary part.

A series $\sum z_n$ is said to converge to w if $\sum_1^n z_j \to w$, that is, partial sum sequence converges. Easy to see this happens iff the real series madeup of real/imaginary parts converge.

Let us quickly recall. $f : [0, 1] \to C$ then for each x, $f(x)$ is a complex number and hence is $f_1(x) + if_2(x)$ where $f_1(x)$ and $f_2(x)$ are real numbers. Thus given a complex valued function, there are two real valued functions f_1 and f_2 so that $f = f_1 + if_2$. The function f_1 is called the real part of f and f_2 is called the imaginary part of f. You must note that these functions are real valued.
Conversely, given two real valued functions, the above equality provides you a complex function. Unless stated to the contrary, when we write $f = f_1 + if_2$ we mean that these are the real and imaginary parts. Also, bringing in the notion of convergence in C, you see that f is continuous iff its real part and imaginary part are continuous.
For later use, let us also make an observation. Suppose you have a sequence $(f^n = f_1^n + if_2^n)$ of complex valued functions and a function $f = f_1 + if_2$. Then f^n converges to f uniformly iff the real parts and imaginary parts converge uniformly. This follows from a simple observation. For a complex number $z = a + ib$

$$\max\{|a|, |b|\} \quad \leq \quad |z| \quad \leq \quad |a| + |b|.$$

Thus if $f^n \to f$ uniformly, then $||f_1^n - f_1|| \leq ||f^n - f||$ and hence $f_1^n \to f^1$ uniformly. Similarly $f_2^n \to f_2$ uniformly.
conversely, if the real and imaginary parts converge uniformly, then

$$||f^n - f|| \leq ||f_1^n - f_1|| + ||f_2^n - f_2||$$

showing that $f^n \to f$ uniformly.

We define (bad definition) for any real number θ,

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Then by induction and using the sine and cosine formulae, we can show

$$e^{in\theta} = \cos n\theta + i\sin n\theta.$$

This is called De Moivre's formula and you must have seen it in high school. Actually, the correct way is to define for every complex number z, the series

$$\exp\{z\} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots + \frac{z^n}{n!} + \cdots$$

This series converges. Indeed take the disc $\{|z| \leq M\}$. Let $S_n(z)$ be the n-th partial sum of the above series. Then the sequence $\{S_n(z)\}$ is uniformly Cauchy in the disc because for $n \geq m$

$$|S_n(z) - S_m(z)| \leq \sum_{m+1}^{n} \frac{M^k}{k!} \to 0 \text{ as } m, n \to \infty.$$

If you take θ to be real number, then the above definition along with the fact that $i^2 = -1$ tells us

$$\begin{aligned}\exp\{i\theta\} &= 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \cdots + \frac{(i\theta)^n}{n!} + \cdots \\ &= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \cdots + (-1)^n \frac{\theta^{2n}}{(2n)!} + \cdots \\ &+ i\left\{\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots + (-1)^n \frac{\theta^{2n+1}}{(2n+1)!} + \cdots\right\}\end{aligned}$$

Do not worry about my rearranging, we have absolutely convergent series.

$$= \cos\theta + i\sin\theta$$

where the last identification is from the expressions we obtained in our first course for the sine and cosine functions. Thus the definition we used above is same as the natural one defined just now. Further, if z has zero imaginary part, that is, z is real, then this definition coincides with what we learnt earlier in real analysis.

One can prove Cauchy theorem on products of series and show, exactly as in the real case, that

$$\exp\{z + w\} = \exp\{z\}\exp\{w\}$$

justifying the use of the notation e^z for $\exp\{z\}$. This gives another proof of the De Moivre formula.
Recall that integration is done by using the real and imaginary parts. For a continuous function $f = f_1 + if_2$ on $[0, 1]$

$$\int_0^1 f(t)dt = \int_0^1 f_1(t)dt + i\int_0^1 f_2(t)dt$$

The left side integrand is complex valued but on right side both integrands are real valued. If you wish you can go via Riemann sums etc, but that gives this same answer. Thus for $k \neq 0$,

$$\int_0^1 e^{2\pi ikt}dt = \int_0^1 \cos(2\pi kt)dt + i\int_0^1 \sin(2\pi kt)dt = 0.$$

you would tempted to use your intuition from real integrals and say

$$\int e^{i5t}dt = \frac{e^{i5t}}{i5}$$

This is alright because

$$\int e^{i5t}dt = \int \cos(5t)dt + i\int \sin(5t)dt = \frac{\sin(5t)}{5} + i\frac{-\cos(5t)}{5}$$

$$= \frac{1}{i}[i\frac{\sin(5t)}{5} + \frac{\cos(5t)}{5}] = \frac{e^{i5t}}{i5}$$

Since we deal with complex valued functions defined on R, you can use definition of derivative as usual and see $f' = f_1' + if_2'$.

standardization:

There are several ways of fixing 'domain': some consider periodic functions on R of period 2π. For such a function if you know it on any interval of length 2π you know the function. Some take basic setup as functions f on $[0, 2\pi]$ with $f(0) = f(2\pi)$. Some take as functions on $[-\pi, +\pi]$ with $f(-\pi) = f(+\pi)$.
These in turn is same as considering the unit circle

$$T = \{(x, y) \in R^2 : x^2 + y^2 = 1\} = \{z \in \mathfrak{C} : |z|^2 = 1\}$$

and complex valued functions on T. This T is same as any closed bounded interval on R with end points identified. It is a compact metric space. Since

conjugate of the function z^n (on T) is the function z^{-n}, the collection of all functions of the form: finite linear combinations of $\{z^n, n = 0, \pm 1, \pm 2, \cdots\}$ is dense in the collection of complex valued continuous functions on T, namely, $C(X, \mathfrak{C})$ — Stone-Weierstrass.

We shall consider functions on R of period one; We take interval $[0, 1]$ and all functions taking same value at end points. Unit interval is identified with T via the map $t \mapsto e^{2\pi it}$. Thus z^n on T corresponds to the function $e^{2\pi int}$ on $[0, 1]$.

The reason for the choice $[0, 1]$ is that we integrate without any factor (because length is one) where as if you take interval $[0, 2\pi]$ you need to use factor $1/(2\pi)$. More precisely, the function z^n of T: which translates to $e_n(t) = e^{2\pi int}$ on $[0, 1]$ has the property that $\int_0^1 e_n(t)\overline{e_n(t)}dt = 1$; where as on $[0, 2\pi]$ the same function z^n on T translates to $f_n(t) = e^{int}$ and $\int_0^{2\pi} f_n(t)\overline{f_n(t)}dt = 2\pi$. If you want this 'vector' f_n to be normal, you should use not dt but $dt/(2\pi)$.

The statement of Fourier amounts to saying that any periodic function of period one is a sum of linear combinations of the functions: $\sin 2\pi kx$ and $\cos 2\pi kx$. Thus mathematically, the statement amounts to saying that given periodic function of period one, there are numbers (a_n) and (b_n) such that

$$f(x) = a_0 + (a_1 \sin 2\pi x + b_1 \cos 2\pi x) + (a_2 \sin 4\pi x + b_2 \cos 4\pi x) + \cdots$$
$$+(a_k \sin 2k\pi x + b_k \cos 2k\pi x) + \cdots .$$

Equivalently, $f(t) = \sum_{-\infty}^{\infty} c_n e^{2\pi int}$. Such a series is called Fourier series.

$C[0, 1]$ as $\mathfrak{C}^n$ for Huge n:

Let us from now consider the space of complex valued continuous functions on $[0, 1]$, instead of just real functions. This will be $C[0, 1]$ during the remaining part of our discussion. To start with we start considering $C_{per}[0, 1]$ the space of complex valued continuous functions f on $[0, 1]$ with $f(0) = f(1)$. This is 'same as' the collection of continuous functions on R of period one, that is $f(t + 1) = f(t)$ for all t. This will replace $\mathfrak{C}^n$, as you will see.
We define some special functions on $[0, 1]$ as follows.

$$e_n(t) \quad = \quad e^{2\pi int}, \quad n = 0, \pm 1, \pm 2 \cdots .$$

The calculation of integral above can be recast in terms of these functions.

$$\int_0^1 e_n(t)\overline{e_m(t)}dt = 0; \quad if \;\; n \neq m; \quad = 1 \;\; if \;\; n = m.$$

We defined on $\mathfrak{C}^n$ inner product

$$\langle z, w\rangle = \sum z_k \overline{w_k}; \quad z = (z_1, \cdots, z_n); \;\; w = (w_1, \cdots, w_n).$$

Analogously let us define an inner product on the vector space $C[0,1]$ (remember complex valued continuous functions)

$$\langle f, g\rangle = \int_0^1 f(t)\overline{g(t)}dt$$

This is indeed an inner product: linear in the first argument f, conjugate linear in the second argument g; $\langle f, f\rangle = \int |f|^2 \geq 0$; $\langle f, f\rangle = 0$ iff $f = 0$ (remember continuity of f); $\langle f, g\rangle = \overline{\langle g, f\rangle}$ because $\int \overline{\varphi} = \overline{\int \varphi}$.
Moreover the integrals of exponentials we evaluated above, tells us that the family $\{e_n : -\infty < n < \infty\}$ is an orthonormal system.
The only thing is that in $\mathfrak{C}^n$ we have a finite orthonormal basis. If you found n orthonormal vectors, then there are no more to extend this system. If you found only m orthonormal vectors and $m < n$ you can find more. You should actually find more because you can not express every vector as linear combination of just these m vectors.
In the present vector space we have found an infinite system of orthonormal vectors. It is not clear if there are any more that can be added to this list. Just as we expanded an vector in $\mathfrak{C}^n$ in terms of an orthonormal system, we can try to expand a function in terms of this orthonormal system. Hopefully

$$f = \sum_{-\infty}^{\infty} c_k e_k.$$

Fourier Series is the exploration of this theme, with useful and interesting consequences, in Analysis, Number Theory, Physics and so on.

For $f \in C_{per}[0,1]$, and integer n, we define

$$\widehat{f}(n) = \int_0^1 f(t)e^{-2\pi int}dt = \langle f, e_n \rangle$$

Here e_n is the function, $e_n(t) = e^{2\pi int}$. These are called Fourier coefficients of f. Keep your finite dimensional $\mathfrak{C}^n$ in mind (except when we differentiate etc). The main difference is the following. In case of $\mathfrak{C}^n$, the vectors are n tuples of numbers and the coefficients – w.r.t. an orthonormal basis – are also n tuples of numbers. Here vectors f are periodic continuous functions on $[0,1]$ where as coefficients $\widehat{f}$ are functions on the set Z of integers.
The expression

$$\sum_{-\infty}^{\infty} \widehat{f}(n)e^{2\pi inx}$$

is called Fourier series of f. At this moment this infinite sum may not be meaningful for many or all points x. We define partial sums

$$S_N(x) = \sum_{-N}^{N} \widehat{f}(n)e^{2\pi inx} \qquad N = 0, 1, 2, 3 \cdots$$

We say Fourier expansion is valid or Fourier series converges to the given function if $S_N(x) \to f(x)$; may be point-wise, may be uniformly, may be some other sense; then you say convergence holds in that sense.
Here are some properties of Fourier Coefficients and consequences.

(1) (Linearity) $\widehat{f+g} = \widehat{f} + \widehat{g}$. This means, for each $n \in Z$

$$(\widehat{f+g})(n) = \widehat{f}(n) + \widehat{g}(n)$$

(2) (linearity) For any c (real or complex) $\widehat{cf} = c\widehat{f}$. This means for each $n \in Z$

$$\widehat{(cf)}(n) = c\ \widehat{f}(n)$$

These follow from linearity of integrals.

(3) (conjugation) Denoting by $\overline{f}$, the function $\overline{f}(x) = \overline{f(x)}$; the function $\overline{f}$ is again in $C_{per}[0,1]$ and for each n,

$$\widehat{\overline{f}}\ (n) = \overline{\widehat{f}(-n)}$$

because

$$\int_0^1 \overline{f}(x)e^{-2\pi inx}dx = \int_0^1 \overline{f(x)e^{2\pi inx}}dx = \overline{\int_0^1 f(x)e^{2\pi inx}dx} = \overline{\widehat{f}(-n)}$$

(4) (differentiation) If f continuously differentiable as a periodic function on R, so that $f' \in C_{per}[0,1]$, then

$$\widehat{f'}\,(n) = 2\pi in\widehat{f}(n)$$

this follows from integration by parts:

$$\int_0^1 f'(t)e^{-2\pi int}dt = f(t)e^{-2\pi int}\,\big|_0^1 - (-2\pi in)\int_0^1 f(t)e^{-2\pi int}dt$$

and the first term on right side is zero because both f and $e^{-2\pi int}$ take same value at the end points.

(5) (translation) If a is any real number then $g(t) = f(a+t)$ is also in $C_{per}[0,1]$ and $\widehat{g}(n) = e^{2\pi ina}\widehat{f}(n)$.
Continuity is clear. Periodicity is true because $g(t+1) = f(a+(t+1)) = f(a+t+1) = f(a+t) = g(t)$.

(6) (Uniqueness) If $\widehat{f}(n) = 0$ for all n, then $f \equiv 0$.

One easy way to prove is the following. Use Stone-Weierstrass theorem. Recall the space $C_{per}[0,1]$ is identified with continuous functions on T, — via the map $f \in C_{per}[0,1]$ with $f^*(e^{2\pi int}) = f(t)$ on unit circle — the linear span of the functions $(e^{2\pi inx} : n \in Z)$ in $C_{per}[0,1]$ corresponds to span of $\{z^n : n = 0, \pm 1, \pm 2 \cdots\}$ which is an algebra, closed under conjugation, containing constants, separating points).
If $\int fe_n = 0$ for all n, then $\int fg = 0$ for all g in the linear span of the $(e_n; n \in Z)$ and hence $\int fg = 0$ for all g in our space. Here we use: if $g_n \to g$ uniformly then $fg_n \to fg$ uniformly; also if $h_n \to h$ uniformly then $\int h_n \to \int h$. Taking $g = \overline{f}$ you see $\int |f|^2 = 0$. Since $|f|^2$ is real continuous positive we conclude that $f \equiv 0$. see how Analysis 1 helps.

But here is a way avoiding Stone-Weiestrass (from Stein-Shakarchi). Since our functions are period one functions, integrating over any interval of length one is same. Thus we assume

$$\int_{-1/2}^{1/2} f(t)e_n(t)dt = 0 \qquad n = 0, \pm 1, \pm 2, \cdots$$

Since $\sin(2\pi nt), \cos(2\pi nt)$ functions are (possibly complex) linear combinations of these exponentials we conclude

$$\int_{-1/2}^{1/2} f(t)\cos(2\pi nt)dt = 0 = \int_{-1/2}^{1/2} f(t)\sin(2\pi nt)dt \qquad n = 0, \pm 1, \pm 2, \cdots$$

Either by DeMoivre or otherwise you see that these $\sin(2\pi nx)$ and $\cos(2\pi nx)$ functions are all polynomials in $\sin(2\pi x)$ and $\cos(2\pi x)$ (that is finite linear combinations of powers of these functions). In particular, Hypothesis implies

$$\int_{-1/2}^{1/2} f(t)P(t)dt = 0; \qquad P(t) = \sum c_n[\cos(2\pi t)]^n \qquad (\spadesuit)$$

To show $f \equiv 0$ let us first show $f(0) = 0$. Suppose $f(0) = c > 0$. $(\clubsuit)$
Fix $0 < \delta < 1/4$ so that $f(x) > c/2$ for $-\delta \leq x \leq \delta$. $(\bullet 1)$
Observe $\cos 2\pi x$ is strictly below $+1$ during $[-1/2, -\delta]$ and $[\delta, 1/2]$. Fix $\epsilon > 0$ so that $-1 \leq \cos 2\pi x \leq 1 - 2\epsilon$ for $\delta \leq |x| \leq 1/2$. Let us define $p(x) = \epsilon + \cos 2\pi x$. Thus we have
$-1 + \epsilon \leq p(x) \leq 1 - \epsilon$ for $|x| > \delta$. that is $|p(x)| \leq 1 - \epsilon$. $(\bullet 2)$
As $x \to 0$ we see $p(x) \to \epsilon + 1$. So fix η small so that $0 < \eta < \delta$ and
$p(x) > 1 + (\epsilon/2)$ for $|x| < \eta$. $(\bullet 3)$.

$$[-1/2, 1/2] = [-1/2, -\delta] \cup [-\delta, -\eta] \cup [-\eta, \eta] \cup [\eta, \delta] \cup [\delta, 1/2]$$
$$= (A1) \cup (B1) \cup (C) \cup (B2) \cup (A2) \qquad \text{say } (\bigstar 0)$$

Let us fix a bound for f, say $|f| \leq M$ on $[-1/2, 1/2]$. Let us see what happens to $\int fp^k$ as k (integer) becomes large.
Over $(A1)$ and $(A2)$, $\int fp^k$, in Modulus, is at most $M(1-\epsilon)^k$, in view of $(\bullet 2)$. Thus

$$\int_{A1 \cup A2} fp^k \to 0 \qquad (\bigstar 1)$$

Over $B1$ and $B2$, we know $f > 0$ by $(\bullet 1)$. Also by choice $0 < \delta < 1/4$ and so $\cos 2\pi x \geq 0$ and so $p \geq 0$. Thus

$$\int_{B1 \cup B2} fp^k \geq 0 \qquad (\bigstar 2)$$

Length of the interval C is 2η and here $f > c/2$ by $(\bullet 1)$ and $p > (1 + \epsilon/2)$ by $(\bullet 3)$. Thus

$$\int_C fp^k \;\; \geq \;\; 2\,\eta\,\frac{c}{2}\,\left(1 + \frac{\epsilon}{2}\right)^k \;\; \to \;\; \infty \qquad (\bigstar 3)$$

Finally ($\bigstar 0,1,2,3$) show $\int fp^k \to \infty$ contradicting ($\spadesuit$), completing proof of ($\clubsuit$), namely $f(0)=0$.
All this was done assuming $f(0)>0$. If you assume $f(0)<0$ same contradiction is arrived because $(-f)$ satisfies earlier condition and hypothesis implies that its Fourier coefficients are all zero. Thus $f(0)=0$. If a is any other point with $f(a)\neq 0$ then considering translate $f_a(t)=f(t+a)$, we see $f_a(0)\neq 0$ and hypothesis implies that all Fourier Coefficients of f_a are zero, leading to $f(a)=0$, by the case established already. Finally if $f=f_1+if_2$ is complex valued, note that hypothesis implies Fourier coefficients of f_1 as well as f_2 are all zero and hence both are zero functions. This completes the proof.

Should you complain: What was done with Stone Weierstrass theorem in few lines is done with lot of work and too much work. No. You should appreciate: What was killed with a Big axe is now killed with just high-school knowledge. Go through the proof again and convince yourself. Secondly, such 'hands-on' calculations save you in difficult situations when big axes are not available. Remember, if you know many ways to reach the beach, it helps you to reach even when one road is not accessible.

(7) Does S_N capture 'all of f' which is in span $(e_n : |n|\leq N)$ Yes $f-S_N \perp g$ for any g in the span of $(e_n : |n|\leq N)$.
Direct computation. Say $g=\sum\{c_n e_n : |n|\leq N\}$

$$\langle f-S_N, g\rangle = \langle f, \sum c_n e_n\rangle + \langle \sum \widehat{f}(n)e_n, \sum c_n e_n\rangle$$

$$= \sum \overline{c_n}\widehat{f}(n) \; - \sum_m \sum_n \widehat{f}(n)\overline{c_m}\langle e_n, e_m\rangle \; = \; 0$$

because in the last sum the $m\neq n$ terms are zero.

(8) S_N is the closest to f in span $(e_n : |n|\leq N)$
If g is in span $(e_n : |n|\leq N)$, want to show $||f-g|| \geq \|f-S_N\|$. Since $f-g=(f-S_N)+(S_N-g)$ and S_N-g is in span $(e_n : |n|\leq N)$, we conclude from above $f-S_N \perp S_N-g$ and hence

$$\|f-g\|^2 = \|f-S_N\|^2 + \|S_N-g\|^2 \qquad ||f-g|| \geq \|f-S_N\|$$

Here we used Pythogaras Theorem (also in Sulbasutras of India, centuries before Pythogaras); if $v\perp w$ then $\|v+w\|^2=\|v\|^2+\|w\|^2$. This follows by expanding $\langle v+w, v+w\rangle$ and using orthogonality.

(9) Bessel's inequality

$f = (f - S_N) + S_N$ and $f - S_N \perp S_N$. So as above

$$\|S_N\|^2 + \|f - S_N\|^2 = \|f\|^2; \qquad \|S_N\|^2 \leq \|f\|^2$$

Noting that S_N is sum of orthogonal vectors, we get, whatever be N,

$$\sum_{-N}^{N} |\widehat{f_n}|^2 \leq \int_0^1 |f|^2$$

This is called Bessel's inequality. This being true for every N, we conclude

$$\sum_{-\infty}^{\infty} |\widehat{f_n}|^2 \leq \int_0^1 |f|^2$$

(10) convergence Let $f \in C_{per}[0,1]$ and $\sum_n |\widehat{f}(n)| < \infty$ Then the Fourier series of f converges uniformly to f.
The fact that FS converges follows from the Weierstrass M-test: FS is a series of functions $\sum \widehat{f}(n)e_n$ and $|\widehat{f}(n)e_n| \leq |\widehat{f}(n)|$ and hypothesis says that $\sum |\widehat{f}(n)|$ converges. The only issue is that in Weierstrass M-test we had one sided series and this is 'two sided' series. But the same proof works. Or, you directly argue that $(S_N, N \geq 1)$ is uniformly Cauchy.
Suppose S_N converges to g. To show that $g = f$, using uniform convergence argue that

$$\widehat{g}(n) = \int g\overline{e_n} = \lim_N \int S_N \overline{e_n} = \widehat{f}(n)$$

because the last integral itself equals $\widehat{f}(n)$ for $N > |n|$ – just remember that S_N is a finite sum and (e_n) are orthonormal. Thus f, g have same Fourier coefficients (or $f - g$ has zero Fourier coefficients). Thus $f = g$

(11) convergence for C^1 functions If $f \in C_{per}[0,1]$ and is continuously differentiable as a periodic function on R, then FS of f converges to f uniformly.
Indeed, from earlier result, for $n \neq 0$, $|\widehat{f}(n)| = \frac{1}{2\pi n}|\widehat{f'}(n)|$ so that

$$\sum_{n\neq 0} |\widehat{f}(n)| = \frac{1}{2\pi} \sum_{n\neq 0} \frac{1}{n} |\widehat{f'}(n)| \leq \frac{1}{2\pi} \sqrt{\sum_{n\neq 0} 1/n^2} \sqrt{\sum_{n\neq 0} |\widehat{f'}(n)|^2}$$

Since the last sum, by Bessel, bounded by $\|f'\|^2$ finite and since $\sum(1/n^2)$ (over $n \neq 0$) is finite we can apply earlier result to complete the proof.

In the above proof we used (10) which uses uniqueness. We shall soon give a proof that does not depend on the uniqueness.

(12) Plancherel special case If $f \in C_{per}[0,1]$ and is continuously differentiable as a periodic function on R, then

$$\sum |\widehat{f}(n)|^2 = \int |f|^2$$

This is called Plancherel formula. soon we remove the differentiability restriction on f. Proof of above is easy. If f is con tinuously differentiable, then so is $\overline{f}$. Hence $S_N \to f$ uniformly and $\overline{S_N} \to \overline{f}$ uniformly and so we conclude $|S_N|^2 = S_N\,\overline{S_N} \to f\overline{f} = |f|^2$ uniformly. Hence their integrals converge, that is

$$\sum_{-N}^{N} |\widehat{f}(n)|^2 \to \int |f|^2$$

proving the statement.

We shall first discuss two HA problems that I said are not too easy.

det one matrices:
The problem is to show that the space of matrices of determinant one is connected. We fix a k and consider only matrices of order $k \times k$. Recall the concept of convergence is entry wise; that is, a sequence of matrices M_n converges to a matrix iff for each i, j their (i, j)-th entries converge. This is same as identifying the space with an appropriate subset of Euclidean space of dimension k^2.

The plan is to show that it is path connected. Probably you can do easily using the Lie group formalism, or simple group theory and connected components, I have not thought about it. Probably, you can also do it by any one of several canonical reductions, like echelon form etc. But there are several hands on calculations, via: orthogonalize, normalize, (Gram-Schmidt) and rotate — go to identity matrix from any where. This is lengthy but you can easily walk through. Another one is to go to diagonal matrices and then go to identity. I shall outline the last one.

I said that 2×2 matrices will give you all the ideas needed, so you think of this simple case throughout. Let A be a $k \times k$, given matrix.

(i) No loss to assume that $a_{11} \neq 0$.
This is because you can connect A to a matrix where the first entry is non-zero (and then proceed to the next stage). Why? if a_{11} is already nonzero, there is nothing for you to do. if $a_{11} = 0$, then do the following. Pick j such that $a_{1j} \neq 0$ The matrix being nonsingular, the first row can not be zero. add j-th column to the first column. In other words consider the $k \times k$ matrix B with j th column as in A and other columns zero (do not worry this is singular). Consider the path $t \in [0, 1] \mapsto A + tB$. You must observe continuity before you accept this as a path. Since the determinant is not altered if you add multiple of a column to another column we see $\det(A + tB) = 1$ and thus we have a path in our space (matrices of determinant one) from A to a matrix with first entry non-zero. Now we forget the notation $A + B$ and assume that $a_{11} \neq 0$.

(ii) No loss to assume $a_{11} \neq 0$ and $a_{1j} = 0$ for $j > 1$.

subtract a multiple of the first column from the others. For example if $a_{12} \neq 0$, then consider the matrix B with second column as in A and other columns zero. consider the map: $t \in [0,1] \mapsto A - t\,\frac{a_{11}}{a_{12}}B$. As earlier this is a path in our space (matrices of det 1) and its (1,2)th entry is zero. Successively repeat with other columns where first entry is non-zero. remember when you do with other columns the remaining columns are not altered.

(iii) No loss to assume $a_{11} \neq 0$ and $a_{1j} = 0 = a_{j1}$ for $j > 1$.
This is because you now subtract appropriately the first row from others, one after the other, as done for columns in step (ii) above. thus we can as well pretend that our matrix has $a_{11} \neq 0$ and all other entries in first row and column are zero.

(iv) No loss to assume $a_{11} \neq 0$ and all other entries in first row and column are zero and $a_{22} \neq 0$ and all other entries in second row and column are zero.
This is as in earlier steps. First to make $a_{22} \neq 0$. If $a_{22} \neq 0$ there is nothing for you to do. If $a_{22} = 0$, observe that there must be a $j > 2$ such that $a_{2j} \neq 0$. Otherwise, second row would be zero (keep in mind we already made $a_{21} = 0$). Non-singularity says this is not possible. Now repeat step (i) to make sure $a_{22} \neq 0$ by adding a column $j > 2$ to the second. Observe this does not change first row, because $a_{1j} = 0$ for all $j \geq 2$. Then subtract appropriate multiple of second column from the columns $j > 2$ as in step (ii); then subtract second row from rows below appropriately as in step (iii). Since $a_{1j} = 0 = a_{j1}$ for all $j > 1$ this will not change first row and column.

(v) No loss to assume that A is a diagonal matrix.
In other words given any matrix $\det A = 1$ we can connect it by a path to a diagonal matrix of determinant one (within the space of determinant one matrices). Do the same way for other diagonal entries as was done in steps (i) (ii) (iii) for first row-column and in step (iv) for second row-column. Strictly speaking this should not be accepted as proof, because 'do the same way' has not been justified until we prove 'can be done the same way'. The correct way is to prove by induction: let 'Type A_r' be matrices B with first $r \times r$ corner being diagonal matrix with nonzero entries and $\det B = 1$. Prove by induction ($r = 1, 2, \cdots, k$): Given matrix A can be path connected to a matrix of 'type A_r'. For $r = 1$ first three steps will do. For the induction step, you need to reword step (iv). I am not spelling out the details. Of course if you do upto $r = k - 1$, you achieved $r = k$ too.

From now on let us pretend that A is a diagonal matrix $\mathrm{Diag}(d_1, d_2, \cdots, d_k)$

with $\det A = 1$. Clearly all d_i are nonzero. Since determinant is plus one, either all are positive or if there are negative entries they appear in pairs (that is even number of negative entries).

(vi) we can assume all diagonal entries are ± 1.
Suppose an entry is not ± 1, say d_1, then there must be another entry which is again not ± 1 because the product is one. say d_2. consider the following paths. First observe all the intervals below are non-degenerate and contained in $(0, \infty)$, so the the matrices (division by t) make sense and all have determinant one.

$$0 < |d_1| < 1 : \qquad t \in [1, 1/|d_1|] \mapsto \text{Diag}(td_1, \frac{d_2}{t}, d_3, \cdots d_k)$$

connects A to Diagonal det 1 matrix where first entry ± 1 (depending on whether $d_1 > 0$ or $d_1 < 0$.

$$|d_1| > 1 : \qquad t \in [1/|d_1|, 1] \mapsto \text{Diag}(td_1, \frac{d_2}{t}, d_3, \cdots d_k)$$

Connects a Diagonal det 1 matrix where first entry ± 1 to A.
If now another entry entry is not ± 1 then do the same thing as above. Since every time you are considering things both of which not ± 1, earlier changes are not effected.

(vii) to complete the proof we only need to connect a matrix A, diagonal, entries ± 1 and det 1 to I. Let us see what is the issue by specailizing to the simple 2×2 matrices. as noted already there are only two possibilities $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. In the first case nothing need to be done. In the second case you can successively connect

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \to \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \to \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix} \to \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

[subtract first column from second — add first row to second — subtract second row from first — add second column to first — subtract second row from first] All these row additions etc can be done as in step (i) and (ii). As I said this is just using row column operations. Soham suggested a simple way:

$$t \in [0, \pi] \quad \mapsto \quad \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}$$

You must check this is continuous and determinant one etc.

To complete the general case, now that we know that diagonal entries are ± 1, the entries -1 should occur even number of times and take two and convert as above. (Think what this means) this completes the proof. ∎

Baire category and polynomials:
Let f be an infinitely differentiable function on $[0,1]$.
$(\exists\ k)\ (\forall\ x)\ f^{(k)}(x) = 0$ implies f is a polynomial.

In fact it is a polynomial of degree at most k that appears in the existential quantifier. This you already know. It is a simple 'backward' argument. Take such k. since derivative of $f^{(k-1)}$ is zero conclude $f^{(k-1)}$ is a constant a_0. conclude $f^{(k-2)}$ is $a_0 + a_1 x$. Then conclude $f^{(k-3)}$ must be $a_0 + a_1 x + a_2 x^2$ etc. instead of etc you use induction. We have done it.

The problem to be settled is
$(\forall\ x)\ (\exists\ k)\ f^{(k)}(x) = 0$ implies f is a polynomial.

Thus for each x there is some number $n(x)$ such that the $n(x)$-th derivative at x equals zero. I had already mentioned that this is not an easy problem. Actually the idea is simple. Let us start with some observations. The first two observations spell out the idea and the rest is just an elaboration using Baire.

1°. Suppose we know $f^{(5)} = 0$ on $(1/3, 2/3)$ and $f^{(8)} = 0$ on $(1/4, 1/2)$, two overlapping intervals. Then actually $f^{(5)} = 0$ on the union $(1/4, 2/3)$.

This is because f is a polynomial $P(x)$ in $(1/3, 2/3)$ and a polynomial $Q(x)$ in $(1/4, 1/2)$. Thus in the common part f equals both P and Q. In other words $P = Q$ on a non-degenerate interval, thus the two polynomials must be same.

2°. Suppose (i) at every point x of C Cantor set, $f^{(8)}(x) = 0$ and (ii) f in each open interval complementary to C, we know f is a polynomial (depending on the interval). Thus in one deleted interval it may be poly of degree at most 20 and in other deleted interval it may be poly of degree at most 100. Then we claim that actually f is a polynomial of degree at most 8.

You see the interesting formulation. If $f^{(8)}(x)$ is zero for all points of an open interval you can say that f is a poly of degree at most 8 on that open interval. But if $f^{(8)}(x)$ is zero for all points of Cantor set, it does not make sense to say that f is a poly of degree at most 8 at all points of the Cantor set.

To get clear idea whether such a thing is possible at all, take the function $h(x) = d(x, C)$ which is continuous and is zero exactly at points of C. (We can give many examples of such continuous functions which are zero exactly at points of C). If

$$f(x) = \int_0^x h(y)dy$$

then derivative of f is zero exactly for points of the Cantor set. You can keep taking indefinite integrals. pause and think.

Let us prove the assertion 2°. The proof needs a couple of steps.

First we claim that not only $f^{(8)}(x)$ but $f^{(k)}(x)$ is zero for all $k \geq 8$ at all $x \in C$. The crucial point is that f is differentiable any number of times and every point of C is a limit point of C. Given $x \in C$, the second property enables you to take $p_n \in C$, $p_n \neq x$ and $p_n \to x$. The first property enables you to calculate

$$f^{(9)}(x) = \lim \frac{f^{(8)}(p_n) - f^{(8)}(x)}{p_n - x} = 0.$$

Thus $f^{(9)}(x) = 0$ for all $x \in C$. Now use induction. You see how set theory comes into force, that ever point of C is a limit point of C.

The plan is to show that at every point $f^{(8)}(x) = 0$. Of course this is granted to you for points in C. Need to show for points outside C, that is, in each of the deleted intervals.

So let us fix any interval (a, b) complementary to C. Thus a and b are in C. Let us say f is a poly of degree k on this interval. In case $k \leq 8$, then there is nothing for you to do; by differentiation, you get $f^{(8)}(x) = 0$ on this interval. Suppose $k > 8$. For instance say $k = 9$. Then $f^{(8)}(x)$ is a continuous function on $[a, b]$ and equals zero at end points and its derivative, namely $f^{(9)}$ is zero inside the interval should tell you, by integration (or mean value theorem?), $f^{(8)} \equiv 0$ on all of $[a, b]$.

In the general case you repeat the same argument. Suppose $k > 8$. Remember: we know higher derivatives are all zero at points of C. Thus $f^{(k-1)}$ is zero at the end points a and b of the interval $[a, b]$ and it is continuous and $f^{(k)}$ is zero inside. Thus $f^{(k-1)} \equiv 0$ on $[a, b]$. Now argue $f^{(k-2)} \equiv 0$ on $[a, b]$ and continue till you reach 8. This completes the proof of 2°.

3^0. Let us return to our problem. Set

$$A_k = \{x : f^{(k)}(x) = 0\} \qquad k =, 1, 2, 3, \cdots .$$

Then each A_k is closed because each derivative is continuous and $\cup A_k = [0, 1]$ by hypothesis. The sets A_k, as of now, may neither increase nor decrease. For example, $f(x) = x^3 + x$ has $f'(0) \neq 0$ but $f''(0) = 0$. For $g(x) = x^2$ we have $g'(0) = 0$ but $g''(0) \neq 0$.
However, their interiors A_k^o are increasing. This is because, if the k-th derivative is zero on an open interval, then the later derivatives are zero on that open interval.
But interestingly, A_k^0, being open, is disjoint union of non-empty open intervals, say, $I_p = (a_p, b_p) : p = 1, 2, \cdots$. Then none of these end points can be interior points at any time later. This is precisely the content of observation 1°. Thus any future interiors that get added are disjoint to what you already have. This observation has the following implication. Let

$$V = \cup A_k^o$$

say V is the disjoint union of open intervals $I_n : n \geq 1$. Remember any (non-empty) open set is union of disjoint non-empty open intervals in only one way. Then each of these I_n is already contained in some A_k^o. This in turn has the following implication. In each I_n our f is a polynomial.

Let us denote $P = [0, 1] - V$. There are two possibilities: either P is empty or not. Suppose it is empty, then V is all of $[0, 1]$. In other words the increasing union A_k^o covers all of the compact $[0, 1]$ and so must already equal this for some k and this then completes what we wanted to set out to prove. (unimportant, but note: From the analysis above, one A_k must be all of the interval).

Let us now assume that $P \neq \emptyset$. We claim that it must be a perfect set. Indeed it is closed being complement of open set V. If it has a point, say b, which is not a limit point, then, we have an interval (a, c), $a < b < c$ such that $(a, b) \subset V$ and $(b, c) \subset V$ but $b \notin V$. But then as mentioned above, each interval contained in V is actually contained in A_k for some k depending on that interval. Thus $f^{(i)} \equiv 0$ in (a, b) and $f^{(j)} \equiv 0$ in (b, c). If $m = i \vee j$ then $f^{(m)} = 0$ on both the intervals and hence, by continuity, on all of (a, c). Contradiction because then $(a, c) \subset A_m^o$.

We shall now repeat an argument which we used while proving that R is not a countable infinite union of non-empty disjoint closed sets. from above,

P is a non-empty perfect set (closed, every point is a limit point). And

$$P = (A_1 \cap P) \cup (A_2 \cap P) \cap (A_3 \cap P) \cup \cdots .$$

But P is a polish space in its own right and these are closed subsets of P and thus one of them must contain an open subset of P. Remember what are open subsets of P. Thus there is an k and open interval (a, b) such that

$$(a, b) \cap P \subset A_k \cap P.$$

Now argument 2° leads to a contradiction. Every point of P in (a, b) is a limit point of P and so not only $f^{(k)}(x) = 0$ for all $x \in (a, b) \cap P$ we actually have

$$f^{(m)}(x) = 0; \quad \forall m \geq k \quad \forall x \in (a, b) \cap P.$$

This will help you to show (either by differentiation or by integration, as the case may be) to show $f^{(k)}(x) = 0$ for all x in any of the deleted intervals. This ultimately shows that $f^{(k)}(x) = 0$ for all $x \in (a, b)$. In other words $(a, b) \subset A_k^o$ and would have no point in P. This is a contradiction.

We shall now return to Fourier Series.
Fix $f \in C_{per}[0, 1]$ with $f' \in C_{per}[0, 1]$. In other words f and f' are periodic continuous functions on R (when we say periodic we mean period one). We knew that $\sum |\widehat{f}_n|$ converges and so S_N is uniformly Cauchy. To show that it converges actually to f, we used uniqueness by showing that Fourier coefficients of limit function equal those of f. We now prove it without using uniqueness.

(13) (Dirichlet Theorem)
If $f \in C_{per}[0, 1]$ with $f' \in C_{per}[0, 1]$, then $S_n \to f$ uniformly.
Since S_N is uniformly Cauchy, it suffices to show that it converges to f pointwise.

To this end, let us understand the partial sums. Fix a point x.

$$S_N(x) = \sum_{-N}^{N} \hat{f}_k e_k = \sum \int f(y) e^{-2\pi iky} dy \ e^{2\pi ikx}$$

$$= \int f(y) \sum_{-N}^{N} e^{2\pi ik(x-y)} dy \ = = \int f(x-y) \sum_{-N}^{N} e^{2\pi iky} dy.$$

We have not shown range of integration, it is any interval of length one; this is because the integrand is of period one. Note that this does not affect the

change of variable we made for the last equality. Let us denote

$$D_N(\theta) = \sum_{-N}^{N} e^{2\pi ik\theta}.$$

Then we have

$$S_N(x) = \int_0^1 f(y) D_N(x-y)dy = \int f(x-y) D_n(y)dy$$

Using formula for sum of finite geometric series,

$$D_N(\theta) = e^{-2\pi iN\theta} \frac{1 - e^{2\pi i(2N+1)\theta}}{1 - e^{2\pi i\theta}}.$$

When $\theta = 0$, this is to be interpreted as $(2N+1)$.

$$D_N(\theta) = \frac{e^{-2\pi iN\theta} - e^{2\pi i(N+1)\theta}}{1 - e^{2\pi i\theta}} = \frac{e^{-2\pi i(N+1/2)\theta} - e^{2\pi i(N+1/2)\theta}}{e^{-\pi i\theta} - e^{\pi i\theta}}$$

$$= \frac{\sin \pi(2N+1)\theta}{\sin \pi\theta}.$$

This $D_N(\theta)$ is called Dirichlet Kernel.
As a consequence of the orthogonality of the exponential functions, we also get from the above summation,

$$\int_0^1 \frac{\sin \pi(2N+1)\theta}{\sin \pi\theta} d\theta = 1.$$

Thus

$$S_N(x) = \int_{-1/2}^{1/2} f(x-y) \frac{\sin \pi(2N+1)y}{\sin \pi y} dy.$$

Need to show $S_N(x)$ converges to $f(x)$. Remembering that D_N integrates to one we see

$$S_N(x) - f(x) = \int_{-1/2}^{1/2} [f(x-y) - f(x)] \frac{\sin \pi(2N+1)y}{\sin \pi y} dy.$$

$$= \int_{-1/2}^{1/2} \frac{[f(x-y) - f(x)]}{\sin \pi y} \sin \pi(2N+1)y \, dy.$$

Observe that the function

$$\varphi(y) = \frac{f(x-y) - f(x)}{\sin \pi y}$$

is a nice continuous function on $[-1/2, 1/2]$. Its value at zero is

$$\varphi(0) = f'(x) \cdot \frac{1}{\pi} \cdot (-1).$$

Thus we can write

$$S_N(x) - f(x) = \int \varphi(y) \left[e^{\pi i(2N+1)y} - e^{-\pi i(2N+1)y}\right] \frac{1}{2i} dy$$

$$= \int \varphi(y) e^{\pi iy} \frac{1}{2i} \ e^{2N\pi iy} dy - \int \varphi(y) e^{-\pi iy} \frac{1}{2i} \ e^{-2N\pi iy} dy$$

The first term on right side is $(-N)$-th Fourier coefficient of some continuous function and second term is N-th Fourier coefficient of a continuous function. We show soon that for continuous functions Fourier coefficients go to zero as $N \to \pm\infty$ (Earlier we did this for periodic C^1 functions). This shows that for each x, $S_N(x) - f(x) \to 0$. Since (S_N) is already shown to be uniformly Cauchy, we conclude that S_N converges to f uniformly. This completes proof of the theorem. ∎

Since our motivation came from Fourier's statement that periodic function is a superposition of sine and cosine functions we were considering periodic functions. Mathematically, there was no need to do that. Let us make a trivial generalization of what we have been doing so far – later we do a really nontrivial generalization!
Remember we were initially trying to imitate the finite dimensional case. Let $f \in C[0,1]$ not necessarily period one function, not even defined outside $[0,1]$, not necessarily differentiable, we define Fourier coefficients:

$$\widehat{f}_n = \int_0^1 f(x) e^{-2\pi inx} dx \qquad n = 0, \pm 1, \pm 2, \cdots .$$

Here are properties whose proofs are same as earlier.

(1B) (Linearity) $\widehat{f+g} = \widehat{f} + \widehat{g}$, i.e. $(\widehat{f+g})(n) = \widehat{f}(n) + \widehat{g}(n)$

(2B) (linearity) For $c \in \mathfrak{C}$, $\widehat{cf} = c\widehat{f}$; i.e. $(\widehat{cf})(n) = c\ \widehat{f}(n)$.

(3B) (conjugation) $\widehat{\overline{f}}\ (n) = \overline{\widehat{f}(-n)}$; here $\overline{f}$ is the function $\overline{f}(t) = \overline{f(t)}$.

(5B) (translation) If $a \in R$ and $g(t) = f(a+t)$ (addition modulo one) $\widehat{f}(n) = e^{2\pi ina}\widehat{f}(n)$. Warning: g need not be continuous, but bounded and has at most one discontinuity point and the integral defining $\widehat{g}$ makes sense

as a Riemann integral. Observe that in the values of g, the number $f(1)$ may not show up at all! For example take $f(x) = x$ on $[0,1]$ and $a = 1/2$. thus $g(x) = f(x + 1/2)$ when $0 \leq x < 1/2$ and $g(x) = f(x + (1/2) - 1)$ if $1/2 \leq x \leq 1$. Because of addition modulo one $f(1)$ does not show up. Also this g not continuously at $1/2$. Do not just believe me, Draw graph. Thus we are already using Fourier coefficients for non-continuous, but Riemann integrable, functions to prove this statement (5B).

As usual; define Fourier series and partial sums:

$$\sum_{-\infty}^{\infty} \widehat{f_n}\, e^{2\pi i n x} \quad \text{and} \quad S_N(x) = \sum_{-N}^{N} \widehat{f_n}\, e^{2\pi i n x}$$

Recall that the space $C[0,1]$ has metric d, via inner product;

$$d^2(f,g) = \langle f-g, f-g\rangle = \int_0^1 |f-g|^2; \quad d(f,g) = \sqrt{\int_0^1 |f-g|^2}$$

(7B) Does S_N capture 'all of f' which is in span $(e_n : |n| \leq N)$?
Yes $f - S_N \perp g$ for any g in the span of $(e_n : |n| \leq N)$.
Direct computation. Say $g = \sum\{c_n e_n : |n| \leq N\}$

$$\langle f - S_N, g\rangle = \langle f, \sum c_n e_n\rangle - \langle \sum \widehat{f}(n) e_n, \sum c_n e_n\rangle$$

$$= \sum \overline{c_n}\widehat{f}(n) \; - \sum_m \sum_n \widehat{f}(n)\overline{c_m}\langle e_n, e_m\rangle \; = \; 0$$

because in the last sum the $m \neq n$ terms are zero.

(8B) S_N is the closest to f in span $(e_n : |n| \leq N)$
If g is in span $(e_n : |n| \leq N)$, want to show $||f-g|| \geq \|f - S_N\|$. Since $f - g = (f - S_N) + (S_N - g)$ and $S_N - g$ is in span $(e_n : |n| \leq N)$, we conclude from above $f - S_N \perp S_N - g$ and hence

$$\|f-g\|^2 = \|f - S_N\|^2 + \|S_N - g\|^2 \qquad ||f-g|| \geq \|f - S_N\|$$

Here we used Pythogaras Theorem (also in Sulbasutras of India, centuries before Pythogaras); if $v \perp w$ then $\|v+w\|^2 = \|v\|^2 + \|w\|^2$. This follows by expanding $\langle v+w, v+w\rangle$ and using orthogonality.

(9B) Bessel's inequality
$f = (f - S_N) + S_N$ and $f - S_N \perp S_N$. So as above

$$\|S_N\|^2 + \|f - S_N\|^2 = \|f\|^2; \qquad \|S_N\|^2 \leq \|f\|^2$$

Noting that S_N is sum of orthogonal vectors, we get, whatever be N,

$$\sum_{-N}^{N} |\widehat{f}_n|^2 \leq \int_0^1 |f|^2$$

This is called Bessel's inequality. This being true for every N, we conclude

$$\sum_{-\infty}^{\infty} |\widehat{f}_n|^2 \leq \int_0^1 |f|^2$$

Later we show that actually equality holds above. This immediately leads to the following:

(14) Riemann-Lebesgue Lemma (a version): If $f \in C[0,\ 1]$ then $\widehat{f_n} \to 0$ as $n \to \pm\infty$.

this fulfils the promise made in the proof of (13).

Can we prove, by possibly different methods, that S_N converges to f even when f is not C^1? At least pointwise, if not uniformly? NO.

(15) There is a large set of $C_{per}[0,1]$ functions such that for each f in this set the partial sums are unbounded on a large set.
Here large means complement of a first category set. We shall see this later, a double application of Baire .

But what we can show is that partial sums do converge in the Cesaro sense.
(16) (Fejer's Theorem) Let $f \in C_{per}[0,1]$ and $S_N(x) = \sum\{\widehat{f}_n e_n : |n| \leq N\}$ partial sums for $N = 0, 1, 2, 3, \cdots$. define their successive averages

$$\sigma_n(x) = \frac{S_0(x) + S_1(x) + \cdots + S_{n-1}(x)}{n} \quad n = 1, 2, 3, \cdots$$

Then $\sigma_n \to f$ uniformly.

Let us see some interesting consequences before proving this theorem.

(17) (Denseness of Trigonometric polynomials) The set of Trigonometric polynomials is dense in $C(T)$ under sup norm (complex valued function space).
A trigonometric polynomial on $[0,1]$ is a finite linear combination of powers of $\sin 2\pi x, \cos 2\pi x$, just as ordinary polynomial is a linear combination of

powers of x. Since $\sin nx$ and $\cos nx$ can be expressed in terms of powers of $\sin x$ and $\cos x$ we can conclude that each $e^{2\pi inx} = \cos + i \sin$ is a trigonometric polynomial. Thus S_N is a trigonometric polynomial and so is $\sigma_n(x)$. So Fejer's theorem completes the proof.

(18) (Weierstrass) The set of polynomials in variable x is dense $C[1/4, 3/4]$ with sup metric. Polynomials are dense in $C[a, b]$ for any closed bounded interval.
The second statement follows from the first because we have a nice linear map between any two given closed bounded intervals and hence takes polynomials to polynomials. We shall prove the first.
Given f, extend it as a periodic function on $[0, 1]$ and get a trigonometric polynomial close to it (using above result), say finite linear combinations of 100 terms. each of the $\cos(2\pi nx)$ and $\sin(2\pi nx)$ have power series expansions that converge uniformly on any compact interval. So cut off the power series at a finite stage it is close to the terms $\cos(2\pi nx)$ and $\sin(2\pi nx)$ upto $\epsilon/100$ on the above interval. This does.
(I am a little vague if I am talking about real or complex linear combinations of powers of x. Think). this proof does appear silly since we had proved it using high school methods and Bernstein polynomials. But it is an exercise in using power series and cut-offs.

(19) (Weyl's equidistribution Theorem)
Let α be an irrational number in $[0, 1]$. Consider the sequence $\{x_n = n\alpha \mod 1 : n \geq 1\}$. This sequence is uniformly distributed. That is

$$\frac{\#\{k \leq n : x_k \in (a, b)\}}{n} \to (b - a) \qquad 0 < a < b < 1$$

Also for any continuous function $f \in C[0, 1]$ with $f(0) = f(1)$ the following holds.

$$\frac{1}{n} \sum f(x_k) \to \int_0^1 f(y)dy$$

More generally if z is any number in $[0, 1]$ and $x_k = z + k\alpha(\mod 1)$, then the above two statements are true.
First let us prove the function part. Suppose f is one of our exponential functions: $f(x) = e^{2\pi imx}$. If $m = 0$ this is the function one and integral, as well as averages are one. If $m \neq 0$, then the integral is zero where as the average is

$$\frac{1}{n} \sum_{k=1}^{n} e^{2\pi imk\alpha} = \frac{1}{n} \frac{e^{2\pi im(n+1)\alpha} - e^{2\pi imx}}{e^{2\pi im\alpha} - 1}$$

For the second factor on the right, observe denominator is independent of n and makes sense because, α being irrational, $m\alpha$ is never an integer and so denominator is never zero. Further numerator is at most two in modulus. thus the second factor is bounded and hence $(1/n)$ times that goes to zero.

Observe that if equality holds for two functions then it holds for linear combinations. In other words the equality holds for trigonometric polynomials. Let f be any periodic continuous function and $\epsilon > 0$. Choose, by Fejer, g a trigonometric polynomial so that $|f(x) - g(x)| < \epsilon/4$ for all x. Now

$$|\frac{1}{n}\sum f(x_k) - \int_0^1 f(y)dy| \quad \leq \quad |\frac{1}{n}\sum f(x_k) - \frac{1}{n}\sum g(x_k)| \quad +$$

$$|\frac{1}{n}\sum g(x_k) - \int_0^1 g(y)dy| \quad + \quad |\int_0^1 g(y)dy - \int_0^1 f(y)dy|$$

Note that

$$|\frac{1}{n}\sum f(x_k) - \frac{1}{n}\sum g(x_k)| = |\frac{1}{n}\sum [f(x_k) - g(x_k)]| \leq \epsilon/4$$

and

$$|\int_0^1 g(y)dy - \int_0^1 f(y)dy| \leq \int_0^1 |g(y) - f(y)|dy \leq \epsilon/4$$

and g being a trigonometric polynomial, we can choose n_0 so that for $n \geq n_0$

$$|\frac{1}{n}\sum g(x_k) - \int_0^1 g(y)dy| \leq \epsilon/4$$

Thus for $n \geq n_0$ we have

$$|\frac{1}{n}\sum f(x_k) - \int_0^1 f(y)dy| \leq \epsilon \quad n \geq n_0.$$

To prove the part regarding intervals, fix $0 < a < b < 1$. Note that the statement is exactly same as the above for indicator function. But $I_{(a,\,b)}$ is not continuous. We will again approximate. Fix $\epsilon > 0$ small so that $a - \epsilon > 0$ and also $b + \epsilon < 1$ and $2\epsilon < b - a$. define continuous h as follows: zero on $[0, a - \epsilon]$; increases to one by the time you reach a; stays one during $[a, b]$; decreases to zero by the time you reach $b + \epsilon$ and remains zero after $b + \epsilon$. Similarly define continuous g as follows: zero on $[0, a]$; increases to one by the time you reach $a + \epsilon$; stays one during $[a + \epsilon, , b - \epsilon]$; decreases to zero by the time you reach b and remains zero after b. Clearly g, h are periodic (value zero at end points) and

$$g(y) \leq I_{(a,b)}(y) \leq h(y) \quad \forall y$$

$$\frac{1}{n}\sum g(x_k) \le \frac{1}{n}\sum I_{(a,b)}(x_k) \le \frac{1}{n}\sum h(x_k)$$

$$\frac{1}{n}\sum g(x_k) \le \frac{\#\{k \le n : x_k \in (a,b)\}}{n} \le \frac{1}{n}\sum h(x_k)$$

so that

$$\liminf \frac{1}{n}\sum g(x_k) \le \liminf \frac{\#\{k \le n : x_k \in (a,b)\}}{n}$$

$$\le \limsup \frac{\#\{k \le n : x_k \in (a,b)\}}{n} \le \limsup \frac{1}{n}\sum h(x_k)$$

Using the result proved already for periodic functions

$$\int g(y)dy \le \liminf \frac{\#\{k \le n : x_k \in (a,b)\}}{n}$$

$$\le \limsup \frac{\#\{k \le n : x_k \in (a,b)\}}{n} \le \int h(y)dy$$

Since $|\int h - \int g| \le 4\epsilon$, by direct computation, we conclude that

$$\limsup \frac{\#\{k \le n : x_k \in (a,b)\}}{n} - \liminf \frac{\#\{k \le n : x_k \in (a,b)\}}{n} \le 4\epsilon$$

Since $\epsilon > 0$ is arbitrary, we conclude that

$$\lim \frac{\#\{k \le n : x_k \in (a,b)\}}{n}$$

exists. Thus

$$\int g(y)dy \le \lim \frac{\#\{k \le n : x_k \in (a,b)\}}{n} \le \int h(y)dy$$

Since

$$(a-b) - 2\epsilon \le \int g; \qquad \int h \le (a-b) + 2\epsilon$$

we conclude

$$(a-b) - 2\epsilon \le \lim \frac{\#\{k \le n : x_k \in (a,b)\}}{n} \le (a-b) + 2\epsilon$$

Since $\epsilon > 0$ is arbitrary, we conclude

$$\lim \frac{\#\{k \le n : x_k \in (a,b)\}}{n} = (a-b)$$

(20) We shall now prove

$$\sum \frac{1}{k^2} = \frac{\pi^2}{6}$$

Consider the function

$$f(x) = x, \qquad 0 \leq x \leq 1.$$

This is a continuous function and of course not periodic, takes different values at zero and one. Let us assume that Plancherel is true, we shall show this next.

$$\sum |\hat{f}_k|^2 = \int_0^1 |f|^2.$$

Let us see what it means.

$$\int_0^1 x^2 dx = 1/3.$$

$$\hat{f}_0 = \int_0^1 x dx = 1/2.$$

For $k \neq 0$, integration by parts gives

$$\int_0^1 x e^{-2\pi i k x} dx = -\frac{1}{2\pi i k}$$

Thus we have

$$\frac{1}{4} + 2\sum_1^\infty \frac{1}{4\pi^2 k^2} = \frac{1}{3}$$

In other words

$$\sum_1^\infty \frac{1}{k^2} = \frac{\pi^2}{6}.$$

It is very interesting, in the very first course we showed that the series $\sum(1/k^2)$ converges, but we had to wait for an year to find out what the sum is.

(19) continued:
The equidistribution theorem is sometimes stated in a different, equivalent, way in conformity with probabilistic concepts. As above, let α be an irrational number in $[0,1]$.Then

$$\frac{\#\{k \leq n : x_k \leq b\}}{n} \to b \qquad 0 \leq b \leq 1$$

Also for any continuous function $f \in C[0,1]$

$$\frac{1}{n}\sum f(x_k) \to \int_0^1 f(y)dy$$

Since we used periodic functions in the argument above we allowed a, b strictly between zero and one. Here is the proof of the statement above for any b. We only need to consider $0 < b < 1$. For any $\epsilon > 0$ we have

$$\frac{\#\{k \leq n : x_k \leq b\}}{n} \geq \frac{\#\{k \leq n : \epsilon < x_k < b\}}{n}$$

Thus using what we proved above,

$$\liminf \frac{\#\{k \leq n : x_k \leq b\}}{n} \geq b - \epsilon$$

This being true for any $\epsilon > 0$ we conclude

$$\liminf \frac{\#\{k \leq n : x_k \leq b\}}{n} \geq b \qquad (\bullet 1)$$

Similarly for any $\epsilon > 0$ we have

$$\frac{\#\{k \leq n : x_k > b\}}{n} \geq \frac{\#\{k \leq n : b < x_k < 1 - \epsilon\}}{n}$$

Thus using what we proved earliert,

$$\liminf \frac{\#\{k \leq n : x_k > b\}}{n} \geq 1 - \epsilon - b$$

This being true for any $\epsilon > 0$ we conclude

$$\liminf \frac{\#\{k \leq n : x_k > b\}}{n} \geq 1 - b \qquad (\bullet 2)$$

Now recall $\liminf a_n + \liminf b_n \leq \liminf(a_n + b_n))$ to conclude

$$\liminf \frac{\#\{k \leq n : x_k \leq b\}}{n} + \liminf \frac{\#\{k \leq n : x_k > b\}}{n} \leq 1 \qquad (\bullet 3)$$

Strict inequality in any one of ($\bullet$1) and ($\bullet$2) contradicts ($\bullet$3), we completed the proof.
The statement about functions can also be proved in several ways, for example, going through indicators of intervals, their finite linear combinations and then by using uniform continuity, for any continuous f; Or, reducing the issue to non-negative continuous functions and then approximating such continuous f with periodic functions. We shall not carry out the details.

(13B) Dirichlet a little better:
As usual fix f continuous periodic, not necessarily C^1. Also fix a point x. Let us assume that f is differentiable at the point x. Then The Dirichlet argument showed that

$$S_N(x) - f(x) = \int \varphi(y) \left[e^{\pi i(2N+1)y} - e^{-\pi i(2N+1)y}\right] \frac{1}{2i} dy$$

$$= \int \varphi(y) e^{\pi i y} \frac{1}{2i} \ e^{2N\pi i y} dy - \int \varphi(y) e^{-\pi i y} \frac{1}{2i} \ e^{-2N\pi i y} dy$$

where

$$\varphi(y) = \frac{f(x-y) - f(x)}{\sin \pi y}$$

is a nice continuous function on $[-1/2, 1/2]$. Its value at zero is

$$\varphi(0) = f'(x) \cdot \frac{1}{\pi} \cdot (-1).$$

Riemann-Lebesgue lemma applied to $\varphi(y)e^{\pm \pi i y}$ shows that $S_N(x) - f(x) \to 0$. In Other words, the theorem is local, if f is periodic and differentiable at a point x, then at that point its Fourier series converges to the value of the function. There are better versions but we shall not go to complicated matters and stop here.

(16) We shall now prove Fejer's theorem:
Let $f \in C_{per}[0,1]$ and $S_N(x) = \sum\{\widehat{f}_n \ e_n : |n| \leq N\}$ partial sums for $N = 0, 1, 2, 3, \cdots$. define their successive averages

$$\sigma_n(x) = \frac{S_0(x) + S_1(x) + \cdots + S_{n-1}(x)}{n} \quad n = 1, 2, 3, \cdots$$

Then $\sigma_n \to f$ uniformly.
Note that this is Cesaro convergence of Fourier partial sums S_N. Also observe there is no condition of differentiability on f, just that f be periodic continuous. Recall

$$S_N(x) = \int f(x-y)D_N(y)dy; \qquad D_N(y) = \frac{\sin(2N+1)\pi y}{\sin \pi y}$$

integration is over any interval of length one, we choose $(-1/2, +1/2)$. Thus

$$\sigma_n(x) = \int f(x-y)\frac{1}{n}\sum_{N=0}^{n-1} D_N(y)dy = \int f(x-y)F_n(y)dy$$

where F_n is the Fejer Kernel

$$F_n(y) = \frac{1}{n}\sum_{N=0}^{n-1} D_N(y)$$

First observe that $\int D_N = 1$ tells that $\int F_n = 1$ for all n.,

$$\sum_{N=0}^{n-1} D_N(y) = \frac{1}{\sin \pi y}\sum_{0}^{n-1} \sin(2N+1)\pi y = \frac{1}{\sin \pi y}\Im \sum_{0}^{n-1} e^{(2N+1)i\pi y}$$

where $\Im$ is imaginary part.

$$= \frac{1}{\sin \pi y}\Im \ \frac{e^{(2n+1)i\pi y} - e^{i\pi y}}{e^{2i\pi y} - 1} = \frac{1}{\sin \pi y}\ \Im \ \frac{e^{2ni\pi y} - 1}{e^{i\pi y} - e^{-i\pi y}}$$

$$= \frac{1}{\sin \pi y}\ \Im \ \frac{\cos(2n\pi y) - 1 + i\sin(2n\pi y)}{2i \sin \pi y}$$

$$= \frac{1}{(\sin \pi y)^2}\ \frac{1-\cos(2n\pi y)}{2} \quad = \quad \left[\frac{\sin n\pi y}{\sin \pi y}\right]^2$$

Thus Fejer Kernel is

$$F_n(y) = \frac{1}{n}\ \left[\frac{\sin n\pi y}{\sin \pi y}\right]^2$$

Hence

$$\sigma_n(x) - f(x) = \int [f(x-y) - f(x)]\ \frac{1}{n}\ \left[\frac{\sin n\pi y}{\sin \pi y}\right]^2 dy \qquad (\clubsuit)$$

Let now $\epsilon > 0$ be given. We exhibit n_0 so that for $n \geq n_0$ the quantity is at most ϵ. Our choice of n_0 does not depend on x, showing uniform convergence.

First, using uniform continuity of f, fix $0 < \delta < 1/2$ so that $|u-v| \leq \delta$ implies $|f(u) - f(v)| < \epsilon/2$. Then using $\int F_n = 1$

$$\int_{0 \leq |y| \leq \delta} |f(x-y) - f(x)| \; \frac{1}{n} \; \left[\frac{\sin n\pi y}{\sin \pi y} \right]^2 dy \; \leq \; \frac{\epsilon}{2} \int F_n \; = \; \frac{\epsilon}{2} \qquad (\star 1)$$

Using boundedness of f, fix $M > 0$ so that $|f(x)| \leq M$ for all x. When $\delta < |y| < 1/2$ we see $|\sin \pi y| \geq \sin \delta\pi$ so that

$$\int_{\delta \leq |y| \leq 1/2} |f(x-y) - f(x)| \; \frac{1}{n} \; \left[\frac{\sin n\pi y}{\sin \pi y} \right]^2 dy$$

$$\leq 2M \; \frac{1}{n} \; \frac{1}{(\sin \delta\pi)^2} \; \int_{\delta \leq |y| \leq 1/2} (\sin n\pi y)^2 dy \leq 2M \; \frac{1}{(\sin \delta\pi)^2} \quad \frac{1}{n} \qquad (\star 2)$$

Fix n_0 so that

$$2M \; \frac{1}{(\sin \delta\pi)^2} \quad \frac{1}{n_0} \; < \; \frac{\epsilon}{2}$$

Clearly using $(\star 1)$ and $(\star 2)$ in $(\clubsuit)$ we see that $|\sigma_n(x) - f(x)| < \epsilon$ for $n \geq n_0$. Since choice of n_0 does not depend on x, the proof of Fejer's Theorem is complete.

(21) Both $\int F_n$ and $\int D_N$ equal one. The difference between Fejer and Dirichlet kernels is this. $F_n \geq 0$ whereas D_n takes positive and negative values. Thus we can say $\int |F_n| = 1$ for all n. We can not say $\int |D_N| = 1$ for all N. In fact we have

$$\int |D_N| \to \infty$$

We shall prove this now.

$$\int_0^1 |D_N(y)| dy \;\; = \;\; \int_0^1 \left| \frac{\sin(2N+1)\pi y}{\sin \pi y} \right| dy \;\; \geq \;\; \int_0^1 \left| \frac{\sin(2N+1)\pi y}{\pi y} \right| dy$$

where we used $|\sin u| \leq |u|$ for any real number u; now change $(2N+1)\pi y = x$

$$= \int_0^{(2N+1)\pi} \left| \frac{\sin x}{\pi x} \right| dx \;\; \geq \;\; \sum_{k=0}^{2N} \int_{k\pi}^{(k+1)\pi} \left| \frac{\sin x}{\pi x} \right| dy$$

we ignored the integral from $2N$ to $(2N+1)$, note integrand is non-negative

$$\geq \;\; \sum_{k=0}^{2N} \frac{1}{\pi(k+1)} \int_{k\pi}^{(k+1)\pi} |\sin x| \;\; dx \;\; = \;\; \frac{2}{\pi} \sum_{k=0}^{2N} \frac{1}{(k+1)}$$

and hence $\int |D_N| \to \infty$ as claimed.

(15) We shall now show that for a large class of functions the partial sums S_N of its FS do not converge, in fact the partial sums are unbounded on a large set. More precisely, There is a set $A \subset C_{period}[0,1]$ with A^c first category (thus A is large by Baire category theorem) such that if you take $f \in A$ the set D_f^c is of first category where $D_f = \{t \in [0,1] : \sup |S_N(f,t)| = \infty\}$ (thus D_f is a large set).

Towards this end, fix a point $t \in [0,1]$. For notational convenience we fix $t = 0$, but the same proof works for any t. To start with we show B^c is first category, where

$$B = \{f \in C_{period}[0,1] : \sup_N |S_N(f,0)| = \infty\}$$

For each $k \geq 1$ consider the set

$$A_k = \{f \in C_{period}[0,1] : \sup_N |S_N(f,0)| \leq k\}$$

Each of these sets is a closed set because $f \mapsto S_N(f,0)$ is a continuous function on $C_{period}[0,1]$ and each set $\{f : |S_N(f,0)| \leq k\}$ is a closed set and A_k is just intersection (over N) of these closed sets. If each A_k has empty interior, we are done because then each A_k is nowhere dense and so $\cup A_k = B^c$ is of first category.
Suppose that there is an A_K which has non-empty interior. (♠)
Here is a contradiction: Say A_3 has non-empty interior, hence contains a ball, say $B(\varphi, a) \subset A_3$, no loss to take $0 < a < 1$. But then $B(0,a) \subset A_6$. Reason is this: if $f \in B(0,a)$ then $f + \varphi \in B(\varphi, a)$ and of course $\varphi \in B(\varphi, a)$ so that for any N

$$|S_N(f+\varphi, 0)| \leq 3; \qquad |S_N(\varphi, 0)| \leq 3$$

Now $S_N(f,0) = S_N(f+\varphi,0) - S_N(\varphi,0)$ gives

$$|S_N(f,0)| \leq |S_N(f+\varphi,0)| + |S_N(\varphi,0)| \leq 6$$

But now see the following: if $f \in B(0,2)$, then $(a/2)f \in B(0,a)$ hence for any N, $|S_N((a/2)f, 0| \leq 6$ or $|S_N(f,0)| \leq 6 \times (2/a) = 12/a$.

$$f \in B(0,2) \Rightarrow \sup_N |S_N(f,0)| \leq \frac{12}{a} \qquad (\clubsuit)$$

We show $\int |D_N| \leq 12/a$ for every N contradicting $\sup_N \int |D_N| = \infty$ which was proved earlier. In order to show this, fix N. consider D_N on $[-1/2, 1/2]$, then using expression for D_N in terms of sine functions we get a finite partition,

$$-\frac{1}{2} = t_0 < t_1 < t_2 < \cdots < t_M = +\frac{1}{2}$$

such that in each interval (t_i, t_{i+1}) the function D_N has same sign either positive or negative If h is the function which is $+1$ in an interval where D_N is positive and -1 in an interval where D_N is negative then we see — Remember $S_N(h, 0) = \int h(y)D(y)dy$;

$$S_N(h, 0) = \int hD_N = \int |D_N| \leq 12/a$$

the last inequality is from (♣) using $|h| \leq 1$. But the only catch with this argument is that h is not a continuous function and the entire argument is meaningless. But we make it correct as follows. Take small intervals around each of the partition points,

$$\Big(t_i - \frac{\epsilon}{2M},\ \ t_i + \frac{\epsilon}{2M}\Big)$$

Consider the function h_ϵ which equals the ± 1 as said earlier, but during each of these small intervals it continuously increases from $+1$ to -1 or continuously decreases from -1 to $+1$ as the case may be. Thus this h_ϵ is a nice function, continuous periodic, and bounded by one and so the earlier argument – which was illegal for h – applies to this h_ϵ giving

$$|S_N(h_\epsilon, 0)| = |\int h_\epsilon D_N| \leq 12/a$$

Observe that $h_\epsilon D_N$ and $|D_N|$ differ only on a finite union of intervals whose total length is at most ϵ. So the integrals differ by at most $(2N + 1)\epsilon$ — remember D_N has maximum $(2N + 1)$ and N is now fixed. Since $\epsilon > 0$ is arbitrary we conclude $\int |D_N| \leq 12/a$ This being true for any N we conclude $\sup \int |D_N|$ is finite which is not true.
thus our assumption (♠) is incorrect.
Thus changing notation the set

$$F^0 = \{f \in C_{period}[0, 1] : \sup_N |S_N(f, 0)| < \infty\}$$

is first category set. Since the same is true for any number $t \in [0, 1]$, let us take a dense countable set $D \subset [0, 1]$ and define

$$F^t = \{f \in C_{period}[0, 1] : \sup_N |S_N(f, t)| < \infty\} \qquad F = \cup_{t\in D} F^t$$

Then F, being countable union of sets of first category, is itself of first category. This is the set we promised. If $f \in F^c$, then the set $\{s : \sup_N |S_N(f,s)| = \infty\}$ is a dense set because it includes D. Also it is a countable intersection of open sets and hence Baire tells you its complement is of first category.

(22) Plancherel for $C[0,1]$:
Recall we extended the concept of Fourier coefficients to $C[0,1]$ from $C_{period}[0,1]$ and saw Bessel's inequality. We now show Plancherel equality:

$$\int |f|^2 = \sum |\widehat{f_n}|^2$$

We know that the space l_2 – the collection of complex sequences $a = (a_n : -\infty < n < \infty)$ with $\sum |a_n|^2 < \infty$ – is a complete metric space with $d(a,b) = \sqrt{\sum |a_n - b_n|^2}$. Indeed if you define

$$\langle a, b\rangle = \sum_{-\infty}^{\infty} a_j \overline{b_j}$$

then it makes sense, is an inner product, and $d(a,b) = \langle b-a, b-a\rangle$ — Just as you have in $\mathfrak{C}^n$ and just as we had in $C[0,1]$.

Clearly $C'_{period}[0,1]$ is a dense subset of $C[0,1]$ where our metric is

$$d(f,g) = \sqrt{\int |f-g|^2}$$

this is seen as follows: take f real continuous on $[0,1]$. Given $\epsilon > 0$, by Weierstrass get P polynomial $\epsilon/2$ close to f. This is differentiable but not periodic. So cut it off as follows. Take a C^∞ function φ on R which is zero till $t = 0$ and in creases to one by the time $t = \eta$ then stays one till $1-\eta$ and falls down to zero by the tine $t = 1$. Here $\eta > 0$ is fixed and small. Your Analysis 1 gives you such φ. consider the product $g = \varphi P$. It is C^∞, periodic, it differs from f by at most $\epsilon/2$ during $(\eta, 1-\eta)$. Now you can complete argument.

We know that the map $f \mapsto \widehat{f}$ from $C'_{period}[0,1]$ to l_2 is a distance preserving map, by Plancherel So you can extend it to all of $C[0,1]$ with values in l_2 as a distance preserving map. Here completeness of l_2 is important. What is this extension? is this our already defined hat map on all of $C[0,1]$? Yes because if you take $f^n \in C'_{period}[0,1]$ converging to $f \in C[0,1]$ then for any$k$,

$$\widehat{f_k^n} = \int f^n(t) e^{-2\pi i k t} dt \to \int f(t) e^{-2\pi i k t} dt = \widehat{f_k}$$

Since you know convergence in l_2 implies coordinate-wise convergence (though not conversely) this identifies the topological extension of hat map as the already defined hat map on $C[0,1]$. Thus hat map is distance preserving on $C[0,1]$ to l_2. This is precisely Plancherel.

(23) Final Destination!

The present situation is the following: consider $C[0,1]$ with the metric $d(f,g) = \sqrt{\int |f-g|^2}$ and the explicit hat map defined on it taking values in l_2. Is $C[0,1]$ complete? No, because taking g to be indicator of $(1/4, 3/4)$, you can get, by now familiar argument, $f_n \in C[0,1]$ with $\int |f_n - g|^2 \to 0$. Thus f_n is Cauchy and you can show there is no g in $C[0,1]$ to which it converges in our distance. But then Cantor tells you, you can always complete any metric space: so there is complete metric space (X, ρ) such that $(C[0,1], d)$ is a dense subset. This means X is complete under the metric ρ and $C[0,1]$ is a dense subset of X and the metric d agrees with ρ for points of $C[0,1]$.
Obviously you can extend the hat map as a distance preserving map to all of X with values in l_2. Question: What is this X? What is this map? is X also a collection of functions on $[0,1]$? is the extension also a directly defined integral map?
Beautiful answers to all these is the so called L_2 Fourier Series theory. This completion X can be identified as a collection of functions on $[0,1]$, denoted, $L_2[0,1]$. This extension of hat map can be identified as an integral. This extension map is now a distance preserving bijection of L_2 to l_2.

But this theory has to wait!
If you are feeling tired with what all we have been doing, relax and relax. No more theory. Just pause and reflect on what all this is about and how we ended in this (mess or beautiful garden – depends on you, think without any fear of exams). We do a couple of fun things and stop.

(24) Jacobi identity:
For real number $t > 0$ define

$$\vartheta(t) = \sum_{-\infty}^{\infty} e^{-n^2 \pi t.}$$

This is called theta function. The series converges because $a = \exp\{-\pi t\} < 1$ and geometric series $\sum a^n$ converges and the above series is dominated by

this geometric series. Here is an useful identity due to Jacobi.

$$\vartheta(t) = \frac{1}{\sqrt{t}}\,\vartheta(1/t).$$

Apparently this identity is useful in many contexts: Riemann zeta function, number theory, statistical physics (unfortunately I do not know much). Not only of theoretical importance but also of computational importance. If t is very small the closeness you get to the actual value by summing twenty terms on left side is apparently obtained with just one term of the right side. We shall prove this identity now. Consider the function

$$f(x) = \sum_{-\infty}^{\infty} e^{-(x-n)^2/2t}; \qquad 0 \le x \le 1.$$

This series is uniformly convergent. In fact, fix any $x \in [0,1]$. If $|n| = 2$, then $(x-n)^2 \ge 1^2$; if $|n| = 3$, then $(x-n)^2 \ge 2^2$; and so on; thus this series is dominated (beyond the terms $n = 0, \pm 1$) again by a convergent series of numbers and is hence uniformly convergent.
In particular this series defines a continuous function on $[0,1]$. This is also periodic, $f(0) = f(1)$; it is the same sum, just the terms get shifted. Let us compute its Fourier coefficients.

$$\hat{f}_k = \int_0^1 \sum_{-\infty}^{\infty} e^{-(x-n)^2/2t} e^{-2\pi i k x} dx;$$

Because of uniform convergence you can interchange the order of integration and summation;

$$= \sum_{-\infty}^{\infty} \int_0^1 e^{-(x-n)^2/2t} e^{-2\pi i k x} dx;$$

Now a change of variable and the realization that $\exp\{2\pi i k n\} = 1$ for any integer n will tell us the following. For example take the term $n = -1$

$$\int_0^1 e^{-(x+1)^2/2t} e^{-2\pi i k x} dx = \int_1^2 e^{-y^2/2t} e^{-2\pi i k y} dy$$

Thus

$$\widehat{f}_k = \int_{-\infty}^{\infty} e^{-y^2/2t} e^{-2\pi i k y} dy.$$

Fortunately, this is something well known, characteristic function of the normal distribution. We show

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2} e^{iux} dx = e^{-u^2/2}. \qquad (\bigstar)$$

Changing the variable x to $y/\sqrt{t}$ we get

$$\frac{1}{\sqrt{2\pi t}}\int_{-\infty}^{\infty} e^{-y^2/2t}e^{iuy/\sqrt{t}}dy = e^{-u^2/2}.$$

Remember $t > 0$ is fixed. Use this formula with $u = -2\pi k\sqrt{t}$ to see

$$\frac{1}{\sqrt{2\pi t}}\int_{-\infty}^{\infty} e^{-y^2/2t}e^{-2\pi ikty}dy = e^{-4\pi^2k^2t/2} = e^{-2\pi^2k^2t}.$$

or

$$\int_{-\infty}^{\infty} e^{-y^2/2t}e^{-2\pi ikty}dy = \sqrt{2\pi t}\ \ e^{-2\pi^2k^2t}.$$

Thus returning to our calculation

$$\widehat{f}_k = \sqrt{2\pi t}\ \ e^{-2\pi^2k^2t}.$$

What an achievement! we have been able to evaluate a rather complicated looking integral, the integrand is itself an infinite series.
Will someone tells us that the Fourier expansion is valid for our function, that is

$$f(x) = \sum_{-\infty}^{\infty} \widehat{f}_n\ \ e^{2\pi inx}.$$

Yes, because $\sum |\widehat{f}_k| < \infty$. Thus we see that for every $x \in [0,1]$

$$\sum_{-\infty}^{\infty} e^{-(x-n)^2/2t} = \sum_{-\infty}^{\infty} \sqrt{2\pi t}\ \ e^{-2\pi^2n^2t}e^{2\pi inx}.$$

Read this equation for $x = 0$ to get

$$\sum_{-\infty}^{\infty} e^{-n^2/2t} = \sum_{-\infty}^{\infty} \sqrt{2\pi t}e^{-2\pi^2n^2t}.$$

or

$$\frac{1}{\sqrt{2\pi t}}\sum_{-\infty}^{\infty} e^{-n^2/2t} = \sum_{-\infty}^{\infty} e^{-2\pi^2n^2t}.$$

read this equation for $t/2\pi$ instead of t.

$$\frac{1}{\sqrt{t}}\sum_{-\infty}^{\infty} e^{-n^2\pi/t} = \sum_{-\infty}^{\infty} e^{-n^2\pi t}.$$

This is precisely what we are looking for.
it remains to prove (★) which is we do next.

We shall now prove

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-x^2/2}e^{iux}dx = e^{-u^2/2}. \qquad (\bigstar)$$

One way to argue is to rewrite it as

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-x^2/2}e^{iux}e^{u^2/2}dx = 1; \qquad \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-(x-iu)^2/2}dx = 1$$

Substitute $x - iu = y$ and u being constant we have $dx = dy$ so that the above amounts to proving

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-y^2/2}dy = 1$$

which is true because normal density integrates to one. The trouble is that the substitution using complex constant is illegal. However this can be rectified when you learn contour integration in complex analysis. However simpler route is at our disposal.
Recalling our definition of integral of complex valued functions on R, we need to show

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-x^2/2}\cos(ux)dx + i\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-x^2/2}\sin(ux)dx = e^{-u^2/2}.$$

Since the second integrand is integrable (continuous and bounded in modulus by normal density) and since it is odd function, the integral is zero. Thus need to show

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-x^2/2}\cos(ux)dx = e^{-u^2/2}.$$

Let

$$\varphi(u) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-x^2/2}\cos(ux)dx$$

Then, differentiating under the integral sign which can be justified using what we did in analysis 1 (but we shall not do because it involves recalling definition of integral over unbounded interval, and theorem on changing the order of integration/differentiation of a nice function on a bounded interval and then taking limit as the interval increases to R).

$$\varphi'(u) = -\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-x^2/2}\, x\sin(ux)dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{d}{dx}[e^{-x^2/2}] \sin(ux)dx \;=\; -\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2}\; u \;\; \cos(ux)dx$$

where we integrated by parts (restores minus sign)

$$= -u\varphi(u)$$

Thus the unknown function φ satisfies $\varphi'(u) = -u\varphi(u)$. Hence for some constant c, we must have

$$\varphi(u) = ce^{-u^2/2}$$

(reason: have $\varphi'(u) + u\varphi(u) \equiv 0$, since exponential never vanishes, this is same as $e^{u^2/2}\ \varphi'(u) + u\ e^{u^2/2}\ \varphi(u) \equiv 0$ or $[e^{u^2/2}\varphi(u)]' \equiv 0$. Thus $e^{u^2/2}\varphi(u) = c$ for some constant c).
direct verification show that $\varphi(0) = 1$, so that we must have $c = 1$.

$$\varphi(u) = e^{-u^2/2}$$

This completes the proof.

(25) Parseval equation:
Before our last application, there is an easy and useful equality gto understand. Plancherel says that $\int |f|^2$ same as $\sum |\widehat{f}_n|^2$. Parseval says something more (actually equivalent) namely $\langle f, g\rangle = \langle\ \widehat{f}, \widehat{g}\ \rangle$. Clearly if you take $g = f$, you get Plancherel.

One way is to proceed imitating Plancherel. Let f, g be given in $C^1_{period}[0, 1]$. Denote S_N and T_N the Fourier partial sums for f and g. Then $S_N \to f$ and $\overline{T}_N \to \overline{g}$, uniformly so that $S_N\overline{T}_N \to f\overline{g}$ uniformly. So $\int S_N\overline{T}_N \to \int f\overline{g}$. But $\int S_N\overline{T}_N = \sum_{-N}^{+N} \widehat{f}_n\overline{\widehat{g}_n}$ converges to the infinite sum which is $\langle\widehat{f}, \widehat{g}\rangle$. Then by approximations you can do for functions in $C[0, 1]$.

But linearity considerations give you Parseval from Plancherel as follows, which uses only properties of inner product. A formula similar to $(x+y)^2 - (x-y)^2 = 2(xy+yx)$ is obtained by expanding the inner product on left side below.

$$\langle f+g, f+g\rangle \;-\; \langle f-g, f-g\rangle \;=\; 2[\langle f, g\rangle + \langle g, f\rangle] \quad (1)$$

$$\langle f+ig, f+ig\rangle \;-\; \langle f-ig, f-ig\rangle \;=\; 2[\langle f, ig\rangle + \langle ig, f\rangle]$$

$$= 2i[-\langle f, g\rangle + \langle g, f\rangle]$$

$$i[\langle f+ig, f+ig\rangle \;-\; \langle f-ig, f-ig\rangle] \;=\; 2[\langle f, g\rangle - \langle g, f\rangle] \quad (2)$$

Adding (1) and (2),

$$\langle f+g, f+g\rangle \ - \ \langle f-g, f-g\rangle \ + i[\langle f+ig, f+ig\rangle \ - \ \langle f-ig, f-ig\rangle] \\ = 4\langle f, g\rangle$$

Similar equality holds for $\widehat{f}$ and $\widehat{g}$. Write it. Plancherel tells then left sides of these two equations are same and so right sides must be equal, that is, $\langle f, g\rangle = \langle \widehat{f}, \widehat{g}\rangle$

(26) An Isoperimetric problem:
Our last application is the following: Suppose you have a region in the plane bounded by a smooth curve of fixed length, say, length one. How large can be its area and what shape of the region achieves that maximum area? Here is the beautiful answer.
Among all plane regions bounded by a simple closed smooth curve of length one, the circle encloses the largest area.

First let us understand the terms involved. Shall do enough that is needed for our problem and also to get a feel for the terms, shall not do in full generality.
A curve means a continuous function defined on a finite subinterval of R with values in R^2. Name the interval as $[a, b]$ and for each t in this interval, name the x and y coordinates of the value of our function by $x(t)$ and $y(t)$ respectively. Thus our curve is

$$t \in [a, b] \ \mapsto \ (x(t), y(t)) \in R^2$$

where $x(t)$ and $y(t)$ are real valued continuous functions. Of course you can think of the range of the map also as the curve.
[Soon certain confusions will disappear. For example consider the curves $t \in [0, 2\pi] \ \mapsto \ (\cos t, \sin t) \in R^2$ and $t \in [0, 4\pi] \ \mapsto \ (\cos t, \sin t) \in R^2$. Both are unit circles. First one traces it once and the second traces it twice. As curves they are different, but if you think of the range of the map as the curve, then ranges are same. So at this stage, it is too early to say that you can think of the curve as range of the map. Wait, we put enough conditions o. our curve soon.]

A curve is smooth if both x and y are continuously differentiable functions of t on the interval $[a, b]$. For a smooth curve its length is defined by the formula:

$$\int_a^b \sqrt{\dot{x}^2(t) + \dot{y}^2(t)}dt \qquad (\spadesuit)$$

where dot denotes derivative. The reason is the following. How do you measure length? Since we know length of straight line, we try to approximate the curve by straight lines as follows. take any (finite) partition

$$\pi = \{a = t_0 < t_1 < \cdots < t_k = b\}$$

of the interval $[a, b]$. Consider the chords joining the successive points on the curve corresponding to these partition points and take their total length:

$$L_\pi = \sum_0^{k-1} \sqrt{[x(t_{i+1}) - x(t_i)]^2 + [y(t_{i+1}) - y(t_i)^2]}$$

Straight line distance being the shortest between two points, you see that the length of the curve must be larger than such numbers as above, whatever be the partition π. If the partition becomes finer the length increases (does not decrease) by triangle inequality. Also as the partition becomes finer and finer the polygonal curve obtained by our chords, approaches the original curve. Thus it is reasonable to define length of the curve, name the curve by C, as

$$L(C) = \sup L_\pi = \sup_\pi \sum_0^{k-1} \sqrt{[x(t_{i+1}) - x(t_i)]^2 + [y(t_{i+1}) - y(t_i)^2]}$$

$$= \lim_{\|\pi\| \to 0} \sum_0^{k-1} \sqrt{[x(t_{i+1}) - x(t_i)]^2 + [y(t_{i+1}) - y(t_i)^2]}$$

where the last equality is because as the partition becomes finer the quantity L_π increases (does not decrease). Recall $\|\pi\|$ is the length of the largest interval of the partition π. By mean value theorem there are points θ_i and η_i in $[t_i, t_{i+1}]$ so that

$$[x(t_{i+1}) - x(t_i)] = \dot{x}(\theta_i) \times (t_{i+1} - t_i); \quad [y(t_{i+1)} - y(t_i)] = \dot{y}(\eta_i) \times (t_{i+1} - t_i)$$

$$L = \lim_{\|\pi\| \to 0} \sum_0^{k-1} \sqrt{\dot{x}^2(\theta_i) + \dot{y}^2(\eta_i)} \ (t_{i+1} - t_i). \qquad (\bullet)$$

How nice if we had, instead of $(\bullet)$

$$L = \lim_{\|\pi\| \to 0} \sum_0^{k-1} \sqrt{\dot{x}^2(\theta_i) + \dot{y}^2(\theta_i)} \ (t_{i+1} - t_i). \qquad (\bullet\bullet)$$

Because then, by definition of Riemann integral (bounded interval, continuous function) this last expression is nothing but $(\spadesuit)$ as required. We now

complete the justification of the length formula by arguing that these two limits are same. This is achieved by showing

$$\sum_0^{k-1} \left| \sqrt{\dot{x}^2(\theta_i) + \dot{y}^2(\eta_i)} \;\; - \;\; \sqrt{\dot{x}^2(\theta_i) + \dot{y}^2(\theta_i)} \right| \;\; (t_{i+1} - t_i) \to 0 \qquad (\clubsuit)$$

Take any three numbers a, b, c. Consider the triangle with vertices $(a, 0)$, $(0, b)$ and $(0, c)$ it has side lengths $\sqrt{a^2 + b^2}$, $|b - c|$ and $\sqrt{a^2 + c^2}$. Thus

$$\sqrt{a^2 + b^2} \leq \sqrt{a^2 + c^2} + |b - c|, \quad \sqrt{a^2 + c^2} \leq \sqrt{a^2 + b^2} + |b - c|$$

Boundary cases like $b = c$ will not give you triangle, but direct verification is trivial. The above inequalities are nothing but

$$|\sqrt{a^2 + b^2} - \sqrt{a^2 + c^2}| \leq |b - c|$$

As a result ($\clubsuit$) follows if we show

$$\sum_0^{k-1} |\dot{y}(\eta_i) - \dot{y}(\theta_i)| \; (t_{i+1} - t_i) \to 0$$

But this is immediate as follows. Given $\epsilon > 0$, use uniform continuity of the function $\dot{y}$ on the interval $[a, b]$, choose δ so that $|s - t| < \delta$ implies $|\dot{y}(s) - \dot{y}(t)| < \epsilon/(b - a)$. Now Take any partition π with $\|\pi\| < \delta$. Since θ_i and η_i are in the same partition interval you see that the above sum is at most

$$\frac{\epsilon}{b - a} \sum (t_{i+1} - t_i) = \epsilon.$$

Thus ($\spadesuit$) is justified as length of the curve. Just notice that the curves we discussed earlier – of going around circle once, going around twice – will have lengths 2π and 4π respectively; as it should be.

We say that the curve is simple if the following is true: $(x(t), y(t)) = (x(s), y(s))$ for two different points s, t then $\{s, t\} = \{a, b\}$. Thus the curve can not return to a point on the curve unless it is the end point returning to starting point. Of course it may not return at all, thus for different points t the pairs $(x(t), y(t))$ are different. Let us consider a simple smooth curve C having length L. Then we can reparametrize it with arc length, that is with interval $[0, L]$ so that as you trace the curve starting from zero, the length of the curve traced till any point u equals u.

Suppose the curve is $(x(t), y(t))$ defined on $[a, b]$ with length L. Consider the map

$$t \in [a, b] \mapsto \psi(t) = \int_a^t \sqrt{\dot{x}^2(s) + \dot{y}^2(s)} ds \in [0, L]$$

For future use observe

$$\dot{\psi}(t) = \sqrt{\dot{x}^2(t) + \dot{y}^2(t)}; \quad (\dot{\psi}(t))^2 = \dot{x}^2(t) + \dot{y}^2(t) \qquad (\star)$$

Observe $\psi(a) = 0$ and $\psi(b) = L$. And ψ is strictly increasing because the curve being simple, at least one of $\dot{x}$ and $\dot{y}$ is non-zero in any interval and so the integral strictly increases over any interval. And ψ is differentiable, being indefinite integral of a continuous function. As a result we have inverse map which is differentiable:

$$u \in [0, L] \mapsto \varphi(u) \in [a, b]; \quad \varphi(\psi(t)) = t \quad \psi(\varphi(u)) = u$$

Note $\varphi(0) = a$ and $\varphi(L) = b$ and φ is strictly increasing and differentiable.

$$(\dot{\varphi})(\psi(t))\ (\dot{\psi}(t)) = 1; \qquad (\dot{\psi})(\varphi(u))\ (\dot{\varphi}(u)) = 1$$

all these follow from analysis 1.

We define the curve (x_1, y_1) by

$$x_1(u) = x(\varphi(u)); \qquad y_1(u)) = y(\varphi(u)); \quad u \in [0, L]$$

Clearly as u goes from zero to L; φ goes from a to b; (x_1, y_1) goes from $(x(a), y(a))$ to $(x(b), y(b))$. In other words the same curve (x, y) is traced by (x_1, y_1) in the same order but with 'uniform speed'. This last phrase will be justified now by showing that at any point the vector $(\dot{x}_1, \dot{y}_1)$ is a unit vector. Fix $u \in [0, L]$, denote $\varphi(u) = t$ so that $\psi(t) = u$.

$$\dot{x}_1(u) = \dot{x}(\varphi(u))\ \dot{\varphi}(u), \quad \dot{y}_1(u) = \dot{y}(\varphi(u))\ \dot{\varphi}(u)$$

$$\dot{x}_1(u) = \dot{x}(t)\ \dot{\varphi}(u), \quad \dot{y}_1(u) = \dot{y}(t)\ \dot{\varphi}(u)$$

$$(\dot{x}_1)^2(u) + (\dot{y}_1)^2(u) = [(\dot{x})^2(t) + (\dot{y})^2(t)]\ (\dot{\varphi})^2(u)$$

$$= [(\dot{x})^2(t) + (\dot{y})^2(t)]\ \frac{1}{(\dot{\psi}(t))^2} = 1$$

where we used $(\star)$ for last equality. In other words for any $u \in [0, L]$ the vector $(\dot{x}_1, \dot{y}_1)$ is a unit vector.

A curve is simple closed if it is simple and $(x(a), y(a)) = (x(b), y(b))$. Thus the curve returns to its starting point and never crosses itself.

A simple closed curve (x, y) is smooth if the curve is continuously differentiable as a periodic function on $[a, b]$. In other words $\dot{x}$ and $\dot{y}$ are continuous functions and $\dot{x}(a) = \dot{x}(b)$. and $\dot{y}(a) = \dot{y}(b)$.

The final term we need to understand is 'the area enclosed'. Let C be a simple smooth closed curve. Pictorially it is obvious (but mathematically highly non-trivial to prove) that the plane is divided into three parts (i) the curve traced (ii) the interior of the curve (bounded region) and (iii) the exterior of the curve (unbounded region). The area enclosed by the curve is the area in the interior of the curve. Here is a beautiful formula. if the curve is described by (x, y) then the area enclosed, denoted by A, is given by

$$A = \frac{1}{2} \int_a^b [x(t)\dot{y}(t) - \dot{x}(t)y(t)]dt$$

(there is an important point but I did not want to get into too many details. The point is that the region should be to your left as you move along the curve, remember curve has no direction unless you give it. When you parametrize, you are giving a direction as t goes from a to b, you are tracing the curve from $(x(a), y(a))$ to $(x(b), y(b))$. For example you can describe/parametrize the circle either clockwise or anticlockwise) This formula is simple high school formula if you have a rectangle. For example the rectangle $S = [4, 7] \times [6, 8]$ has area 6. It is described by the formula (not smooth, but only has few corners, you can use any parametrization) as follows. The t-interval is $[0, 8]$ and $(x(t), y(t))$ are
For $t \in [0, 3] \ \mapsto (4 + t, 6)$ giving the side: $(4, 6)$ to $(7, 6)$
For $t \in [3, 5] \ \mapsto (7, t + 3)$ giving the side $(7, 6)$ to $(7, 8)$
For $t \in [5, 8] \ \mapsto (12 - t, 8)$ giving the side $(7, 8)$ to $(4, 8)$
For $t \in [8, 10] \ \mapsto (4, 16 - t)$ giving the side $(4, 8)$ to $(4, 6)$
In this case The integrand above equals -6 and 7 and 8 and -4 respectively in those four intervals. Thus The right side above equals

$$\frac{1}{2}\left[\int_0^3 (-6) + \int_3^5 (7) + \int_5^8 (8) + \int_8^{10} (-4)\right]$$

$$= \frac{1}{2}[-18 + 14 + 24 - 8] = 6$$

as it should be. For general region, you cut into rectangles and argue via limits.

Now let us return to our problem. We have a smooth simple closed curve of length one. Let us use arc length parameter for describing the curve. Thus

we have $t \in [0,1] \mapsto (x(t), y(t))$ and at each point as we saw above $(\dot{x}(t), \dot{y}(t))$ is a unit vector. In particular

$$\int_0^1 (\dot{x})^2 + (\dot{y})^2 = 1 \qquad (\dagger)$$

$$\frac{1}{2}\int_a^b [x(t)\dot{y}(t) - \dot{x}(t)y(t)]dt = A \qquad (\dagger\dagger)$$

The first equation invites Plancherel and the second invites Parseval. Even though you do not see conjugates, our functions being real you can pretend there are. We can rewrite them as follows:

$$\sum_{-\infty}^{\infty} \left[|\widehat{\dot{x}}_n|^2 + |\widehat{\dot{y}}_n|^2\right] = 1; \quad \frac{1}{2}\sum_{-\infty}^{\infty} \left[\widehat{x}_n\overline{\widehat{\dot{y}}_n} - \overline{\widehat{\dot{x}}}_n\widehat{y}_n\right] = A$$

Use $\widehat{\dot{x}}_n = 2\pi i n \widehat{x}_n$ and $\widehat{\dot{y}}_n = 2\pi i n \widehat{y}_n$

$$4\pi^2 \sum n^2 \left[|\widehat{x}_n|^2 + |\widehat{y}_n|^2\right] \;=\; 1 \qquad \pi i \sum n \left[\widehat{x}_n\overline{\widehat{y}_n} - \overline{\widehat{x}}_n\widehat{y}_n\right] = A$$

$$i\left[\widehat{x}_n\overline{\widehat{y}_n} - \overline{\widehat{x}}_n\widehat{y}_n\right] = i \times 2i \;\; \Im(\widehat{x}_n\overline{\widehat{y}_n}) = -2 \;\; \Im(\widehat{x}_n\overline{\widehat{y}_n})$$

$$4\pi^2 \sum n^2 \left[|\widehat{x}_n|^2 + |\widehat{y}_n|^2\right] \;=\; 1 \qquad -2\pi \sum n \;\; \Im(\widehat{x}_n\overline{\widehat{y}_n}) = A$$

Let $\widehat{x}_n = a_n + ib_n$ and $\widehat{y}_n = c_n + id_n$ Then the above equalities reduce to

$$\pi \sum n^2(a_n^2 + b_n^2 + c_n^2 + d_n^2) = \frac{1}{4\pi} \qquad 2\pi \sum n\,(a_n d_n - b_n c_n) = A \quad (\bigstar)$$

We want to show circle encloses maximum area. Since length of curve is one we must have $2\pi r = 1$ or $r = 1/(2\pi)$. Then area is $\pi r^2 = 1/(4\pi)$. Thus our statement amounts to $A \leq 1/(4\pi)$ and if equality holds then it must be a circle. Using $(\bigstar)$ we show $\frac{1}{4\pi} - A \geq 0$ and if equality holds then it is circle. This is just high school algebra which I messed up and wasted your time. Next time we shall quickly do it.